VMWARE NSX INTELLIGENCE AND NSX APPLICATION PLATFORM

Deployment / POC Guide

Contents

Version History

Date	Rev.	Author	Description	Contributors
22 Jul 2020	1.0	Ray Budavari	Initial Document – NSX Intelligence 1.0	
14 Aug 2020	1.1	Ray Budavari	NSX Intelligence 1.1 and Formatting Updates	
2 Feb 2022	2.0	Ray Budavari	Addition of NSX Application Platform	
7 Mar 2022	2.1	Ray Budavari	Add Sections for TKGs and NSX-T plus Tanzu Community Edition	
25 Apr 2022	2.2	Ray Budavari	Include updated Feature Enablement content	Shank Mohan
28 Oct 2022	3.0	Ray Budavari	Update for NSX 4.0.x	

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Overview

This document details the requirements and implementation for either a Production Deployment or Proof of Concept (PoC) evaluation of NSX Intelligence v3.2 or newer released with VMware NSX.

NSX Intelligence, a native and distributed NSX analytics engine, provides deep security visibility, policy recommendations, network traffic analysis and more, as an extension of the NSX Network and Security Virtualization Platform and is deployed and managed integrated directly within the NSX interface.

As of NSX-T Data Center 3.2, VMware has introduced the NSX Application Platform (NAPP). This is a new microservices-based solution that provides a highly available, resilient, scale-out architecture to deliver a set of core platform services which run several new NSX features such as:

- **NSX Intelligence 3.2 (with Scale-Out)**
- **NSX Metrics**
- **NSX Network Detection and Response**
- **NSX Malware Prevention**

For additional details on features and capabilities of NSX and Intelligence (now at version 4.0), please refer to the product documentation on: https://docs.vmware.com/en/VMware-NSX-Intelligence/4.0/install-upgrade/GUID-87C3B75F-6403-4DBB-A27F-6E91FD98CED3.html

The scope of the Deployment and PoC Guide is to provide a detailed, visual deployment document to simplify installations by providing a prescriptive step-by-step process that covers all details including the base infrastructure. It covers everything you need to successfully deploy NSX Application Platform and any features which run on this Platform. In addition, it captures best practices for deployment, common or well-known issues, and troubleshooting steps.

As a comprehensive reference the guide covers the deployment process in extensive detail to ensure you can use this environment in production and have a full awareness of the requirements and steps performed.

Note: For deployment it is **strongly recommended** to use the **NSX Application Platform Automation Appliance** which provides an end to end fully automated deployment of NAPP. This Automation Appliance integrates with NSX as a native UI Plugin, performs detailed environment validation and dramatically simplifies deployment with no experience of the underlying microservices environment needed. There are no CLIs required as it is delivered as a turnkey solution that is an extension of NSX and vSphere. The companion **NSX Application Platform Automation Guide** covers the details on how to obtain and use the automated virtual appliance. The Automation Appliance will apply all the best practices for NAPP as detailed in this deployment guide.

Any IP Addresses, hostnames, networks, datastores, usernames etc. used in this guide are provided as a reference and should be updated to fit your environment as appropriate.

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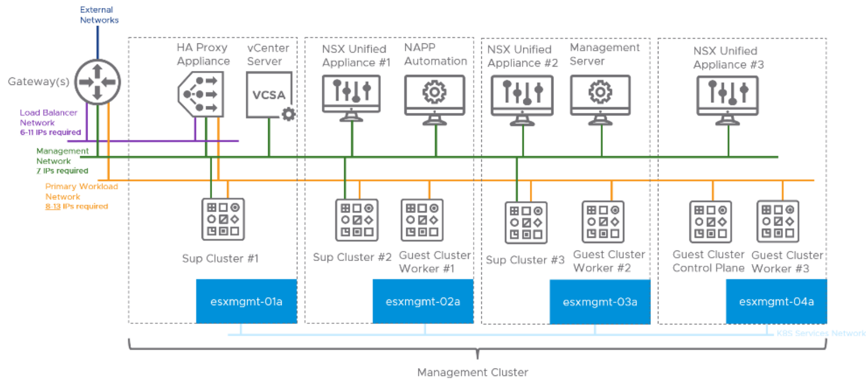
NSX Application Platform Deployment Topologies

Note: This section provides a visual representation of the supported NSX Application Platform topologies and their required network connectivity. Please refer to NSX Application Platform System and Network Requirements sections of this guide for additional details on networks, and connectivity details. This is a representative topology using a 4-host management cluster to run all components supporting NAPP. As long as the documented connectivity requirements are met, you can run these components in different clusters or environments.

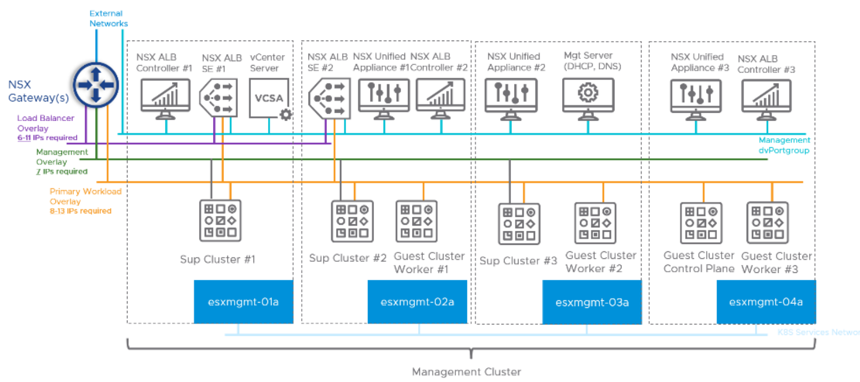
Topology 1: vSphere with Tanzu and VDS Networking

This is the most common and **recommended** deployment option for NAPP. As NAPP is a single microservices cluster and networking is relatively static, the flexible networking capabilities of NSX-T are not required. Also considering NAPP is an extension of the NSX Platform it is typically deployed in a Management Cluster where NSX is not typically available. Similarly, HAProxy meets the Load Balancing needs with minimal footprint. Therefore, there is no requirement to deploy NSX Advanced Load Balancer specifically for NAPP, as the AVI Controller and Service Engines would require significant additional resources. However, if NSX ALB is already available for other workloads in your environment it can also be leveraged for NAPP. This topology will use dedicated networks for Management, Workload and Load Balancer VIPs which is the recommended approach. While it is possible to share the Workload and LB networks this can make managing address space and conflicts more challenging and should only be used if networks or addresses are constrained.



Topology 2: vSphere with Tanzu and NSX-T Networking

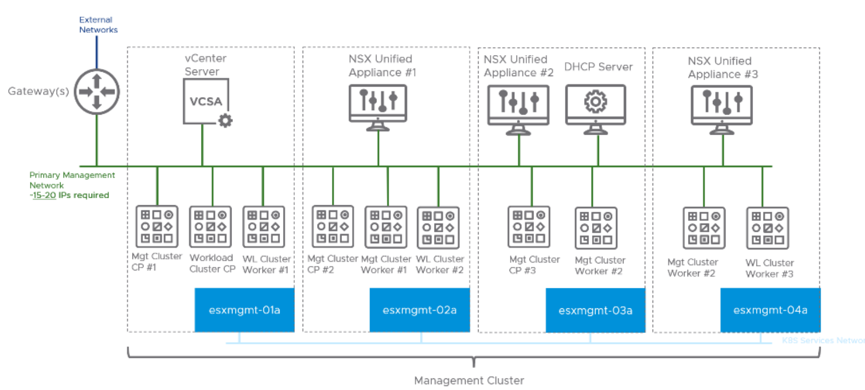
NSX-T Logical Networking and NSX Advanced Load Balancer can also be used to support your NAPP environment. As management clusters (where NAPP will typically be deployed) are often not prepared for NSX, this topology has higher resource requirements, additional components in the data plane between NAPP and the data sources it is receiving information from. Therefore, the simpler TKGs VDS-based networking topology is recommended for NAPP deployments unless NSX-T is already available in the environment.



Topology 3: Upstream Kubernetes

If you cannot run vSphere with Tanzu (TKGs) then Tanzu Community Edition (TCE) or Upstream Kubernetes is a viable option. We will use Tanzu Community Edition in this deployment guide to cover the Upstream Kubernetes topology as this is a VMware-packaged, freely-available open-source distribution of Tanzu that provides a microservices platform which can be used to run NSX Application Platform. **Note:** As Tanzu Community Edition is not being actively maintained the recommended platforms are TKGs or Upstream/Vanilla K8s.

It is the simplest topology as it is possible to share one routed management network for all connectivity.



NSX Application Platform System Requirements

Planning and preparing your environment to meet the Deployment prerequisites and System Requirements for NSX Application Platform and NSX Intelligence is essential for a successful installation.

This section provides a checklist of these requirements to ensure you are ready to procced with deployment. This checklist should be verified beforehand to ensure all

Table 1. NSX Application Platform Infrastructure Requirements

	Requirements	Comments
☐	NSX-T Datacenter running version 3.2.0.1 or newer deployed. A minimum of one NSX Manager with vSphere HA enabled on the underlying cluster is required for evaluations. For a production environment, deploying a management cluster of three NSX Managers is recommended.	
☐	To deploy the NSX Application Platform you must have a valid, non-expired NSX license of one of the following types: • NSX Data Center/NSX-T Enterprise Plus • NSX Data Center/NSX-T Advanced with NSX ATP • NSX Data Center/NSX-T Advanced with NSX TP • NSX Security (Distributed Firewall or Gateway Firewall) • NSX Evaluation	
☐	vSphere 7.0U3c or newer (vCenter Server and ESXi Hypervisor). You will also need a target vSphere Cluster with sufficient resources to meet	

	the NAPP System Requirements listed in Table 2. As NAPP is an extension of the NSX solution, a Management or Shared Cluster is typically used to meet this requirement	
☐	A supported microservices environment for NSX Application Platform. This document covers the required configuration steps for the microservices environment and the NAPP Automation Appliance will perform end to end deployment, so you do not need the microservices environment in advance, but you will need to ensure that the infrastructure is compatible with the listed versions. VMware supports and validates NAPP on the following solutions: • vSphere with Tanzu (TKGs) NSX 3.2: **v1.17.17 to v1.21.6** NSX 4.0: **v1.20.7 to 1.22.9** Note: It is recommended to use a Tanzu Kubernetes Release of version **1.20.7** or newer for important fixes and enhancements • Upstream Kubernetes NSX 3.2: **v1.17 to v1.21** NSX 4.0: **v1.20 to v1.24** There are several other Kubernetes distributions which are known to work, but have not been officially validated and therefore are unsupported by VMware. Contact VMware support if you have questions about interoperability on	

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	different distributions and your options for deployment.	
☐	Ability to add a fully qualified domain name (FQDN) in DNS. This is required during the NSX Application Platform deployment to create Service Names which are the API and messaging endpoints used to connect to the NSX Application Platform.	
☐	Network connectivity between your microservices environment and NSX Manager/vCenter.	
☐	Traffic permitted between all the Ports and Protocols listed in Tables 4 and 5.	
☐	Time synchronized between NSX, vSphere and your Kubernetes environments	
☐	A system which can be used to run the required tools to bootstrap your environment. This should be a supported Windows, MacOS or Linux operating system for accessing the vSphere Web Client, NSX-T Datacenter and NSX Intelligence UIs and also support kubernetes tools. The following Operating System and Browser combinations are recommended: • Google Chrome 89 or later on Windows 10, Mac OS 10.13, 10.14 and Ubuntu 18.04 • Mozilla Firefox 80 or later on Windows 10, Mac OS 10.13, 10.14 and Ubuntu 18.04 • Microsoft Edge 90 or later on Windows 10	

	Refer to the vSphere and NSX Product Documentation for more information on browser requirements	
☐	For optimal performance when viewing the NSX Intelligence user interface via a supported browser, it is recommended that your client system has a minimum of 1.4 GHz CPU core and 16 GB of RAM for using NSX Intelligence. In addition, the minimum supported browser resolution is 1280 x 800 pixels	

Table 2. NSX Application Platform System Requirements

	Requirements	Comments
☐	There are several form factors for NSX Application Platform dependent on the features you plan to enable. Resource requirements and supported features for each form factor are listed below. Please ensure your target vSphere Cluster has sufficient capacity for the selected form factor:	

Form Factor	# of Nodes	vCPU	Memory	Storage	NSX Features
Standard	1 control node and 3 worker nodes*	Control: 2vCPU Worker: 4vCPU	Control: 4-8GB Worker: 16GB	200 GB per node	NSX ATP/NDR NSX Malware Prevention NSX Metrics
Advanced	1 control node and 3 or more worker nodes	Control: 2vCPU Worker: 16vCPU	Control: 4-8GB Worker: 64GB	1 TB per node	NSX Intelligence NSX ATP/NDR NSX Malware Prevention NSX Metrics
Evaluation	1 control node and 1 worker node*	Control: 2vCPU	Control: 4GB Worker: 64GB	1 TB per node	NSX Intelligence NSX ATP/NDR

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<tr><td></td><td></td><td>Worker:
16vCPU</td><td></td><td></td><td>NSX Malware Prevention

NSX Metrics</td></tr>
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*Standard and Evaluation Form Factors do not support scale-out at present.
Also, the Evaluation form factor is currently only supported for use in non-production deployments (POC evaluations or demonstrations). It has limited data retention, no high availability support, and no support for upgrades.
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If you are deploying the NSX Application Platform on TKGs, the control plane and worker nodes for the guest cluster will require an additional storage volume of 64 GB for ephemeral storage.
The deployment guide will cover the steps required to provide this storage volume.
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NSX Application Platform Network Requirements

You will need to ensure the following minimum networking and connectivity requirements for NSX Application Platform and dependent features are met. Network sizes are indicative and cover requirements for NAPP and its dependencies. It is advised to leave headroom when sizing subnets for these networks for potential scale out of NAPP

Table 3. NSX Application Platform Network Requirements

	Requirement	Comments
☐	Management Network used for the following: - ESXi Host VMkernel Management - vCenter Server - NSX Manager Unified Appliances - Load Balancer Management Interface - Supervisor Cluster VMs This network will require at least 15 available IPs dependent on the number of Hypervisors and management workloads. As long as there is IP connectivity and the required ports and protocols are permitted, then these management components can also be on different networks/subnets For reference the minimum IP address usage on this network to support a NAPP deployment is: - 1 IP per ESXi Hypervisor - 1 IP for vCenter Server - 1 IP Per NSX Manager appliance - 1 IP for NSX Manager VIP (if used) - 5 IPs for TKGs Supervisor Cluster VMs	

	- 1 IP for Load Balancer Management - 1 IP for NSX Application Platform Automation Appliance	
☐	Primary Workload Network used for: - Supervisor Cluster VMs - Guest Cluster VMs - Load Balancer connectivity This network will require at least 8 IPs for NAPP and up to 13 IPs dependents on scale out, so a **/27** network or greater is suggested for future growth: - 3 IPs for TKGs Supervisor Cluster VMs - 1 IP per TKGs Guest Cluster Control VM - 1 IP per TKGs Guest Cluster Worker VM (up to 8) - 1 IP for the Load Balancer	
☐	Dedicated Load Balancer Network (Optional but strongly recommended): - VIPs for Load Balanced K8s Services This network will require at least 9 IPs and up to 14 for NAPP. Based on the logic used for VIP Allocation with Tanzu (covered in more detail during the HA Proxy deployment steps) similarly a **/27** network or larger is suggested: - IPs assigned to TKGs services (3) - External IPs assigned to LoadBalancer Services used by NAPP for Interface and Messaging Endpoints (2) - IPs assigned to each Node for messaging (up to 8) - Load Balancer Interface (1)	

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☐	Dependent on your deployment topology these networks can be backed by either: - dvPortgroups on a Virtual Distributed Switch (most common) or - NSX Overlay Segments	
☐	Tanzu Community Edition will use a single network (this can be either a Management or Workload network) and has some additional requirements: - A DHCP server configured to provide a minimum of 11 IPs in a DHCP range for Management and Workload Cluster Nodes - 5 IPs on this network outside the DHCP scope for the load balancer instance (MetalLB)	

Table 4. NSX Application Platform Port and Protocol Requirements

Source	Target	Port	Protocol	Description
NSX Unified Appliance & NSX Application Platform	Kubelet API Server	10250	TCP	K8s API access
NSX Unified Appliance & NSX Application Platform	Kubernetes Cluster Server	10257, 10259	TCP	K8s server access
NSX Unified Appliance & NSX Application Platform	Etcd Kubernetes API Server	2379, 2380	TCP	API access from NSX to Etcd server
NSX Application Platform &	Kubelet API Server	6443	TCP	K8s API access

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NSX Unified Appliance				
NSX Application Platform	NSX Unified Appliance & NSX Transport Nodes	9092	TCP	NAPP outbound communication to NSX Unified Appliance or Transport Nodes
NSX Application Platform	DNS Servers	53	TCP	DNS
NSX Application Platform	DNS Servers	53	UDP	DNS
NSX Application Platform	NTP Servers	123	UDP	NTP
NTP Servers	NSX Application Platform	123	UDP	NTP
Management Clients	NSX Application Platform	22	TCP	Remote Access via SSH
NSX Unified Appliance	NSX Application Platform	443	TCP	NSX Application Platform API services
NSX Unified Appliance & Transport Nodes	NSX Application Platform	9092	TCP	Inbound messages from NSX Unified Appliance or Transport Nodes to NSX Application Platform

Table 5. NSX Application Platform Outbound Internet Requirements

Covers all outbound connectivity requirements including destination URLs for TKGs, NAPP and features running on NAPP:

Source	Target	Port	Protocol	Description
vCenter Server	wp-content.vmware.com	443	TCP	Used to access Tanzu images for Content Library
NSX Unified Appliance	projects.registry.vmware.com	443	TCP	Access Helm Charts for deployment from VMware's official harbor instance
NSX Application Platform (Guest Cluster Nodes)	projects.registry.vmware.com	443	TCP	Access Docker Images for deployment from harbor
NSX Application Platform (Guest Cluster Nodes)	nsx.lastline.com and dependent on the configured Cloud region nsx.west.us.lastline.com or nsx.nl.emea.lastline.com	443	TCP	Access to cloud services for NSX NDR and NSX Malware Prevention
NSX Application Platform (Guest Cluster Nodes)	api.nsx-sec-prod.com & *.amazonaws.com	443	TCP	Access to NSX Thread Intelligence Cloud Services for NSX NDR

For the latest information on NSX ports and protocols please see: VMware NSX-T Data Center Ports and Protocols

NSX Application Platform and Intelligence Scale

This section provides the current NSX Intelligence recommended configuration and scalability limits. **Note:** Please refer to the **configmax** site for authoritative details on scalability limits in VMware products as this is updated more frequently.

Table 6. NSX Intelligence Scale and Configuration Limits

Maximum	Form Factor Size:	
	Standard	Advanced
NAPP K8s Nodes	3	8
ESXi hosts	TBD	250
VMs	TBD	5000
VMs per recommendation	N/A	100
VM members in Group-based Recommendation	N/A	100
Flows per 5-minutes interval	N/A	500,000 (3 nodes) 1,000,000 (8 nodes)
Retention period	1 week	3 months

Deploying NSX Application Platform

The core of this document provides a guided, step-by-step installation process of the 3 validated and supported deployment options for NSX Application Platform (NAPP) and NSX Intelligence.

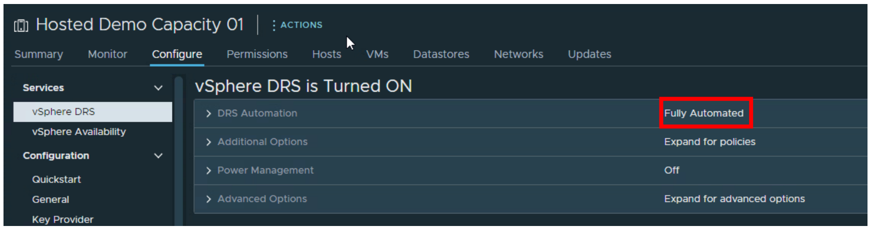
It is divided into Infrastructure Configuration steps followed by the NSX Configuration and Feature Enablement, as these tasks may be performed by different teams or administrators dependent on your environment or operational model. Please ensure you have verified the system and network requirements for installation before proceeding.

Note: All documented deployment options have been validated with NSX Application Platform, however vSphere with Tanzu (TKGs) is recommended for the most seamless and integrated experience with end-to-end VMware support. In addition, as NAPP will run in a single Tanzu Kubernetes Cluster, the networking requirements can be met with VDS Networking and HAProxy. Antrea is used as the Container Network Interface (CNI) to provide network connectivity and security for pod workloads.
This reduces the licensing and footprint requirements and provides the simplest and therefore recommended option for NAPP. Importantly it is not the only option and we are not enforcing the use of TKGs, merely recommending it as NAPP is an extension of a VMware solution.

NSX Application Platform Deployment – vSphere with Tanzu and VDS Networking

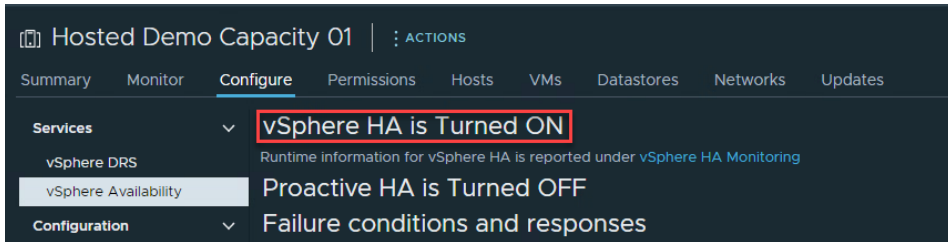
Preparing your vSphere environment for Tanzu

1. First, login to the vCenter Server UI with Administrator privileges
2. Ensure that the vSphere Cluster you will install NSX Application Platform on has DRS set to **Fully Automated** under:
`Cluster -> Configure -> Services -> vSphere DRS`



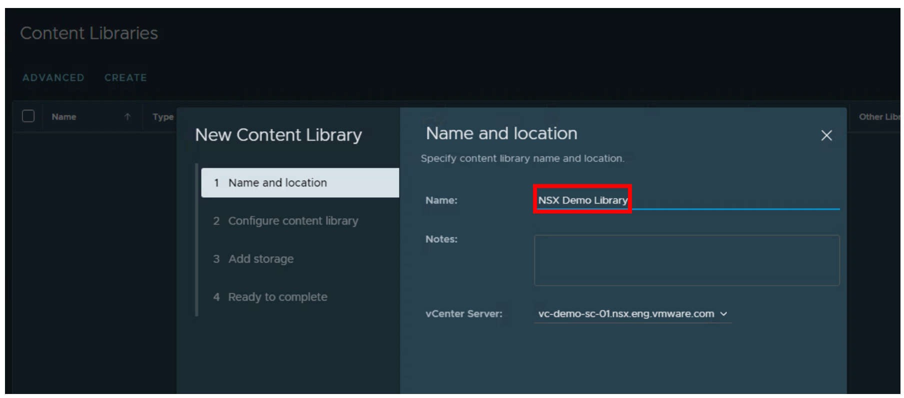
3. TKGs also requires that vSphere HA is enabled on the Cluster that will be enabled for workload Management. This is configured under:
`Cluster -> Configure -> Services -> vSphere`

`Availability`



4. A requirement for enabling vSphere with Tanzu is to create a Content Library that synchronizes the images required for TKGs. Click on:
`Menu -> Content Libraries -> Create`

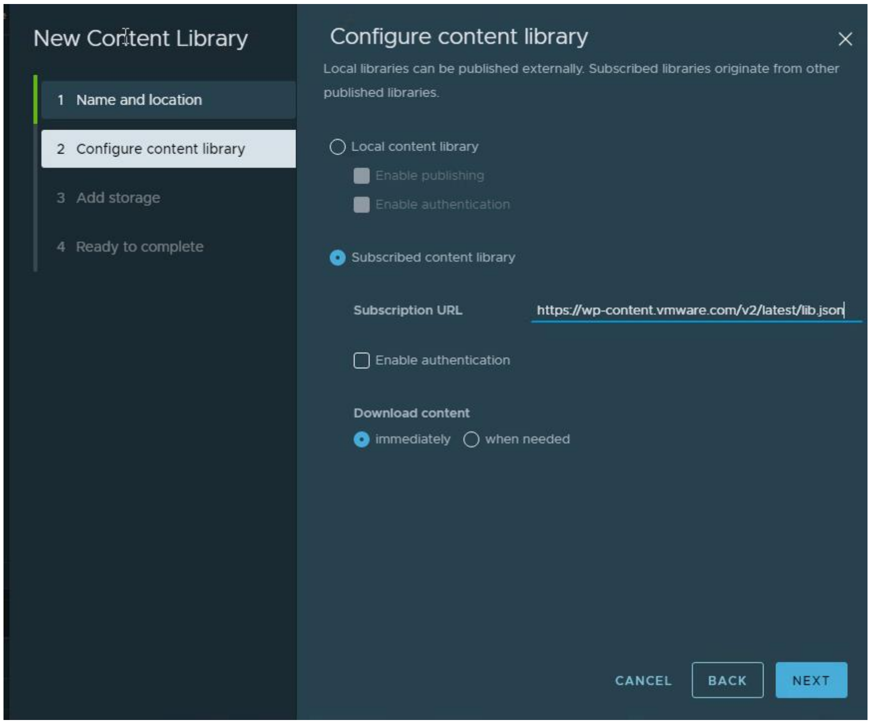
   to add a new Content Library and provide a **Name:**



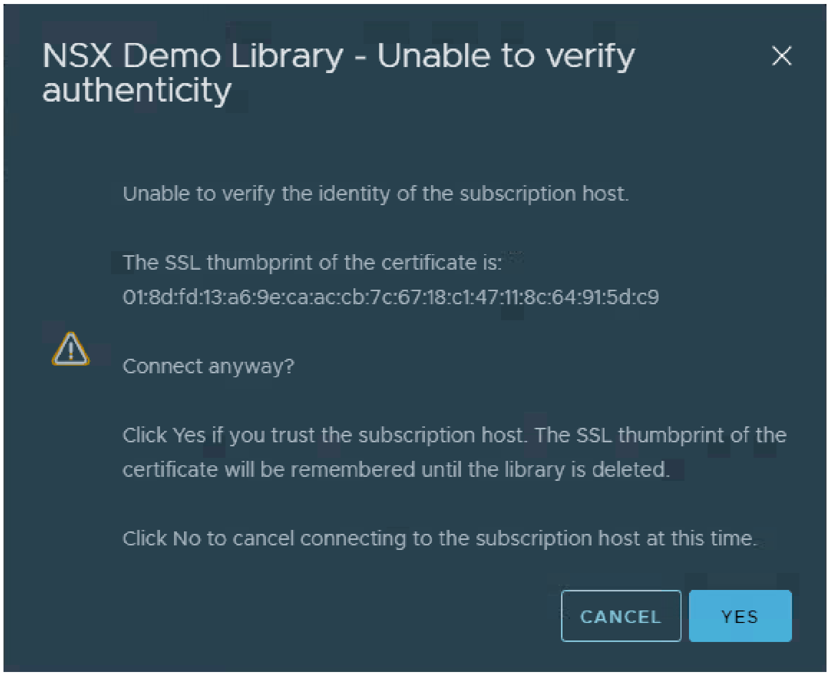
5. Now configure the Content Library settings and select **Subscribed content library**, enter a **Subscription URL** of:

`https://wp-content.vmware.com/v2/latest/lib.json`

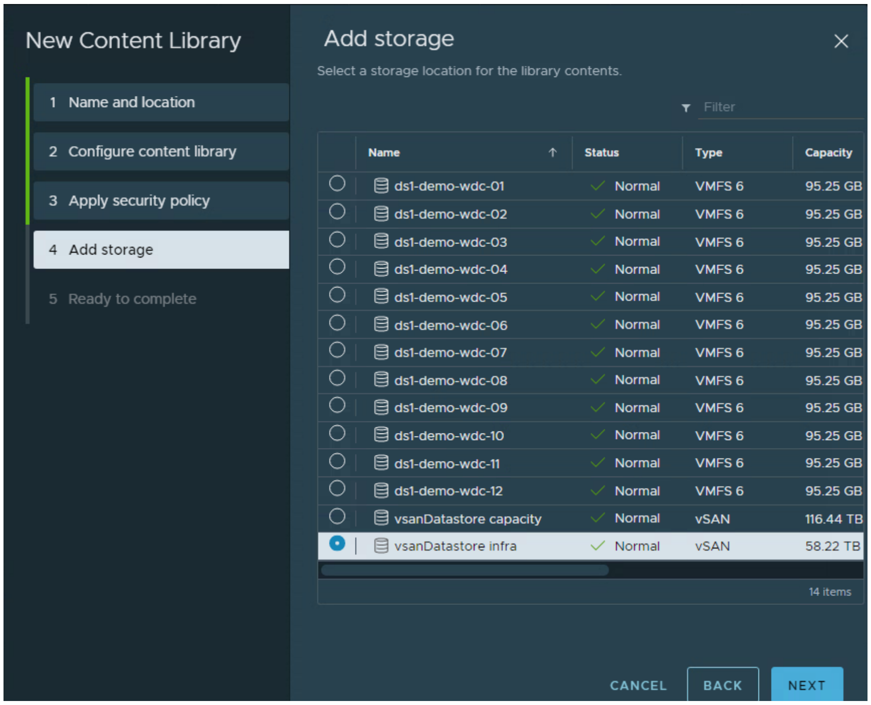
   And **Download content** set to either **immediately** if disk space is not limited, or **when needed** if there are disk or connectivity constraints



6. Accept the certificate for the VMware content library:

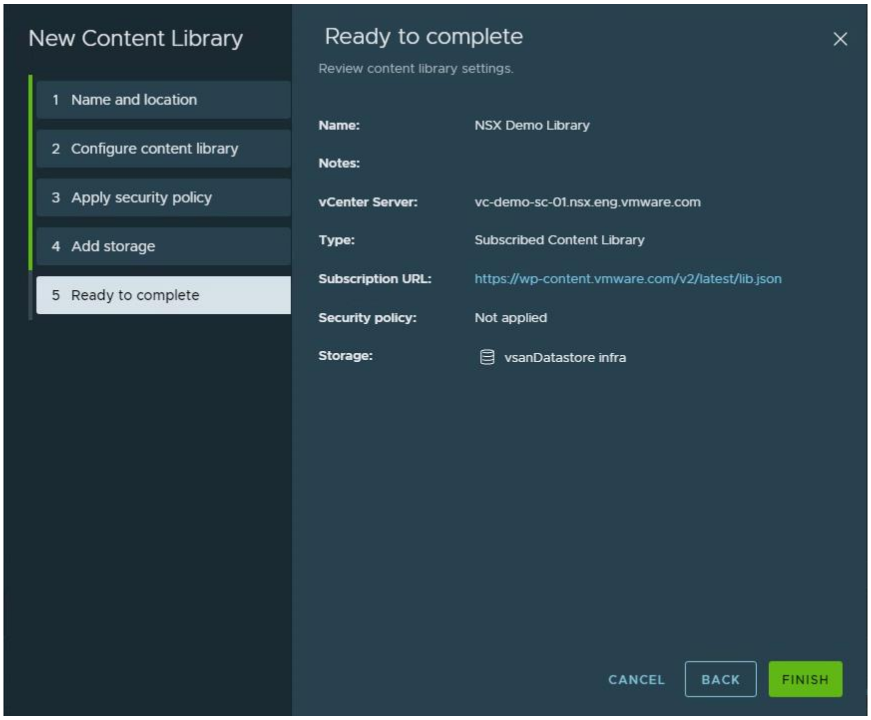


7. Skip the **security policy** and on the **Add storage** screen select the **Datastore** you would like to use for your Content Library:

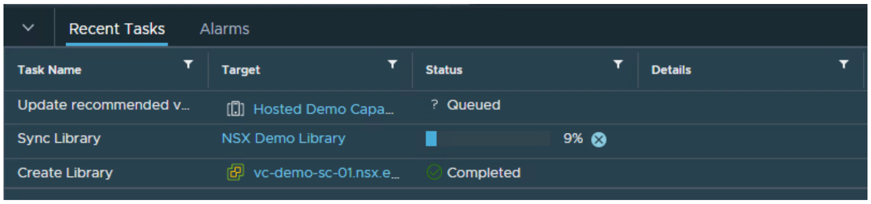


Note: The full Tanzu content library will require over 200GB of disk space. Please ensure the backing datastore has sufficient capacity and dependent on your external connectivity the content library synchronization may take a while to complete.

8. Click **Next** and **Finish** to create the Content Library and begin the synchronization process.

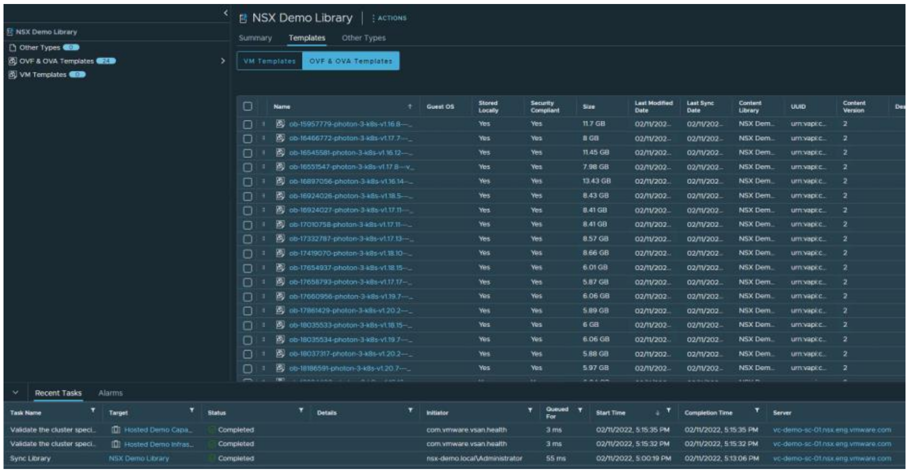


9. You should now wait for the **Sync Library** task to complete:



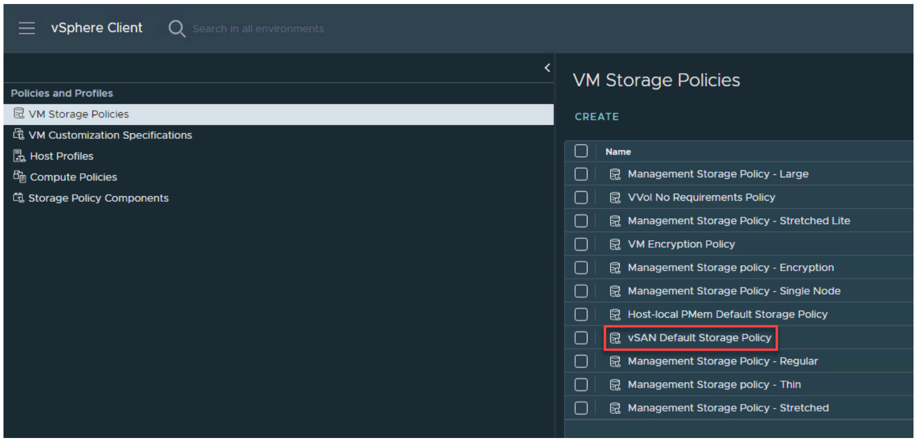
Note: Dependent on your external connectivity the content library synchronization may take some time to complete.

10. And once it is done, verify the required Tanzu Templates are available in your library:

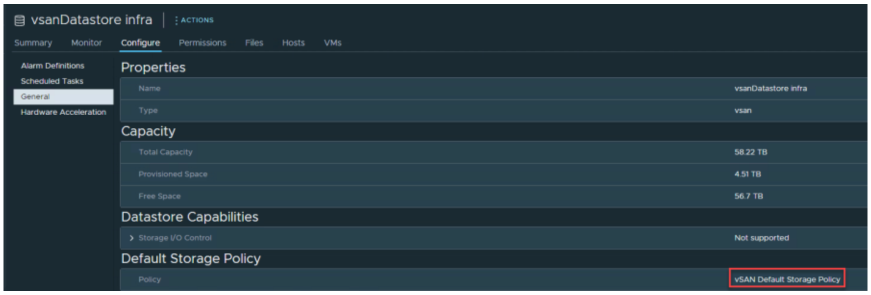


11. Verify you have a **VM Storage Policy** configured and mapped to the datastore you will be using for your NAPP deployment. Storage Policies are mapped to datastores available in the vSphere environment and used to determine the placement of Tanzu Kubernetes objects like control plane VMs, pod ephemeral disks, container images, persistent storage volumes and TKC nodes. VSAN will create and require a storage policy by default.

12. Check that you have an appropriate Storage Policy defined under: `Home -> Policies and Profiles -> VM Storage Policies`

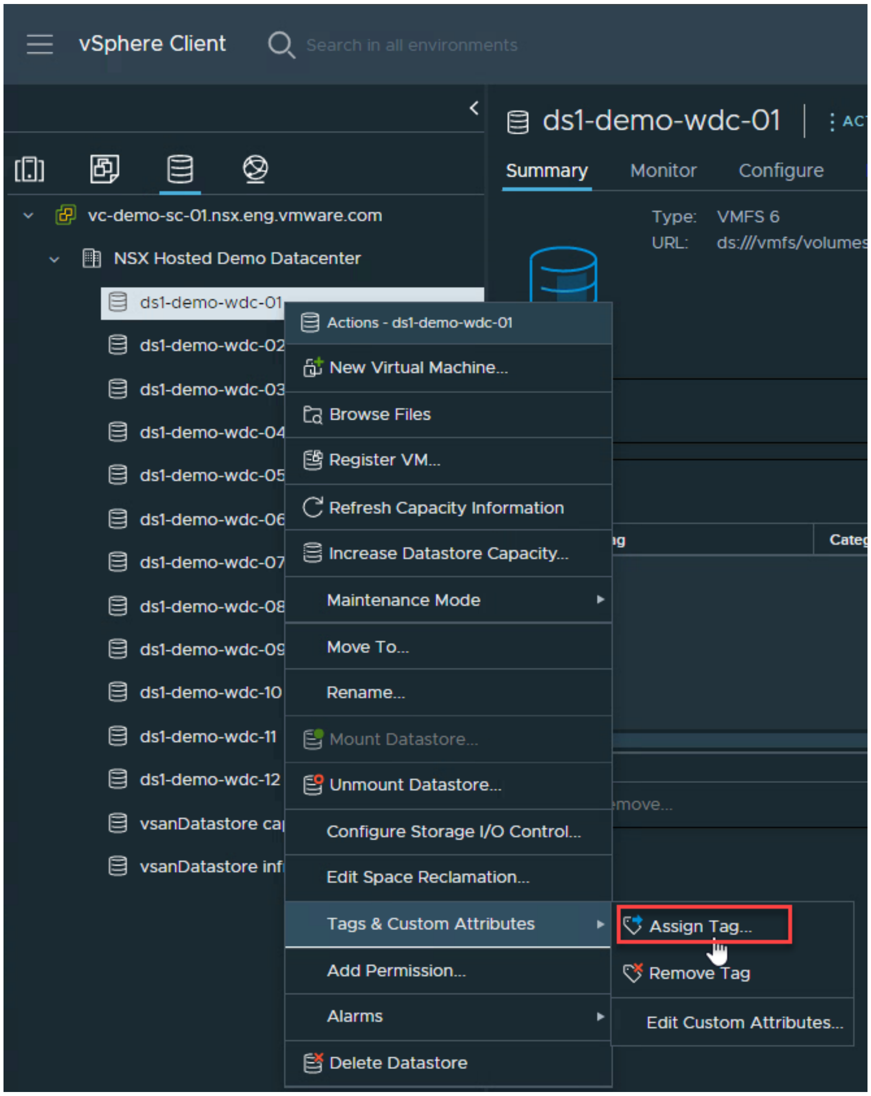


13. Then go to `Home -> Storage -> <target datastore>` to validate your target datastore has this Storage Policy assigned.



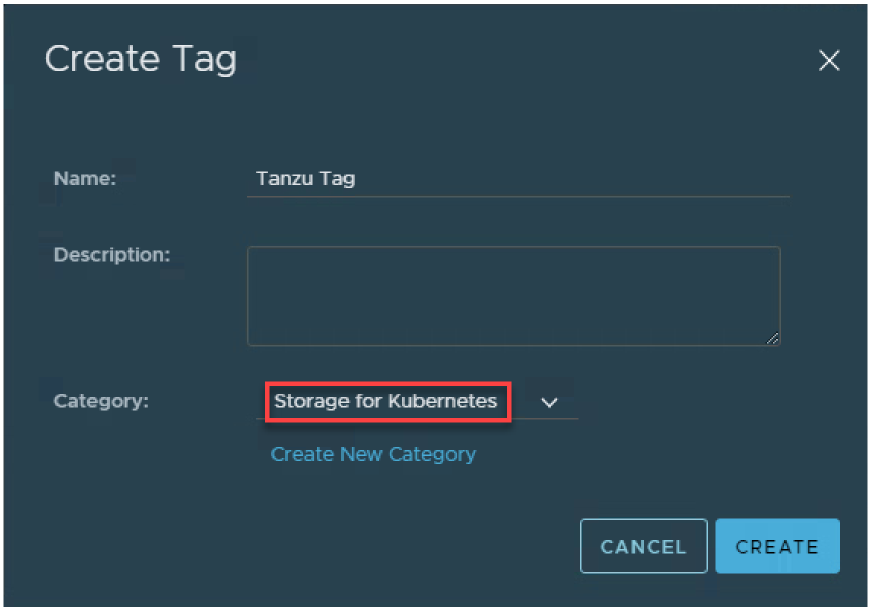
14. A Storage Policy must be configured and assigned to a Datastore for a TKGs deployment. If you are **not** using VSAN (eg, VMFS or NFS Datastores), you will need to perform the following steps before proceeding:

    Navigate to `Storage -> <target datastore>` and **Right-Click** then select `Tags & Custom Attributes -> Assign Tag`



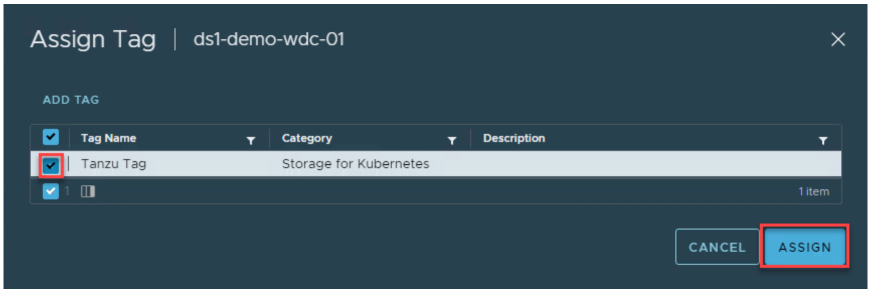
15. Click the link to **ADD TAG**

16. Enter a **Name** for the Tag (e.g., Tanzu Tag), and click **Create New Category**



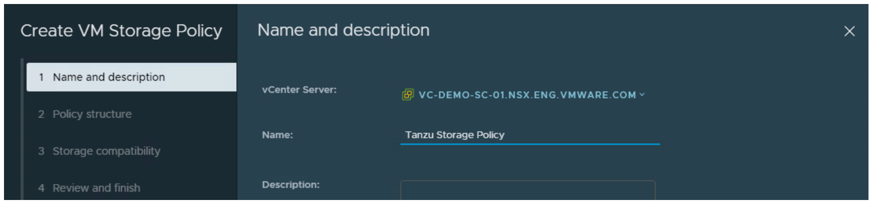
And provide a descriptive name – I have used **Storage for Kubernetes** in the example. You can also limit the **Assignable Object Type** to **Datastore** if you wish

17. Then click on the new Tag Name, followed by **ASSIGN** to link it to the Datastore:

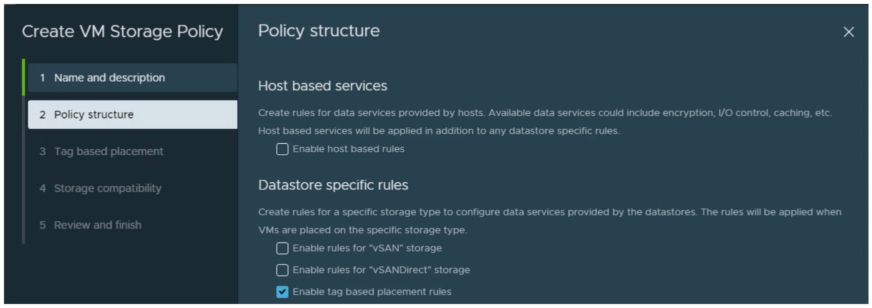


18. Finally return to: `Home -> Policies and Profiles -> VM Storage Policies` and click **CREATE.**

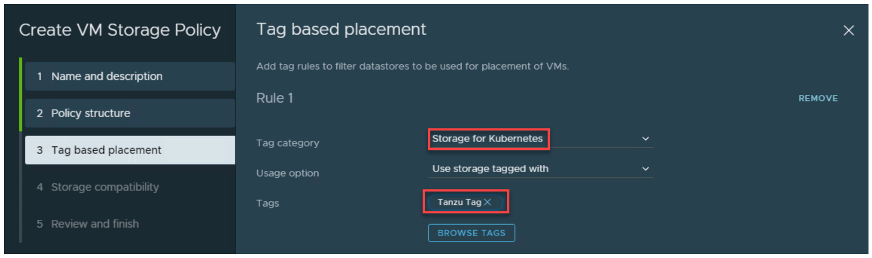
Give the policy a name that makes it clear it will be used by TKGs e.g., **'Tanzu Storage Policy'**



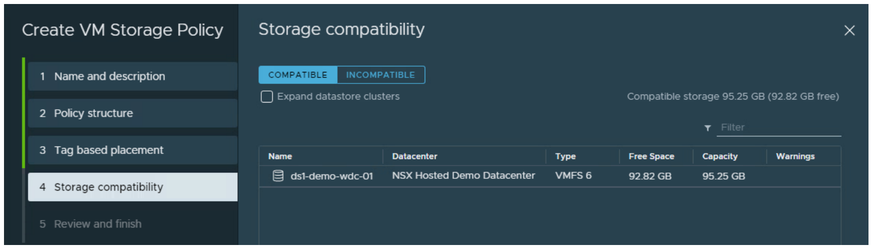
19. Select the option for Datastore specific rules and Enable tag-based placement rules:



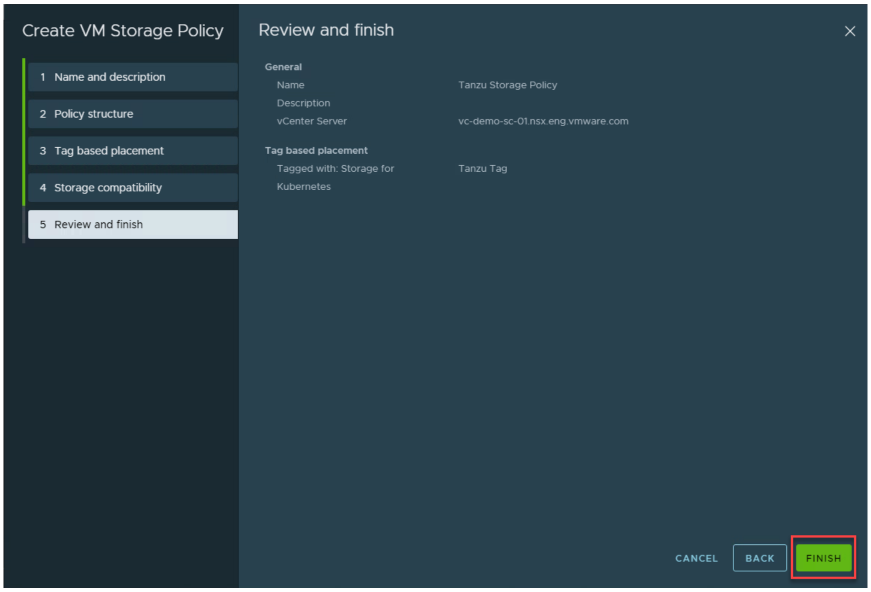
20. Here you will create a Tag-based placement rule. Select the Category and Tag defined earlier in **Tag category** and the **Tags** fields:



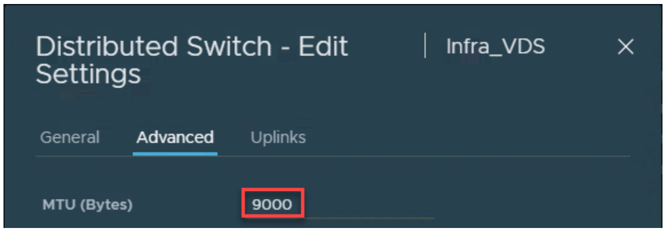
21. As this tag was assigned to your target datastore, this should now automatically show as **COMPATIBLE**:

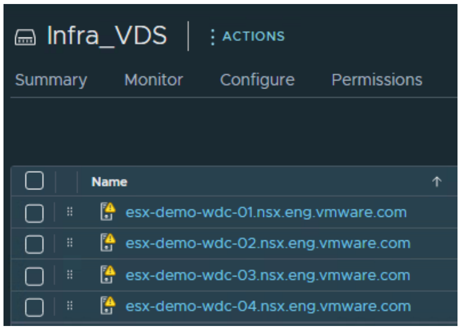


22. Once your Storage Policy is matched to the correct datastore you can click **NEXT** and **FINISH** to complete this step:



23. Next, we will set up VDS networking to support our Tanzu Kubernetes Cluster. From the Networking menu select your Datacenter and start by creating a new **VDS** (if required) and ensure the MTU is set to **1600** or greater and add the hosts from your target cluster to this VDS:

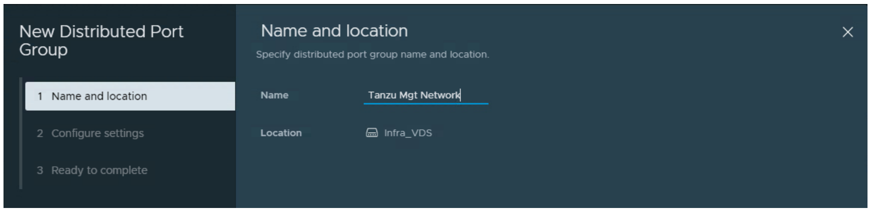




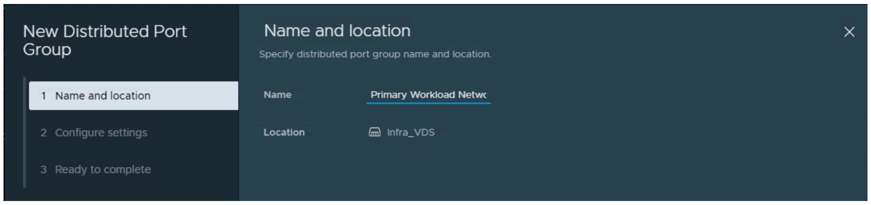
24. Create the dvPortgroups needed for the vSphere with Tanzu Environment. The requirements for these networks are defined in Table

    Note: The NSX-T topology does not require any dvPortgroups to be created as overlay segments are used. While the Tanzu Community Edition requires a single management network only.

    1x Management dvPortgroup. This network must either contain or have IP connectivity to the vCenter Server and VMkernel management interface of your ESXi hypervisors.

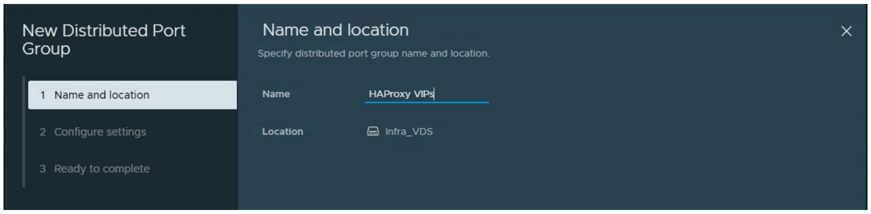


    1x Primary Workload Network dvPortgroup used for your Tanzu Kubernetes Clusters.



    1x HAProxy VIPs dvPortgroup.

    Note: You can use the same network for workload management if there are constraints on the number of IP Addresses or available networks. While this is fully supported, where possible I strongly recommend separating these to make IP allocations easier to track and troubleshoot



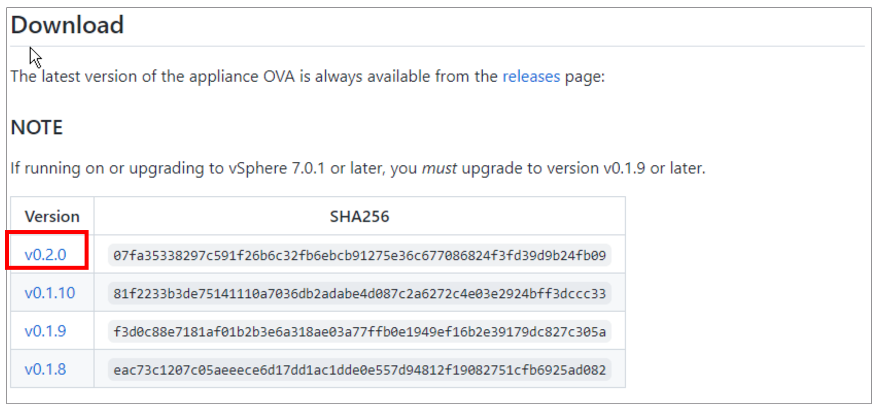
Deploying HAProxy for your vSphere with Tanzu deployment

VMware provides an implementation of the open source HAProxy load balancer that you can use in your vSphere with Tanzu environment along with vSphere Distributed Switch (VDS) networking. This is an end-to-end VMware supported configuration for use with Tanzu.

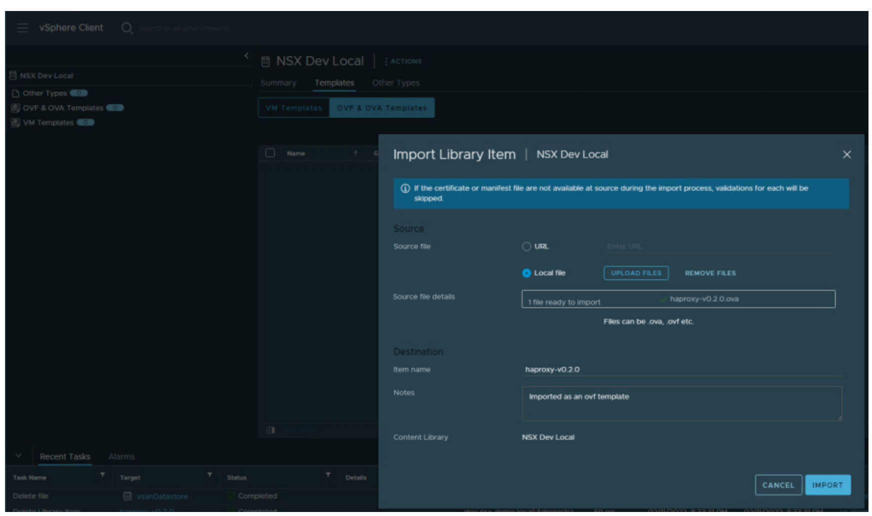
25. Begin by downloading the HAProxy appliance from the following website:

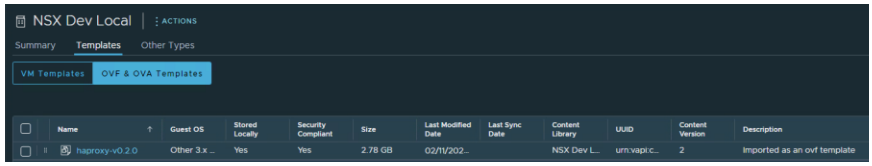
    https://github.com/haproxytech/vmware-haproxy#download

    v0.20 is the latest version at the time of this document and is known to work reliably with TKGs



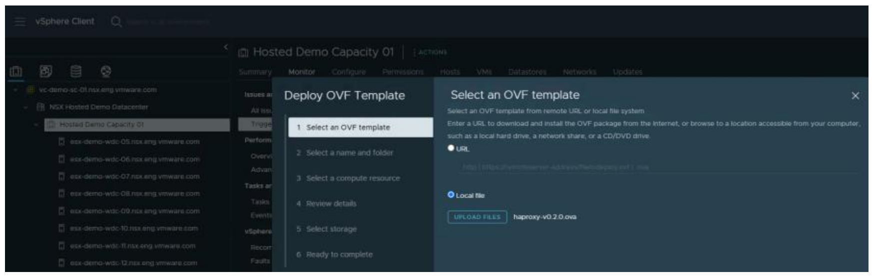
26. To deploy the HAProxy appliance you can either import the OVA directly into a local **Content Library**:



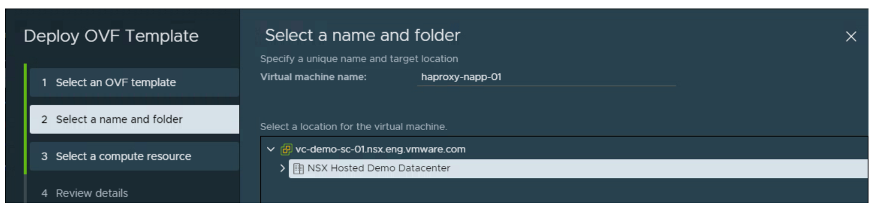


Or deploy the appliance directly by right-clicking on the **target vSphere Cluster for Tanzu** and selecting `Deploy OVF Template`

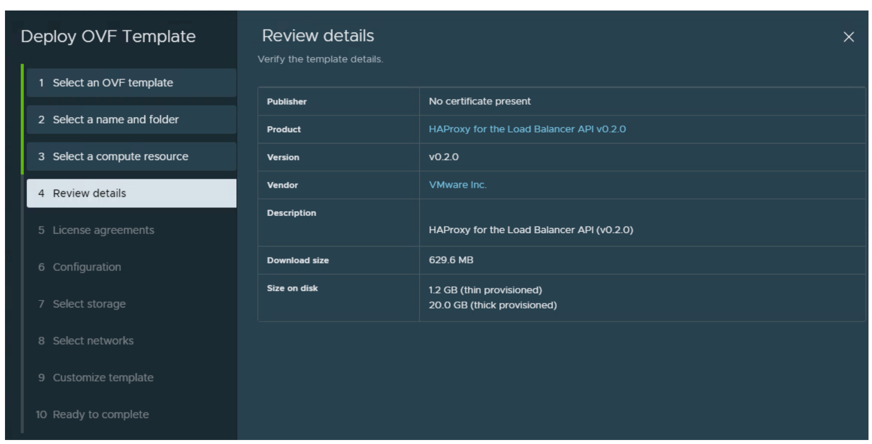
27. Select **Local file**, then **Upload Files** and then choose the **HAProxy OVA** you downloaded earlier:



28. Give the VM a **name** and select the **target cluster**:

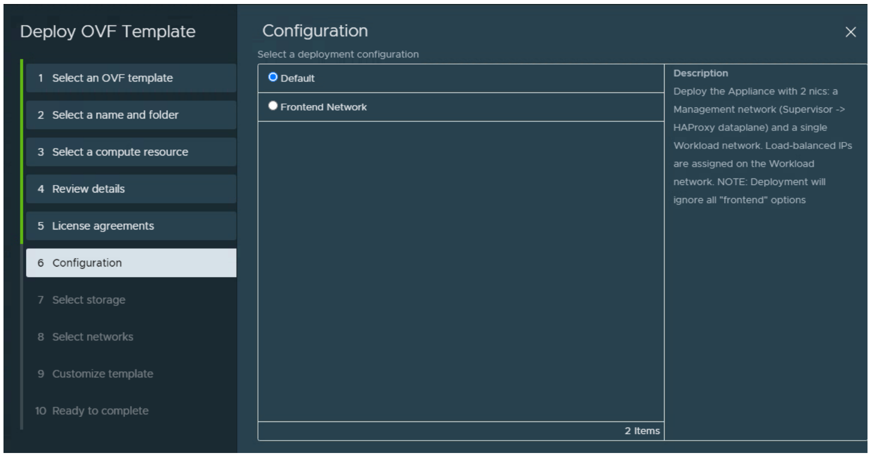


29. Review the vendor details to make sure this is the **VMware Inc.** appliance and **accept** the license agreement:

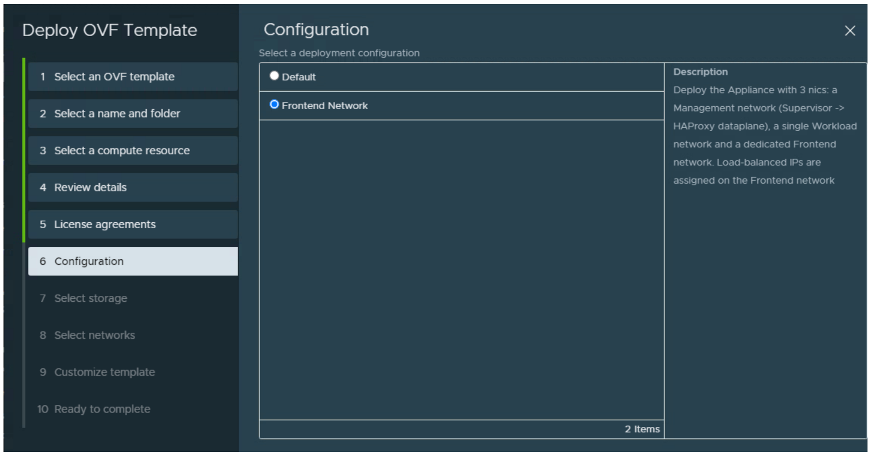


30. Under Configuration you will select a Deployment Configuration of either**:**

a. **Default** (this is a 2 NIC deployment where the workload network and Load Balancer VIPs on the HA Proxy appliance share the same dvPortgroup):



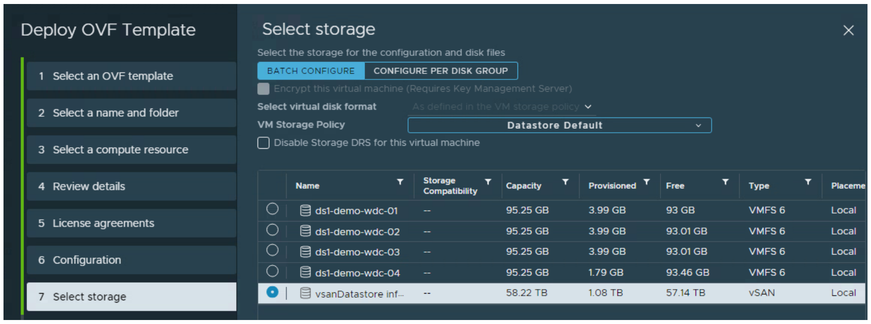
b. Or **Frontend Network** (which is a 3 NIC deployment where the Load Balancer VIPs have a dedicated dvPortgroup):



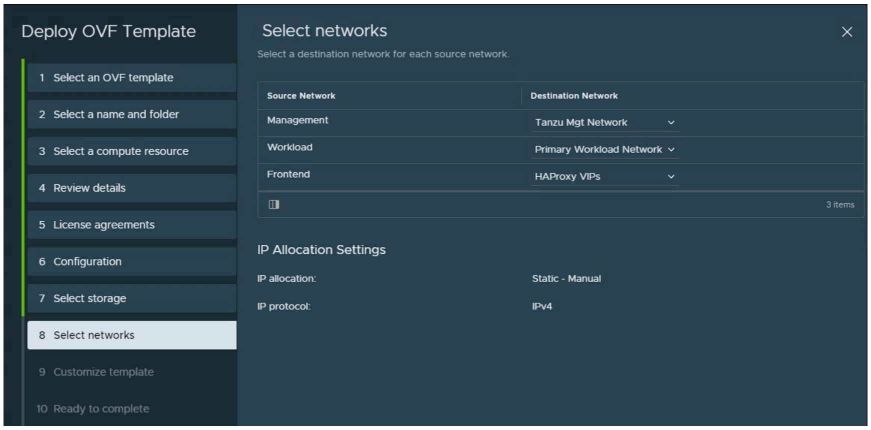
Note: Where possible I strongly recommend the 3 NIC deployment to make IP Allocation easier to manage and reduce the risk of overlapping or issues reclaiming IP Addresses. Refer to the NAPP Deployment Topology section of this document for additional details on the networking configuration and requirements and apply the appropriate Networks, IP Addresses and dvPortgroups for your environment.

31. Then under **Select storage** choose the Datastore you are deploying the

HA Proxy appliance to:



And under **Select networks** map the interfaces of the appliance to the dvPortgroups you created earlier while preparing your vSphere environment:



Finally, in **Customize template** you will enter properties to configure the HA Proxy appliance.
Critical Note: When configuring the value for **Load Balancer IP Range**, please ensure the value:
Is in CIDR format and not an IP range (despite the name of the field)
Must not be the entire subnet CIDR for the Frontend network (otherwise it will result in an IP conflict with the Default GW and HA Proxy interface on that network)
So you must allocate a smaller CIDR that is a subset of the entire subnet that does not conflict with any existing IPs. Here is an example:

If your Frontend Network is: **192.168.140.0/27** with a Default Gateway of **192.168.140.1** and HAProxy interface of **192.168.140.2**, then the correct value for **Load Balancer IP Range** would be **192.168.140.16/27** (IPs from **192.168.140.17** to **192.168.140.30**). It cannot be 192.168.140.0/27 or /28 as that would result in a conflict. This is a very common source of error – pay close attention to this configuration step

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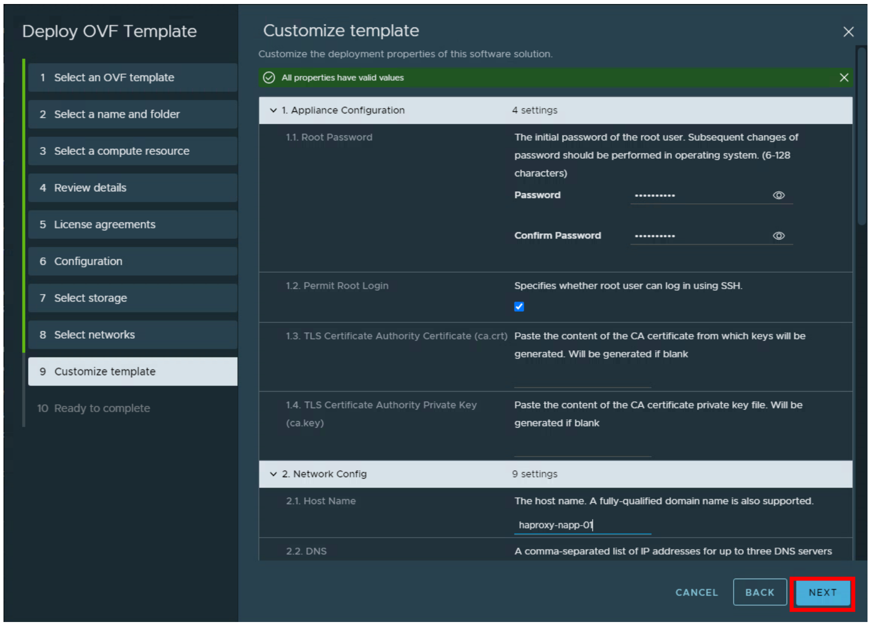
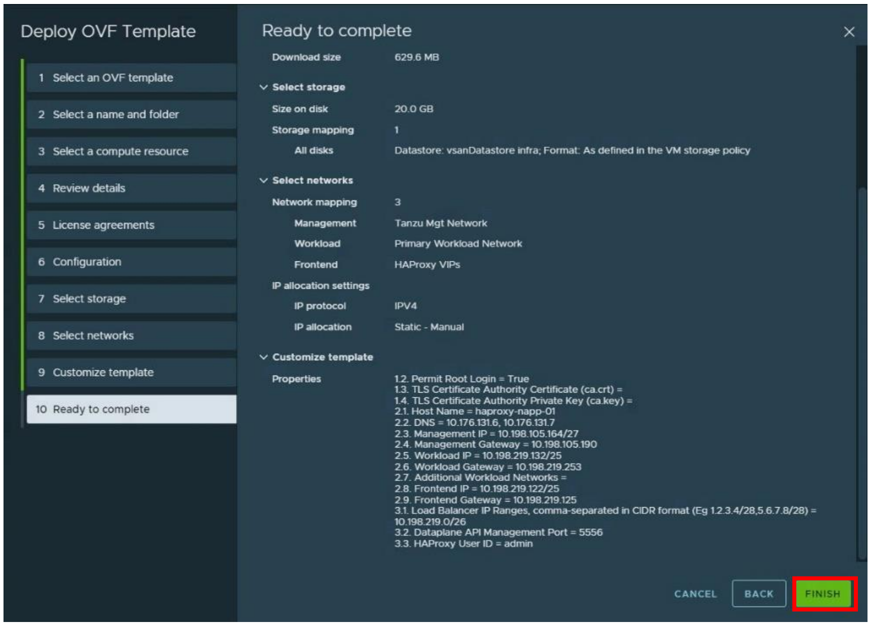


It is important to carefully select and track **Credentials, IP Addresses for Management, Workload and Frontend Networks, Dataplane API Port and Username** as these will be used later in your TKGs configuration. Here is a reference from a working deployment (carefully note the Frontend IP and Load Balancer IP Range inputs are not overlapping):



32. Once the appliance has deployed successfully power it on:

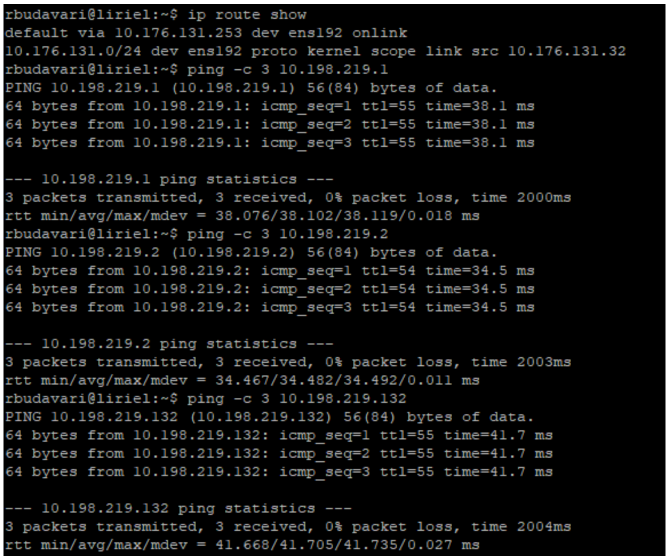
    `Right-Click -> Power -> Power on.`

    Confirm you can login to the management IP of the HA Proxy VM via SSH and while you have logged in run `cat /etc/haproxy/ca.crt` to retrieve the certificate:

```
login as: root
Keyboard-interactive authentication prompts from server:
| Password:
End of keyboard-interactive prompts from server
Last login: Sat Feb 12 06:00:46 2022 from 10.219.86.188
 06:01:44 up 2 min,  1 user,  load average: 0.00, 0.00, 0.00
tdnf update info not available yet!
root@haproxy-napp-01 [ ~ ]# cat /etc/haproxy/ca.crt
-----BEGIN CERTIFICATE-----
MIIDozCCAougAwIBAgIJAOz6jP/aKT5yMA0GCSqGSIb3DQEBBQUAMG8xCzAJBgNV
BAYTAlVTMRMwEQYDVQQIDApDYWxpZm9ybmlhMRIwEAYDVQQHDAlQYWxvIEFsdG8x
DzANBgNVBAoMBlZNd2FyZTENMAsGA1UECwwEQ0FQVjEXMBUGA1UEAwwOMTAuMTk4
LjEwNS4xNjQwHhcNMjIwMjEyMDU1OTE1WhcNMzIwMjEwMDU1OTE1WjBvMQswCQYD
VQQGEwJVUzETMBEGA1UECAwKQ2FsaWZvcm5pYTESMBAGA1UEBwwJUGFsbyBBbHRv
MQ8wDQYDVQQKDAZWTXdhcmUxDTALBgNVBAsMBENBUFYxFzAVBgNVBAMMDjEwLjE5
OC4xMDUuMTY0MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAwdj/B8V5
IXek48DMlop9X0PB7TTgm+aNq+wrl4Qr+Um2imBgljtA+rHpFzyq4CHdRUHyVqzP
A8WXOSnI2vg4P2Mxa+dPEjYiVRjwlhlPhVk5LlHaDUOGPdDaYKf2NuajyAdA7fm7
wsc5ebWYqkXxSDqo3FTZLxnqxhk5DIVdFkrPMrP7qTw+/RchnfBODxQhQqJOh59a
Ew+HvVOILnMklHXN8cpJKwWHwqMXvmAW/J/1FeUj+hxkQv8pYtw2ZsMWQYMErtzq
obgjVgbazLektBqP8q0utRzeDKFrIklrbc5Hgig5Wo+ZNVNfDdksEeXH3NOr4sVE
PzOVxgSZWUtnTwIDAQABo0IwQDAPBgNVHRMBAf8EBTADAQH/MA4GA1UdDwEB/wQE
AwIBhjAdBgNVHQ4EFgQUTb2p4ukoVZhmEzlQpkAWR5H3z8owDQYJKoZIhvcNAQEF
BQADggEBALD3Ax2Cgnc8i6VZve3XzPLRJ+WAaiRJoZHSS9/xO65KftCjLTiU5iFl
iXgHzk5gNz9+HIq5AKFGTrSUQRvc0E/zeZ//cPO+b8AyBpfrlFe4BUMrWztk4oqd
lo9qCko3zatBs6Cs3ni6trNEWdy38Mn3DNv7OPfSXydWwcCALRLR9AUCoIRYdlcU
9ZOiriLuIrJU+wl6TRwCUSTKCetglhPRvKluTe83/HGEDCpkkb9H6iN472FiOgNW
P0FX6F/0zzIpxk/sIgtMtNiSjI6WWodhWMOd0E35bGcHqvOJH0Oe/EHdnzQnkDMk
G8sYUvBVUs66HkUgOSb25AqNgbrYgaI=
-----END CERTIFICATE-----
root@haproxy-napp-01 [ ~ ]#
```

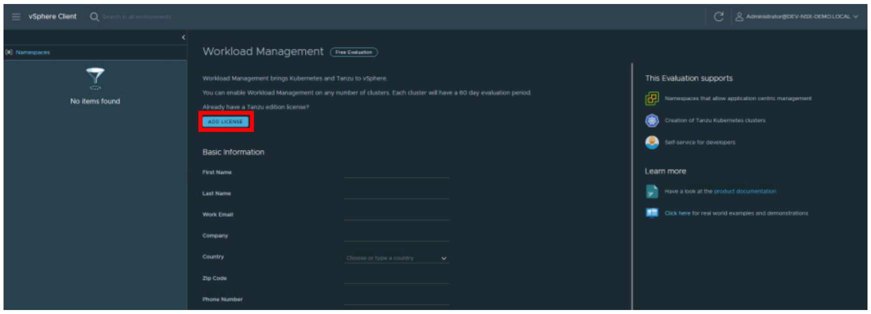
    Alternatively, you can also retrieve the certificate by **right-clicking** on the HAProxy VM and select **Edit Settings**, then `VM Settings -> Advanced -> Configuration Parameters -> Edit Configuration` and copy the CA cert from the `guestinfo.dataplaneapi.cacert` field and convert it from **Base64**

33. The HAProxy appliance will claim all IP addresses allocated as VIPs and respond to ICMP on these IPs, so I strongly recommend you verify IP connectivity to the Workload IP and the LB IPs from an external network before continuing.
    The HAProxy appliance also has multiple routing tables to manage connectivity between interfaces and while this should all be automatically plumbed if you enter the correct details, testing and resolving any connectivity issues now will save you time and troubleshooting later.

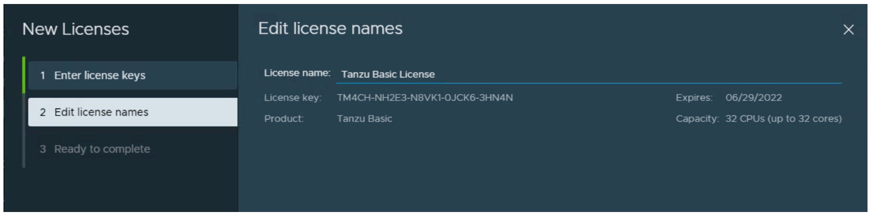


Enabling vSphere with Tanzu and deploying the Supervisor Cluster

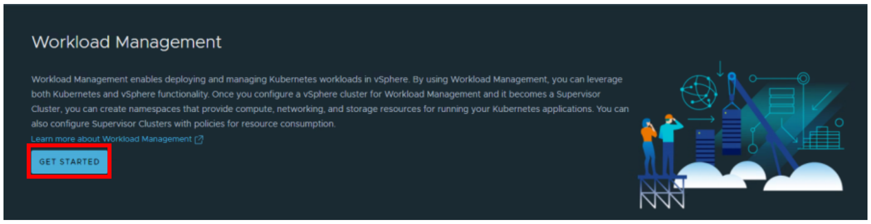
34. You are now ready to enable Tanzu with vSphere. In the vCenter Client navigate to `Home-> Workload Management`



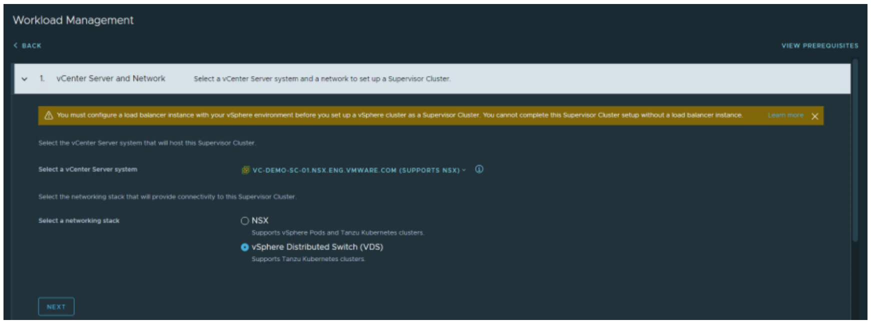
35. Here you either need to click **ADD LICENSE** and enter a valid **Tanzu license** or sign up for the free 60-day evaluation. A Tanzu Basic license is sufficient and your VMware contact for NSX can help provide details on the included licensing options for Tanzu to run NAPP:



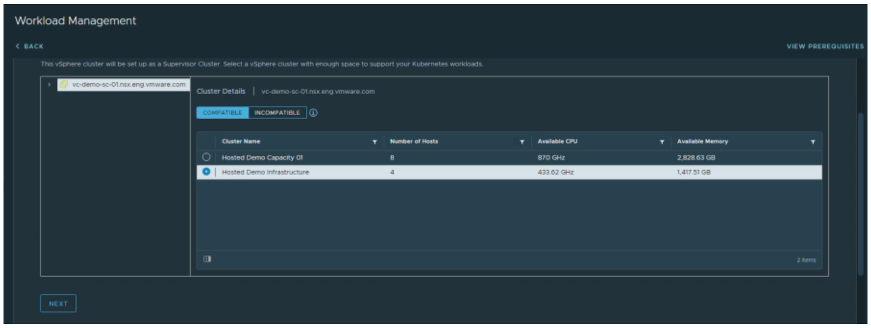
36. Once the license key has been accepted you can click on the **GET STARTED** button under Workload Management



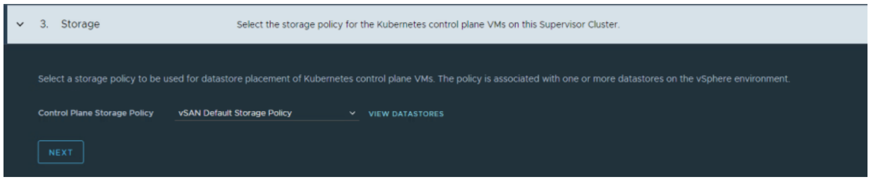
37. For this configuration select networking stack: **vSphere Distributed Switch (VDS)** and click **NEXT**



38. In this screen clusters will be validated against the pre-requirements for TKGs (such as DRS set to Fully Automated, sufficient number of hosts). Here you should select the Compatible cluster you want to enable for TKGs to run your NAPP cluster and click **NEXT**. If you have no compatible clusters listed you will need to remediate and then return to the Workload Management workflow:



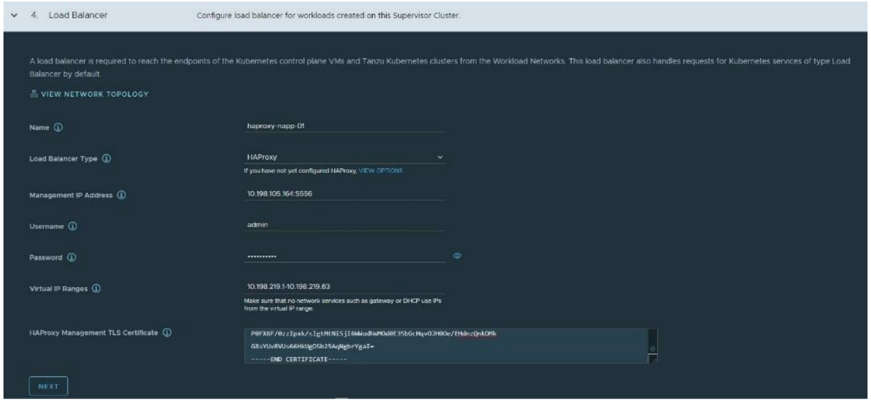
39. Select the appropriate **Control Plane Storage Policy** for your environment:



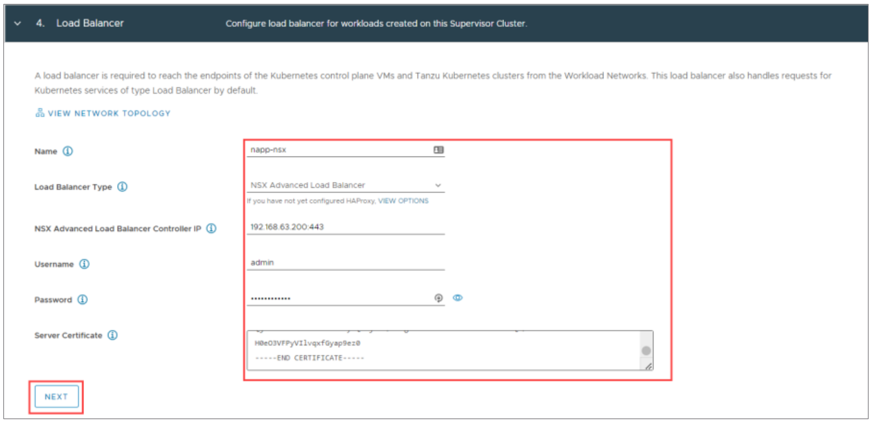
40. Configure HAProxy

On this step enter the same details that were provided during the HAProxy appliance deployment:

- **Name** (VM name)
- **Load Balancer Type** (HAProxy)
- **Management IP Address** (include the Dataplane API port, 5556 in this case)
- **Username & Password**
- **Virtual IP Ranges**
    - **Note:** It is **important** to use an IP Range for VIPs that **does not overlap** with the default gateway or any other workloads. This is one reason why a dedicated network for VIPs is recommended, but even in this case you cannot use entire subnet as your IP range. As you will note in this guide, while my Frontend network is a /25 subnet I am only allocating the first 64 addresses as the Virtual IP Range to avoid any conflicts.
- **HAProxy Certificate** (as retrieved earlier in step 22)

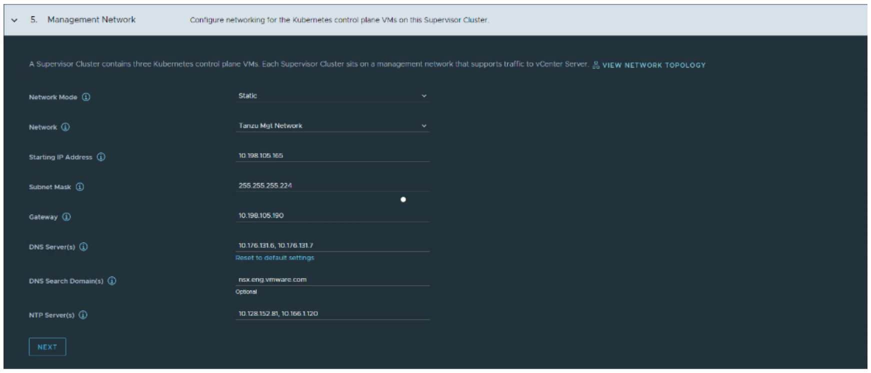


If you already have NSX Advanced Load Balancer in your environment, it is also possible to utilize it instead of HAProxy. Refer to the NSX ALB Documentation for more information on deployment and configuration steps.



41. Configure **Management Network**. This network is used to configure Management IP settings on the Supervisor nodes and you have the option of **Static** or **DHCP** (as infrastructure components **Static** is recommended for the Supervisor VMs). Enter the following inputs for your environment:

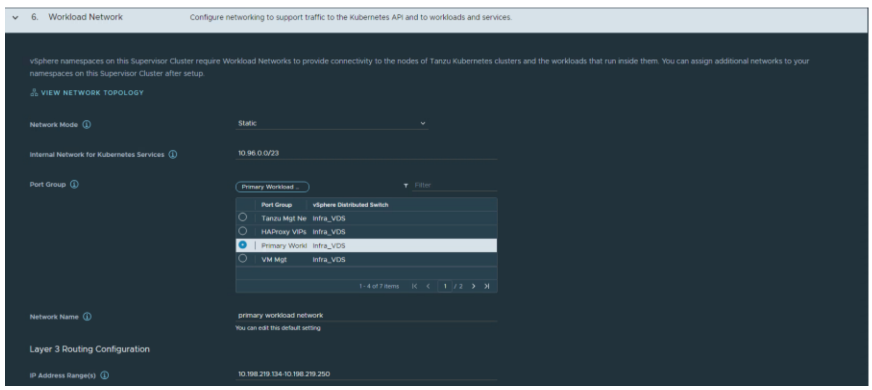
    - **Network Mode** (Static or DHCP)
    - **Starting IP Address**
    - **Subnet Mas**k
    - **Gateway**
    - **DNS and NTP Servers and DNS Search Domain**



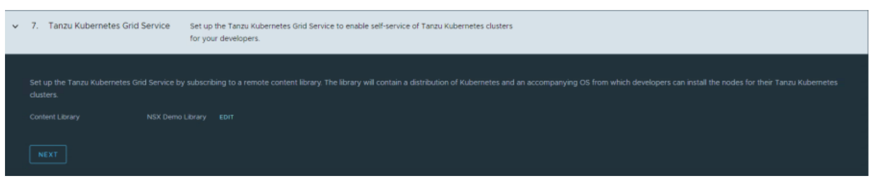
42. Next configure the **Workload Network**. This network is used to communicate between the Supervisor Cluster and any Tanzu Workload/Guest Clusters and as mentioned the workload network must also have IP routing to the Load Balancer VIPs if you are not sharing networks. Select the dvPortgroup created earlier for the Primary

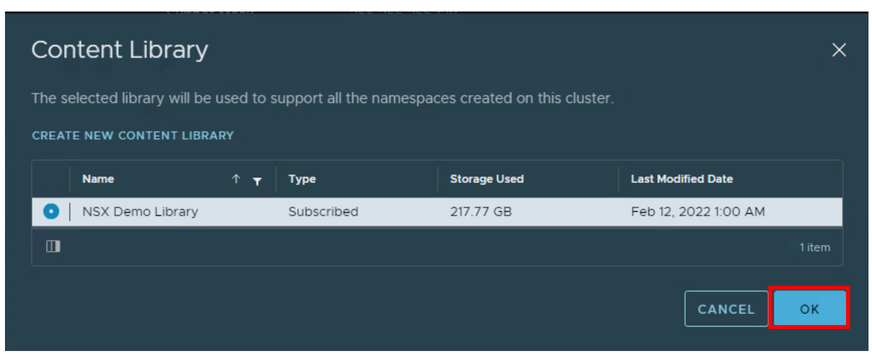
Workload Network and enter the following inputs for your environment:

- **Network Mode** (Static or DHCP)
- **Internal Network for Kubernetes Services** (this can be left as default unless this range overlaps or is in use in your environment)
- **Network Name** (used as a label)
- **IP Address Range(s)**
- **Subnet Mask**
- **Gateway**
- **DNS and NTP Servers**



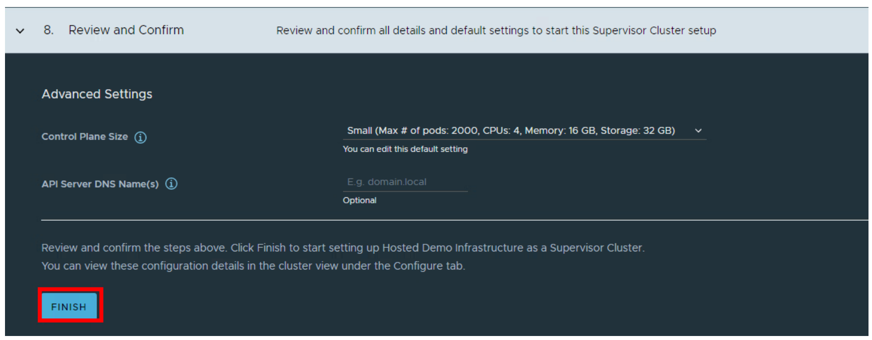
43. Now you will select the **Content Library** created back in step 3:





44. And finally select the **Control Plane Size** and optional **API Server DNS Name** for TKGs. If you are not running additional guest clusters/pods you can use a Control Plane Size of **Tiny** for a NAPP deployment, however **Small** is recommended if resources are not constrained. The API Server DNS Name field is optional as the Load Balancer IP is automatically used if this is left empty.

    Note: Do not add a DNS Server IP, or a hostname that is not correctly defined in DNS to map to the K8s Cluster API Endpoint IP. Generally, unless you have a good reason it is best to leave this field blank.

    Once you are happy, click **FINISH** to complete the Supervisor Cluster setup.

45. During setup the Supervisor Cluster will report a **Config Status** of **Configuring**

46. You can monitor the deployment in
    `Menu -> Workload Management -> Clusters or Menu -> Hosts and Clusters -> Select the Cluster -> Monitor -> Namespaces -> Overview`

    This will begin the deployment of three virtual machines that make up the Supervisor control plane. These Supervisor VMs will be deployed in a namespace and any dependencies will also be configured including VIPs, Resource Pools, TKG UI Plugin, etc. This process will take some time dependent on your environment (estimated at approximately 15 minutes).

    Here you can see Supervisor VMs have been created:

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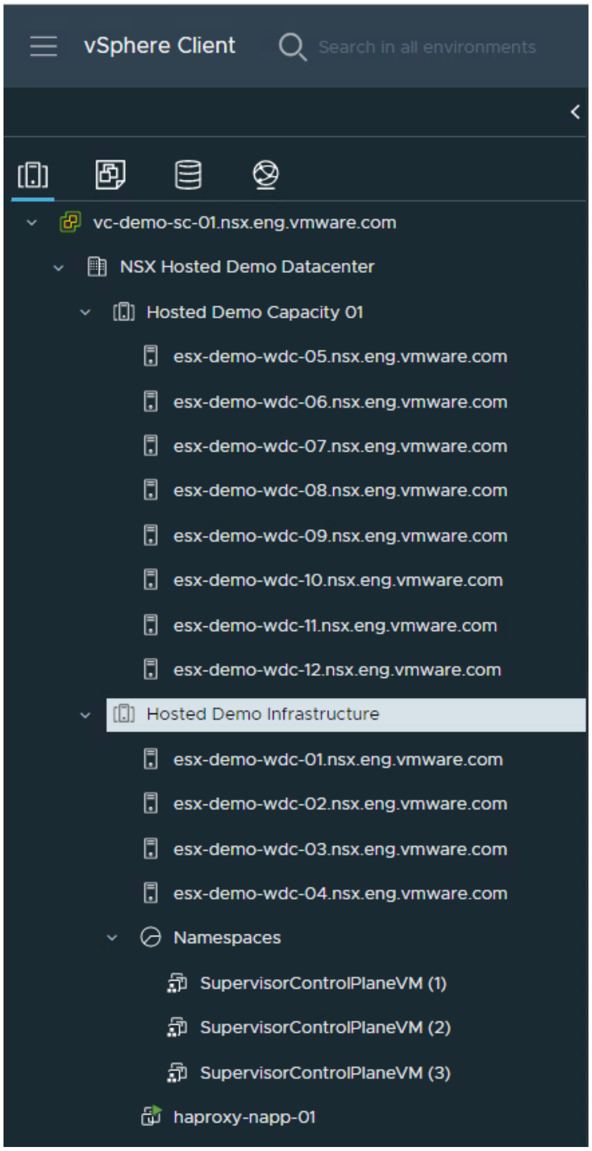
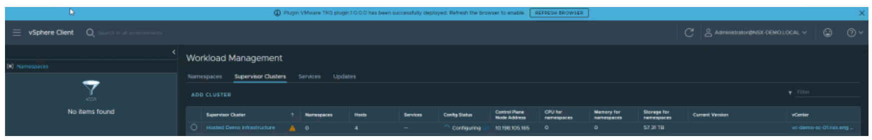


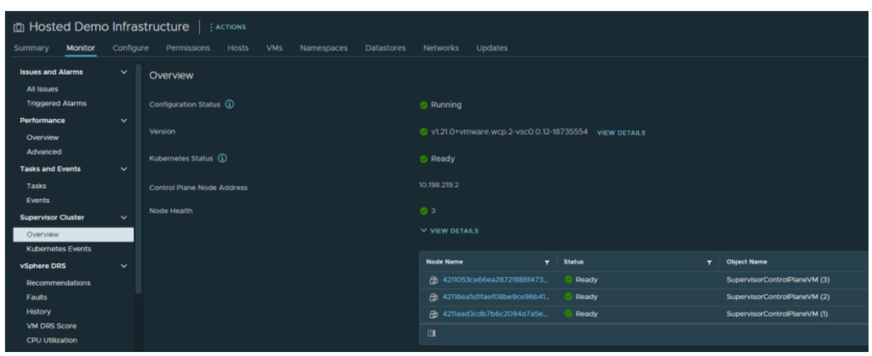
You can also click on **Config Status** for more information on the step-by-step operations and any Errors, Warnings or Informational status messages:

47. During the deployment you will receive a notification to refresh your browser and enable the **TKG UI Plugin**:

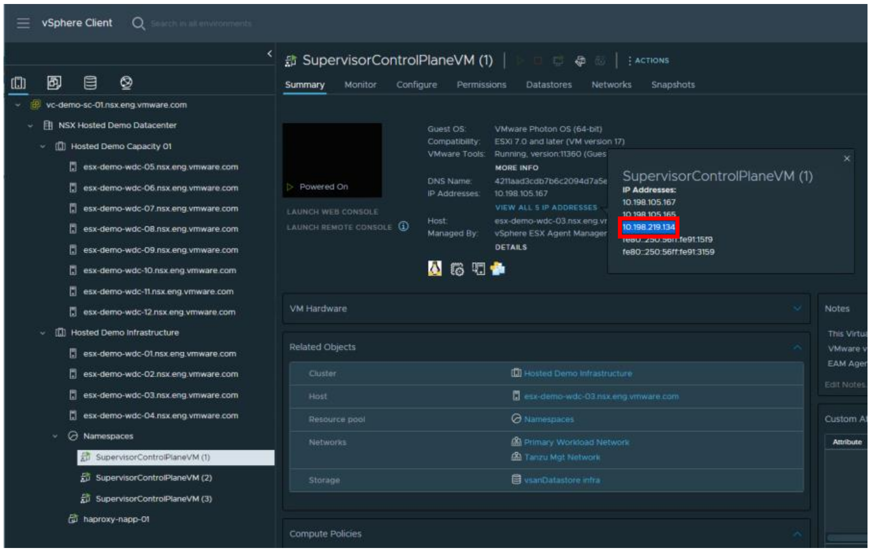


48. Once VM deployment and customization has completed the **Config Status** will change to **Running**:

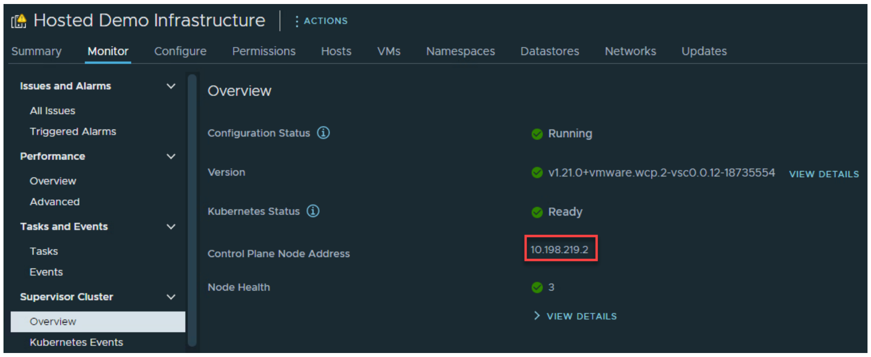
49. For good measure you can also verify the status of your Supervisor Cluster under `Hosts and Clusters->Monitor->Supervisor Cluster->Overview`



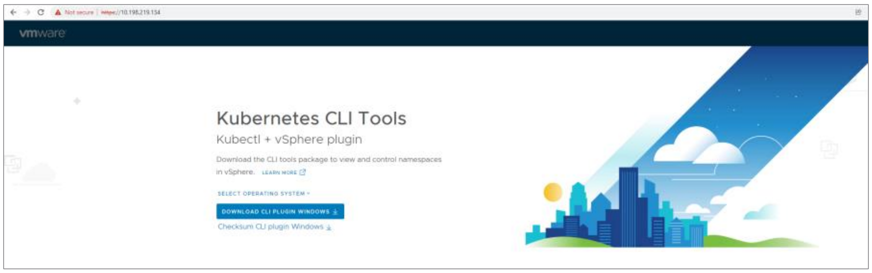
50. And finally connect to the VIP to test connectivity. First retrieve the IP assigned to the **first Supervisor VM** from your Load Balancer VIP range:



Or alternatively you can also use the Control Plane Node Address assigned from the LB VIPs:



51. Then access this IP using a browser via **https** and download the **Kubernetes CLI Tools** as you will use these to connect to your environment:



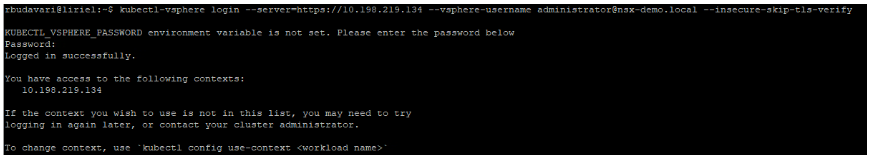
52. Install the tools on either a Windows, Linux or MacOS administrative host

that has connectivity to the management networks in your environment.

To install simply extract the zip file and run the binaries (`kubectl` and `kubectl-vsphere`). You can also add them to your `$PATH`

53. Now test your access by logging in to the Supervisor Cluster using your **vCenter Server** credentials:

```
kubectl-vsphere login --server=https://<sup-cluster-controlplane-ip> --vsphere-username <administrator@your.domain> --insecure-skip-tls-verify
```



```
rbudavari@liriel:~$ kubectl-vsphere login --server=https://10.198.219.134 --vsphere-username administrator@nsx-demo.local --insecure-skip-tls-verify

KUBECTL_VSPHERE_PASSWORD environment variable is not set. Please enter the password below
Password:
Logged in successfully.

You have access to the following contexts:
   10.198.219.134

If the context you wish to use is not in this list, you may need to try
logging in again later, or contact your cluster administrator.

To change context, use `kubectl config use-context <workload name>`
```

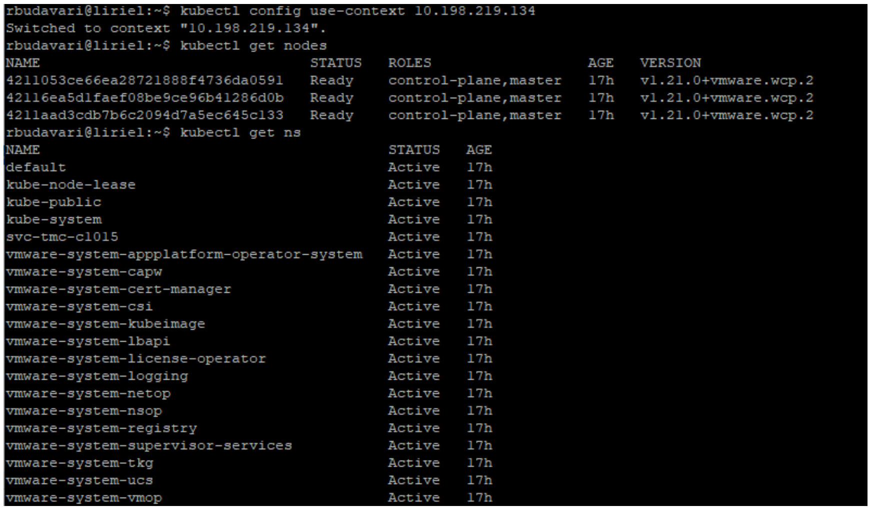
Once you have authenticated successfully, here are some additional `kubectl` commands you can run to verify the environment is operational:

```
kubectl config use-context <sup-cluster-controlplane-ip>

kubectl get nodes

kubectl get ns

kubectl get svc -A
```

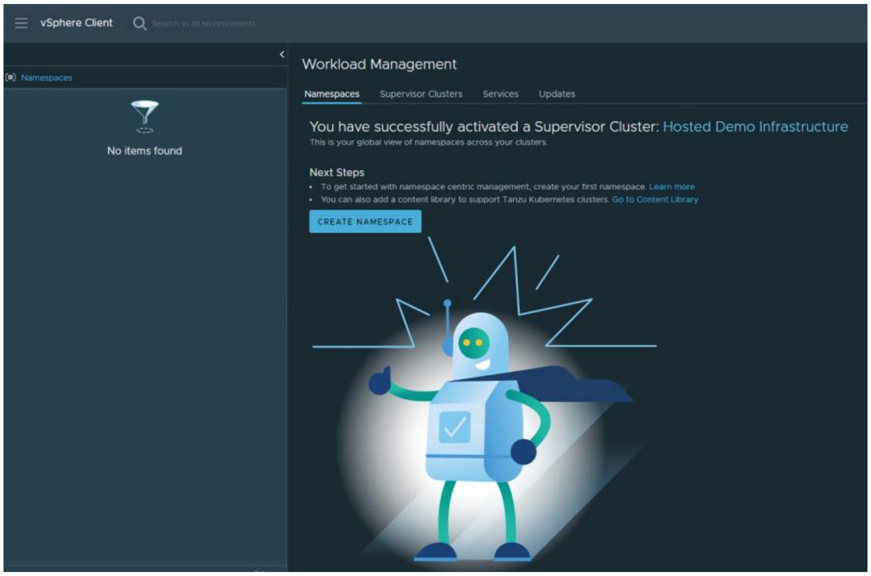


```
rbudavari@liriel:~$ kubectl config use-context 10.198.219.134
Switched to context "10.198.219.134".
rbudavari@liriel:~$ kubectl get nodes
NAME                               STATUS   ROLES                  AGE   VERSION
4211053ce66ea28721888f4736da0591   Ready    control-plane,master   17h   v1.21.0+vmware.wcp.2
42116ea5d1faef08be9ce96b41286d0b   Ready    control-plane,master   17h   v1.21.0+vmware.wcp.2
4211aad3cdb7b6c2094d7a5ec645c133   Ready    control-plane,master   17h   v1.21.0+vmware.wcp.2
rbudavari@liriel:~$ kubectl get ns
NAME                                          STATUS   AGE
default                                       Active   17h
kube-node-lease                               Active   17h
kube-public                                   Active   17h
kube-system                                   Active   17h
svc-tmc-c1015                                 Active   17h
vmware-system-appplatform-operator-system     Active   17h
vmware-system-capw                            Active   17h
vmware-system-cert-manager                    Active   17h
vmware-system-csi                             Active   17h
vmware-system-kubeimage                       Active   17h
vmware-system-lbapi                           Active   17h
vmware-system-license-operator                Active   17h
vmware-system-logging                         Active   17h
vmware-system-netop                           Active   17h
vmware-system-nsop                            Active   17h
vmware-system-registry                        Active   17h
vmware-system-supervisor-services             Active   17h
vmware-system-tkg                             Active   17h
vmware-system-ucs                             Active   17h
vmware-system-vmop                            Active   17h
```

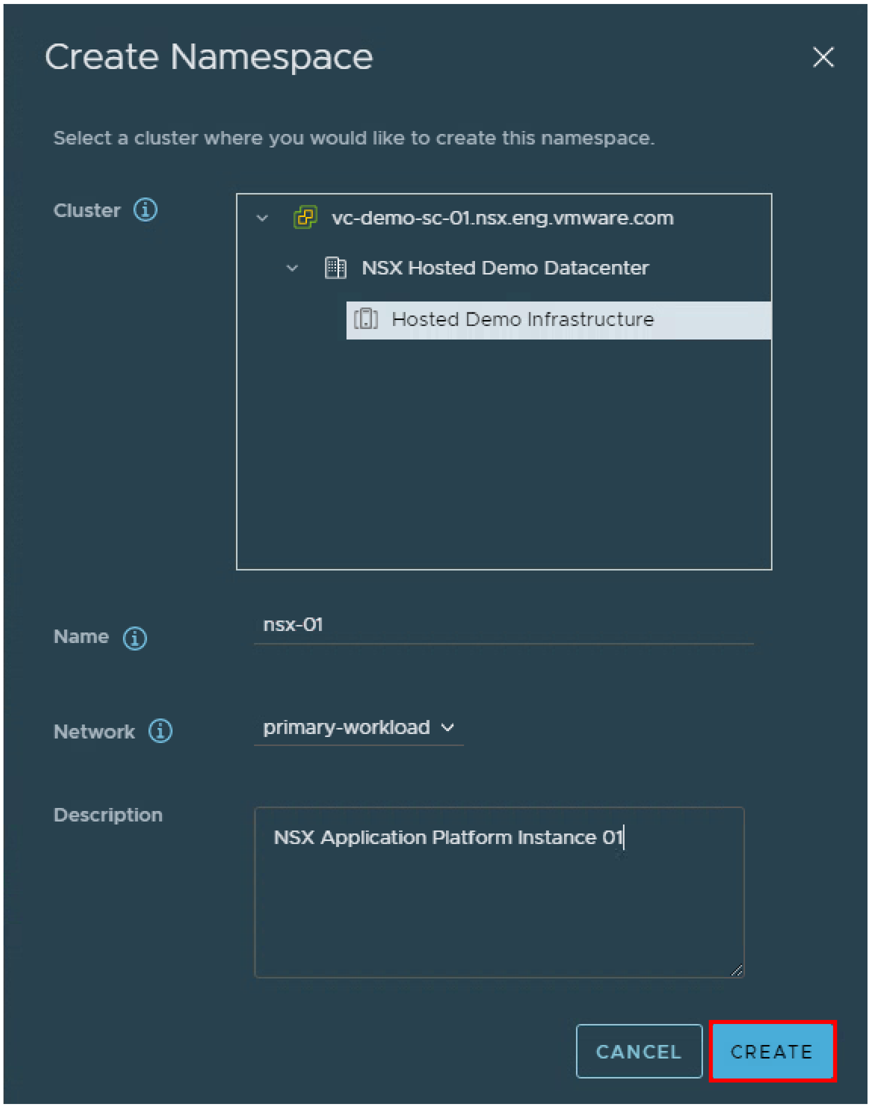
Creating a Tanzu Kubernetes Guest Cluster for NSX Application Platform

54. Now that our supervisor cluster is deployed and returning a status of Ready, we will create a **vSphere Namespace** that the control plane and worker nodes will be deployed in. This namespace maps vSphere resources (Compute, Memory, Storage & Network) and access controls to Tanzu Kubernetes Clusters.

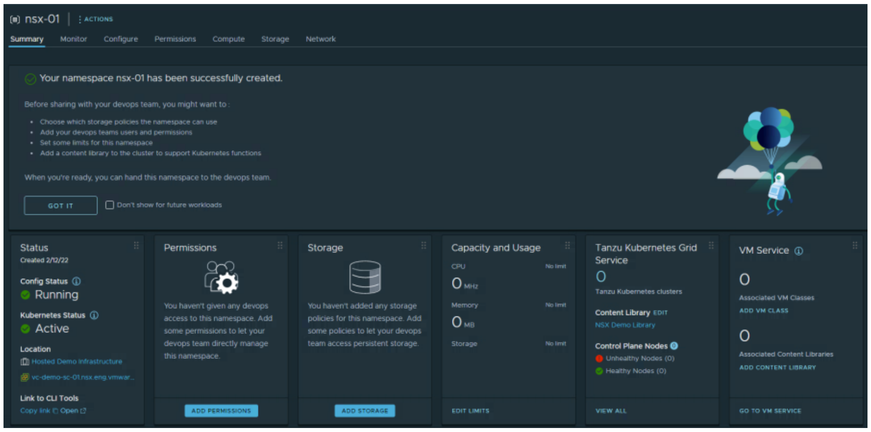
    Navigate to `Menu -> Workload Management – > Namespaces -> Create Namespace`

55. Click **CREATE NAMESPACE**, then select your **target cluster**, provide a descriptive **Name** (make sure you select a unique identifier for name) and ensure the **Primary Workload Network** is correctly populated using the value from your initial Workload Management configuration and finally click **CREATE**



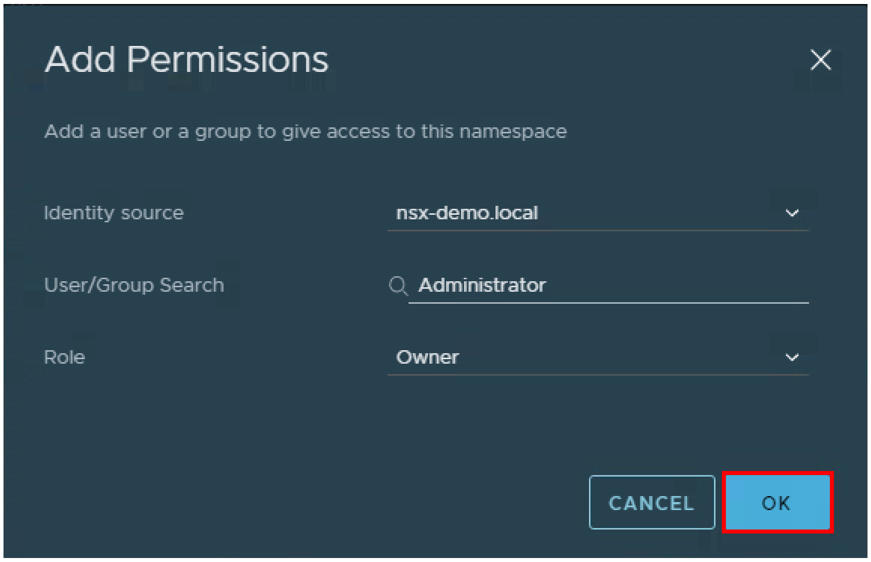
56. Once the **Namespace** has been created you will see the following dashboard:



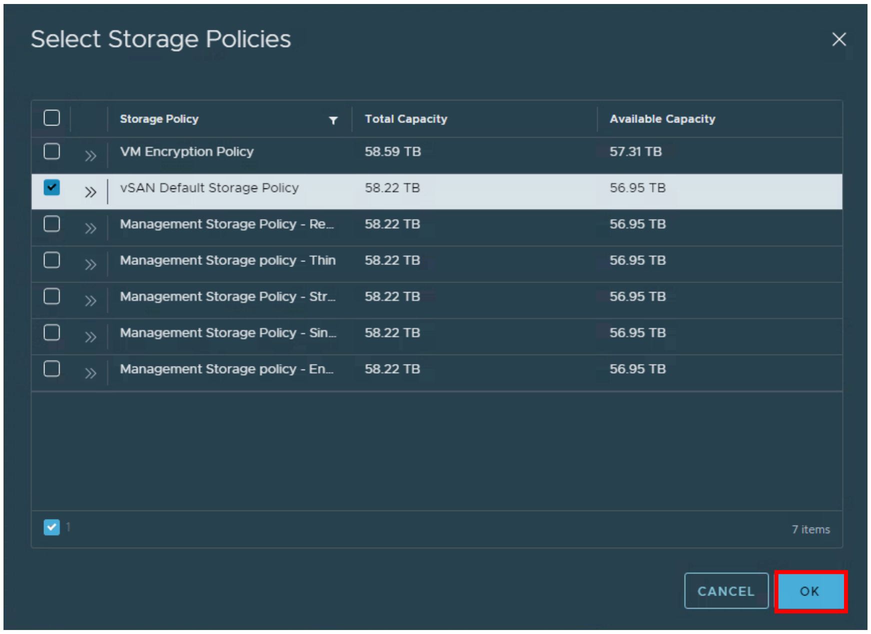
Which will allow you to configure all required namespace settings such as permissions, storage policies, VM classes and add a content library.

57. Start the namespace configuration by clicking **ADD PERMISSIONS**

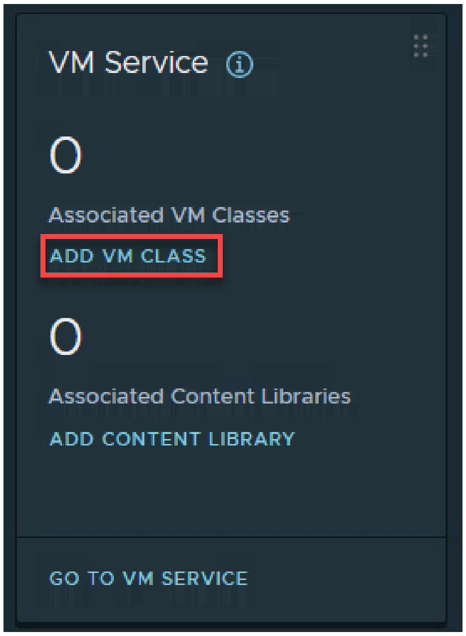
    In this case, as NAPP is a management feature, we are using the **Administrator** user as the assigned owner in this guide. You can select the appropriate user for your environment based on operational policies:



58. Next click **ADD STORAGE** and select the appropriate Storage Policy for your target Cluster



59. Now navigate to the VM Service tile and click on **ADD VM CLASS**. This is where you select VM classes, which determine the VM sizing for the TKC worker and control nodes that are deployed within this namespace:



An NSX Application Platform deployment has different Form Factors based on the features you will run in your environment. Details on

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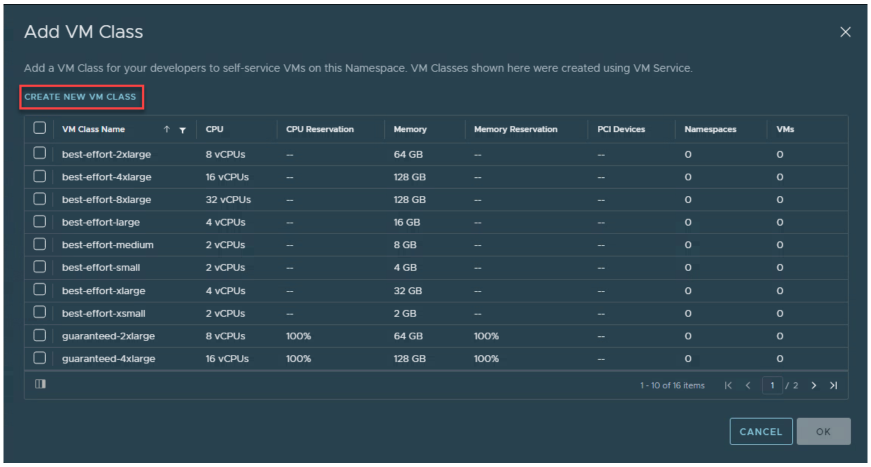
the available Form Factors and supported features can be found here:

https://docs.vmware.com/en/VMware-NSX-T-Data-Center/3.2/nsx-application-platform/GUID-85CD2728-8081-45CE-9A4A-D72F49779D6A.html

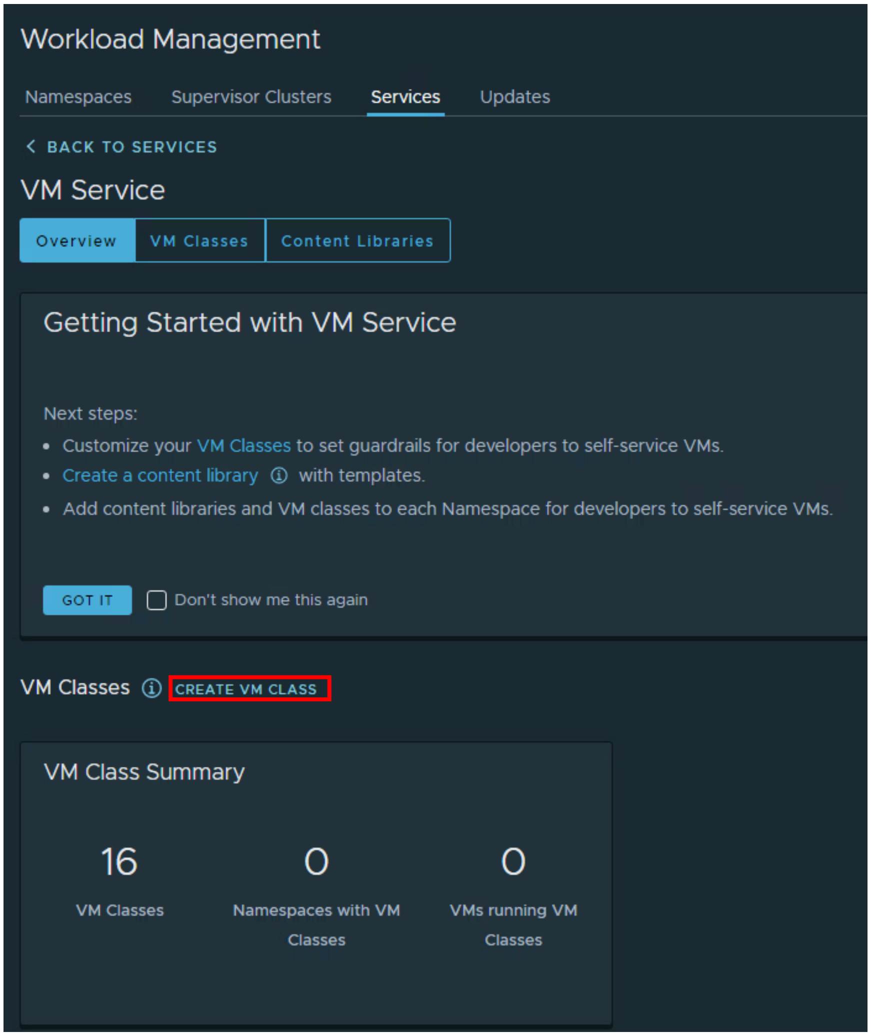
and in summary:

- **Standard** requires **4vCPU** and **16GB RAM**
- **Advanced** is required for NSX Intelligence and needs a Custom VM Class of **16vCPU** and **64GB** RAM
- Control Nodes are recommended to use **2vCPU** and **8GB** (although technically **4GB** will also work)
- If this is a temporary setup and resources are limited you may also consider the **Evaluation** form factor

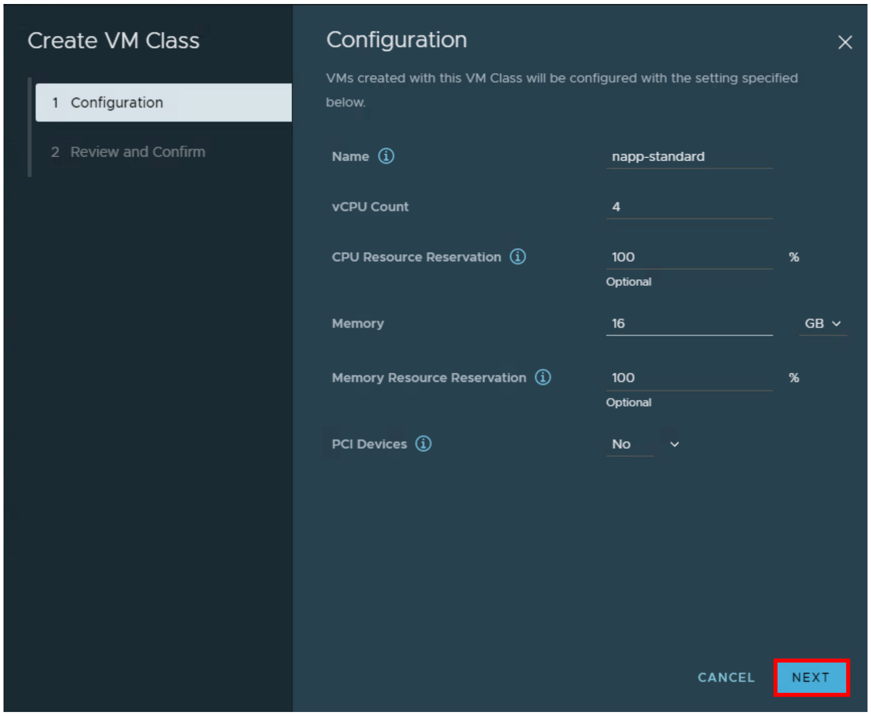
60. If you are planning to run NSX Intelligence on you NAPP cluster, click on **CREATE NEW VM CLASS**:



Where you will be presented with the following screen where you should click on **CREATE VM CLASS**:

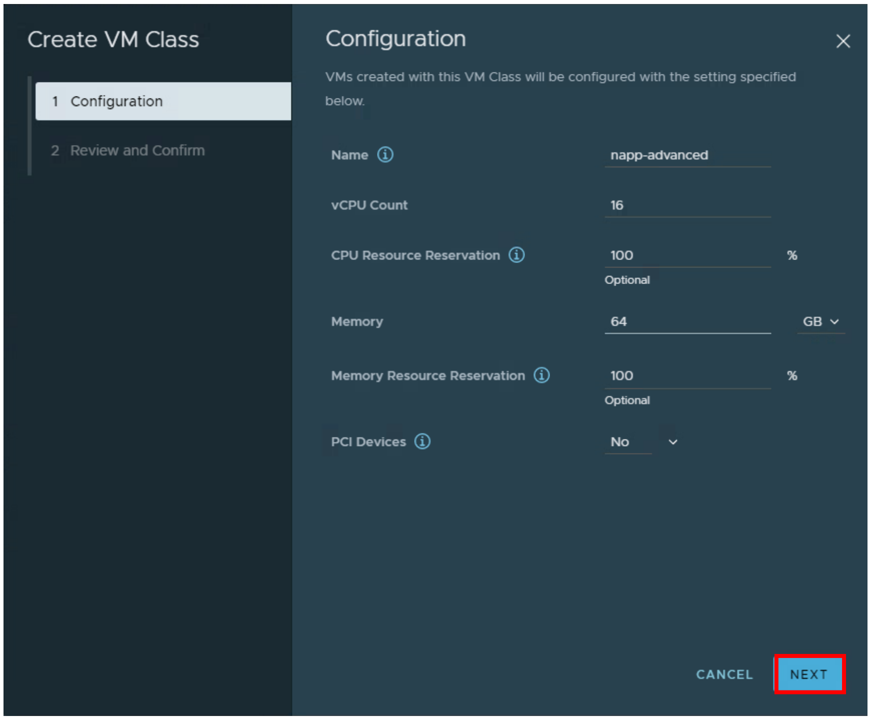


Which will allow you to define a VM class with different specifications than the default options. We will Create a VM Class for napp-standard with 4 vCPU and 16GB of RAM:



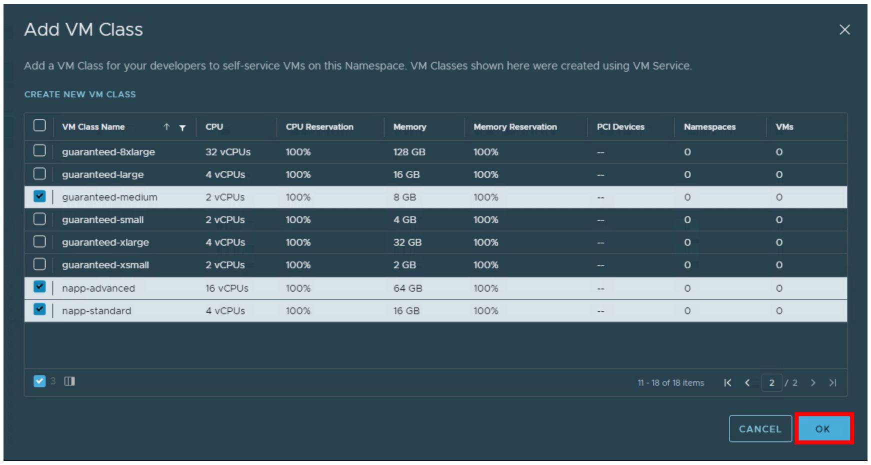
Technically **napp-standar**d could use the existing **guaranteed-large** but I like to maintain consistency in my naming conventions.

61. You will also Create a VM Class for **napp-advanced** with **16 vCPU** and **64GB of RAM**:

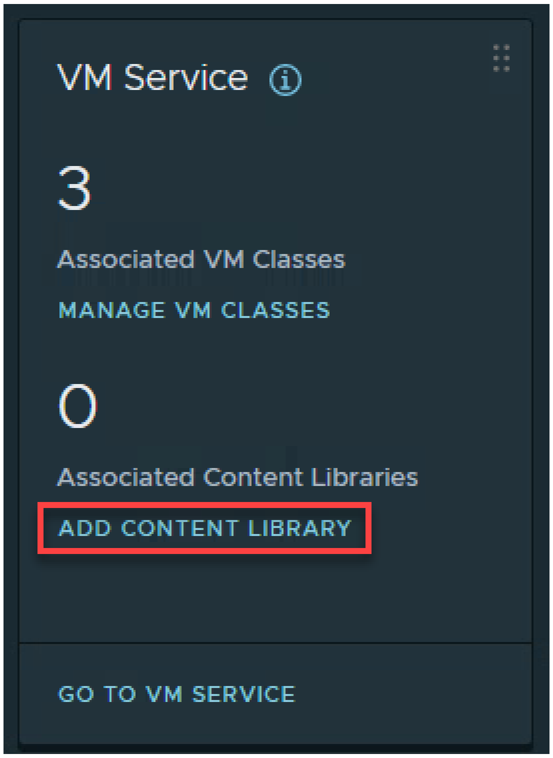


While a guaranteed **Resource Reservation** for **CPU** and **Memory** is recommended, you can remove this if the environment is non-production or you do not have sufficient resources for the reservation

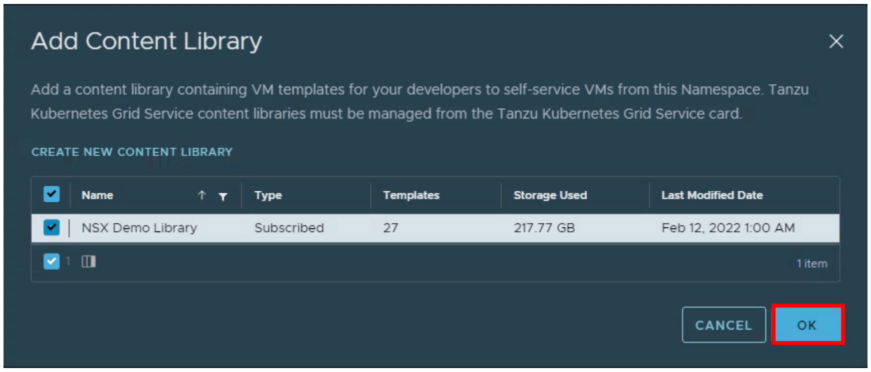
62. Now navigate back to the namespace summary page on the name space and click on **Add VM** Class again. You should now be able to see the VM Classes you just created. For NAPP select **guaranteed-medium** for the control plane node, and the **custom class** that was just added based on your Form Factor for worker nodes:



63. And finally click **ADD CONTENT LIBRARY**



And select the same **Content Library** used when setting up Workload Management:



At this point you have completed Namespace configuration and the **Summary** page should look similar to:

64. You are now ready to Deploy your NSX Application Platform **Workload Cluster** (Control Plane and Worker Nodes). This process will use the `kubectl` CLI and a `yaml` configuration file that describes your cluster configuration.

    To cover the NAPP requirements and additional configuration parameters the following yaml files have been shared based on Form Factor:

    Standard NAPP Configuration

    Advanced NAPP Configuration

    In this Deployment Guide we will use the advanced file to support running NSX Intelligence. If you only plan on using the Metrics, NDR/NTA or Malware Prevention features you can use the standard file. These files will create the correct node sizes and also provision additional ephemeral storage (64GB vs the default 16GB) required for NAPP.

    Download the required `yaml` file to the system you have installed the Kubernetes CLI Tools on using curl:

```
curl -LO https://raw.githubusercontent.com/vmware-nsx/napp-deployment/main/advanced.yaml
```

    You may need to make some changes to the YAML file based on your environment and naming conventions. The first setting to verify is the Tanzu Kubernetes Release, as this Deployment Guide was created using **vSphere 7.0 U3c**. Run the following command to retrieve the available releases of TKG:

```
kubectl get tkr
```

```
rbudavari@liriel:~$ kubectl get tkr
NAME                                VERSION                               READY   COMPATIBLE   CREATED   UPDATES AVAILABLE
v1.16.12---vmware.1-tkg.1.da7afe7   1.16.12+vmware.1-tkg.1.da7afe7   True    True         18h       [1.17.17+vmware.1-tkg.1.d44d45a 1.16.14+vmware.1-tkg.1.ada4837]
v1.16.14---vmware.1-tkg.1.ada4837   1.16.14+vmware.1-tkg.1.ada4837   True    True         18h       [1.17.17+vmware.1-tkg.1.d44d45a]
v1.16.8---vmware.1-tkg.3.60d2ffd    1.16.8+vmware.1-tkg.3.60d2ffd    False   False        18h       [1.17.17+vmware.1-tkg.1.d44d45a 1.16.14+vmware.1-tkg.1.ada4837]
v1.17.11---vmware.1-tkg.1.15f1e18   1.17.11+vmware.1-tkg.1.15f1e18   True    True         18h       [1.18.19+vmware.1-tkg.1.17af790 1.17.17+vmware.1-tkg.1.d44d45a]
v1.17.11---vmware.1-tkg.2.ad3d374   1.17.11+vmware.1-tkg.2.ad3d374   True    True         18h       [1.18.19+vmware.1-tkg.1.17af790 1.17.17+vmware.1-tkg.1.d44d45a]
v1.17.13---vmware.1-tkg.2.2c133ed   1.17.13+vmware.1-tkg.2.2c133ed   True    True         18h       [1.18.19+vmware.1-tkg.1.17af790 1.17.17+vmware.1-tkg.1.d44d45a]
v1.17.17---vmware.1-tkg.1.d44d45a   1.17.17+vmware.1-tkg.1.d44d45a   True    True         18h       [1.18.19+vmware.1-tkg.1.17af790]
v1.17.7---vmware.1-tkg.1.154236c    1.17.7+vmware.1-tkg.1.154236c    True    True         18h       [1.18.19+vmware.1-tkg.1.17af790 1.17.17+vmware.1-tkg.1.d44d45a]
v1.17.8---vmware.1-tkg.1.5417466    1.17.8+vmware.1-tkg.1.5417466    True    True         18h       [1.18.19+vmware.1-tkg.1.17af790 1.17.17+vmware.1-tkg.1.d44d45a]
v1.18.10---vmware.1-tkg.1.3a6cd48   1.18.10+vmware.1-tkg.1.3a6cd48   True    True         18h       [1.19.16+vmware.1-tkg.1.df910e2 1.18.19+vmware.1-tkg.1.17af790]
v1.18.15---vmware.1-tkg.1.600e412   1.18.15+vmware.1-tkg.1.600e412   True    True         18h       [1.19.16+vmware.1-tkg.1.df910e2 1.18.19+vmware.1-tkg.1.17af790]
v1.18.15---vmware.1-tkg.2.ebf6117   1.18.15+vmware.1-tkg.2.ebf6117   True    True         18h       [1.19.16+vmware.1-tkg.1.df910e2 1.18.19+vmware.1-tkg.1.17af790]
v1.18.19---vmware.1-tkg.1.17af790   1.18.19+vmware.1-tkg.1.17af790   True    True         18h       [1.19.16+vmware.1-tkg.1.df910e2]
v1.18.5---vmware.1-tkg.1.c40d30d    1.18.5+vmware.1-tkg.1.c40d30d    True    True         18h       [1.19.16+vmware.1-tkg.1.df910e2 1.18.19+vmware.1-tkg.1.17af790]
v1.19.11---vmware.1-tkg.1.9d9b236   1.19.11+vmware.1-tkg.1.9d9b236   True    True         18h       [1.20.12+vmware.1-tkg.1.b9a42f3 1.19.16+vmware.1-tkg.1.df910e2]
v1.19.14---vmware.1-tkg.1.8753786   1.19.14+vmware.1-tkg.1.8753786   True    True         18h       [1.20.12+vmware.1-tkg.1.b9a42f3 1.19.16+vmware.1-tkg.1.df910e2]
v1.19.16---vmware.1-tkg.1.df910e2   1.19.16+vmware.1-tkg.1.df910e2   True    True         18h       [1.20.12+vmware.1-tkg.1.b9a42f3]
v1.19.7---vmware.1-tkg.1.fc82c41    1.19.7+vmware.1-tkg.1.fc82c41    True    True         18h       [1.20.12+vmware.1-tkg.1.b9a42f3 1.19.16+vmware.1-tkg.1.df910e2]
v1.19.7---vmware.1-tkg.2.f52f85a    1.19.7+vmware.1-tkg.2.f52f85a    True    True         18h       [1.20.12+vmware.1-tkg.1.b9a42f3 1.19.16+vmware.1-tkg.1.df910e2]
v1.20.12---vmware.1-tkg.1.b9a42f3   1.20.12+vmware.1-tkg.1.b9a42f3   True    True         18h       [1.21.6+vmware.1-tkg.1.b3d708a]
v1.20.2---vmware.1-tkg.1.1d4f79a    1.20.2+vmware.1-tkg.1.1d4f79a    True    True         18h       [1.21.6+vmware.1-tkg.1.b3d708a 1.20.12+vmware.1-tkg.1.b9a42f3]
v1.20.2---vmware.1-tkg.2.3e10706    1.20.2+vmware.1-tkg.2.3e10706    True    True         18h       [1.21.6+vmware.1-tkg.1.b3d708a 1.20.12+vmware.1-tkg.1.b9a42f3]
v1.20.7---vmware.1-tkg.1.7fb9067    1.20.7+vmware.1-tkg.1.7fb9067    True    True         18h       [1.21.6+vmware.1-tkg.1.b3d708a 1.20.12+vmware.1-tkg.1.b9a42f3]
v1.20.8---vmware.1-tkg.2            1.20.8+vmware.1-tkg.2            True    True         18h
v1.20.9---vmware.1-tkg.1.a4cee5b    1.20.9+vmware.1-tkg.1.a4cee5b    True    True         18h       [1.21.6+vmware.1-tkg.1.b3d708a 1.20.12+vmware.1-tkg.1.b9a42f3]
v1.21.2---vmware.1-tkg.1.ee25d55    1.21.2+vmware.1-tkg.1.ee25d55    True    True         18h       [1.21.6+vmware.1-tkg.1.b3d708a]
v1.21.6---vmware.1-tkg.1.b3d708a    1.21.6+vmware.1-tkg.1.b3d708a    True    True         18h
```

And note the most recent **VERSION** (1.21.6 in this case)

Then retrieve the available storage classes in your environment using:

```
kubectl get sc
```

```
rbudavari@liriel:~$ kubectl get sc
NAME                          PROVISIONER              RECLAIMPOLICY   VOLUMEBINDINGMODE   ALLOWVOLUMEEXPANSION   AGE
vsan-default-storage-policy   csi.vsphere.vmware.com   Delete          Immediate           true                   4h27m
```

Now open the YAML file and edit the following highlighted parameters which have been highlighted to match your environment:

```
metadata:
            name: napp-advanced-01 #TKGs cluster
            namespace: nsx-01 #TKGs namespace
spec:
           version: v1.21.6   #Latest TKR
topology:
               controlPlane:
                 class: guaranteed-medium #vmclass for CP
                 storageClass: vsan-default-storage-policy
               workers:
                 class: napp-advanced #vmclass for worker
                 storageClass: vsan-default-storage-policy
```

Here is the complete **YAML** file from my environment for reference:

```
rbudavari@liriel:~$ cat advanced.yaml
apiVersion: run.tanzu.vmware.com/v1alpha1       #TKGs API endpoint
kind: TanzuKubernetesCluster
metadata:
  name: napp-advanced-01                        #TKGs cluster name
  namespace: nsx-01                             #TKGs namespace
spec:
  distribution:
    version: v1.21.6                            #Latest TKR
  topology:
    controlPlane:
      count: 1                                  #number of control plane nodes
      class: guaranteed-medium                  #vmclass for control plane nodes
      storageClass: vsan-default-storage-policy #storageclass for control plane
      volumes:
        - name: etcd
          mountPath: /var/lib/etcd
          capacity:
            storage: 64Gi
    workers:
      count: 3                                  #number of worker nodes
      class: napp-advanced                      #vmclass for worker nodes
      storageClass: vsan-default-storage-policy #storageclass for worker nodes
      volumes:
        - name: containerd
          mountPath: /var/lib/containerd
          capacity:
            storage: 64Gi
```

Note - A critical point on IP Addressing using VDS networking

As per the vSphere with Tanzu release notes:

https://docs.vmware.com/en/VMware-vSphere/7.0/rn/vsphere-esxi-vcenter-server-7-vsphere-with-tanzu-release-notes.html?hWord=N4IghgNiBcIIwE4BMA6OA2AHCgDLkAvkA

In vSphere Distributed Switch (VDS) environments, it is possible to configure Tanzu Kubernetes clusters with network CIDR ranges that overlap or conflict with those of the Supervisor Cluster, and vice versa, resulting in components not being able to communicate.

In VDS environments, there is no design-time network validation performed when you configure the CIDR ranges for the Supervisor Cluster, or when you configure the CIDR ranges for Tanzu Kubernetes clusters. As a result, two problems can arise:

1) You create a Supervisor Cluster with CIDR ranges that conflict with the default CIDR ranges reserved for Tanzu Kubernetes clusters.

2) You create a Tanzu Kubernetes cluster with a custom CIDR range that overlaps with the CIDR range used for the Supervisor Clusters.

The reason I mention this is critical for NAPP deployments on TKGs is that the default IP Ranges assigned for K8s by the Antrea CNI are:

192.168.0.0/16 for Service IPs
10.192.0.0/12 for Cluster IPs

And it is very common for internal networks to be allocated from the `192.168.0.0/16` subnet reserved for private use as per RFC 1918 to management workloads.

This means that in particular the Cluster IP range can overlap with networks used in customer environments. These values are not ideal defaults and feedback has been provided to Tanzu team to potentially adjust it in future. However, if you are currently using the `192.168.0.0/16` or `10.192.0.0/12` subnets for any of the Supervisor Cluster networks (management, workload or LB VIPs) in your TKGs deployment it is essential that you apply the steps below **before** you create the guest cluster. These settings will configure the Antrea CNI to use different subnets for Service IPs or Cluster IPs and avoid any conflict. If you do not apply this step and have any overlapping CIDRs, you will see network connectivity errors such as **`no route to host`** or **`connection timeouts`** during NAPP deployment which can be difficult to troubleshoot.

To change these values, edit your guest cluster YAML file and find the following section:

```
spec:
  distribution:
    version: v1.21.6
```

then insert at the same indentation level as the `distribution` entry the following text:

```
  settings:
    network:
      cni:
        name: antrea
      services:
        cidrBlocks: ["10.208.0.0/12"]
      pods:
        cidrBlocks: ["192.169.0.0/16"]
      serviceDomain: "cluster.local"
```

replacing the values used for `cidrBlocks` with networks appropriate for your environment (I have changed both in this example).

Here is another complete **YAML** file for a guest cluster with the IP changes applied as a reference:

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```
apiVersion: run.tanzu.vmware.com/v1alpha1       #TKGs API endpoint
kind: TanzuKubernetesCluster
metadata:
  name: napp-advanced-01                        #TKGs cluster name
  namespace: nsx-01                             #TKGs namespace
spec:
  distribution:
    version: v1.21.6                            #Latest TKR
  settings:
    network:
      cni:
        name: antrea
      services:
        cidrBlocks: ["10.208.0.0/12"]
      pods:
        cidrBlocks: ["192.169.0.0/16"]          # Ensure Pod IPs do no overlap
      serviceDomain: "cluster.local"
  topology:
    controlPlane:
      count: 1                                  #number of control plane nodes
      class: guaranteed-medium                  #vmclass for control plane nodes
      storageClass: tanzu-storage-policy        #storageclass for control plane
      volumes:
        - name: etcd
          mountPath: /var/lib/etcd
          capacity:
            storage: 64Gi
    workers:
      count: 1                                  #number of std worker nodes
      class: nappvmclass                        #vmclass for std worker nodes
      storageClass: tanzu-storage-policy        #storageclass for worker nodes
      volumes:
        - name: containerd
          mountPath: /var/lib/containerd
          capacity:
            storage: 64Gi
```

65. Then, if you aren't already logged in to your supervisor cluster, use `kubectl-vsphere` login to authenticate:

```
rbudavari@liriel:~$ kubectl-vsphere login --server=https://10.198.219.134 --vsphere-username administrator@nsx-demo.local --insecure-skip-tls-verify

KUBECTL_VSPHERE_PASSWORD environment variable is not set. Please enter the password below
Password:
Logged in successfully.

You have access to the following contexts:
   10.198.219.134
   nsx-01

If the context you wish to use is not in this list, you may need to try
logging in again later, or contact your cluster administrator.

To change context, use `kubectl config use-context <workload name>`
```

Also note that the newly-created **namespace** is listed as a context now.

You will then use the following `kubectl` command to deploy your guest cluster providing the YAML file as an input:
`kubectl apply -f advanced.yaml`

```
rbudavari@liriel:~$ kubectl apply -f advanced.yaml
tanzukubernetescluster.run.tanzu.vmware.com/napp-advanced-01 created
```


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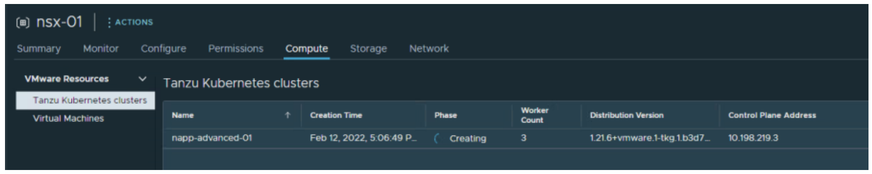
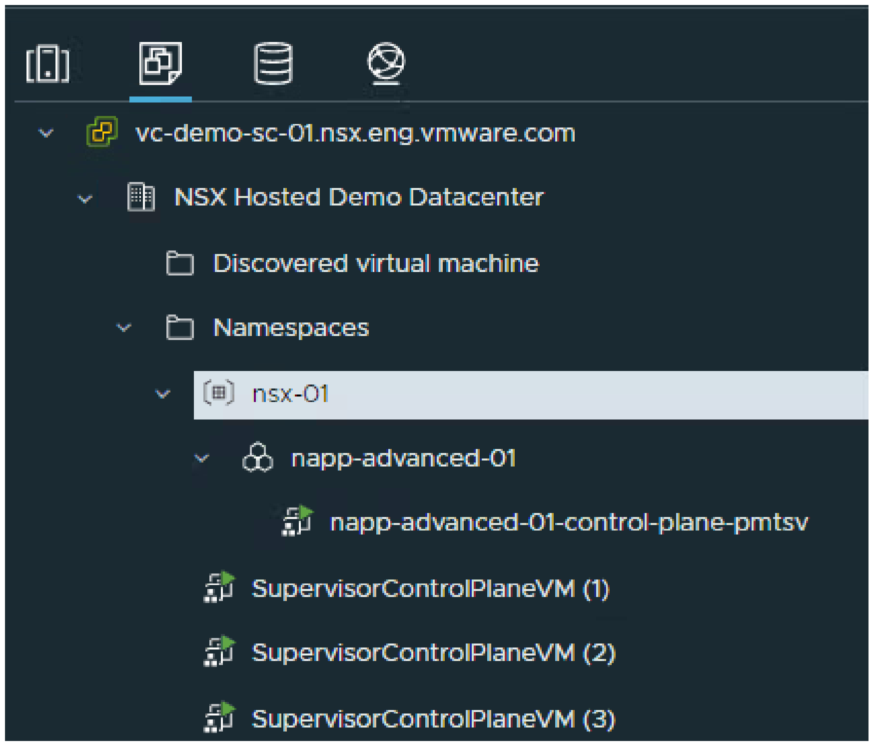
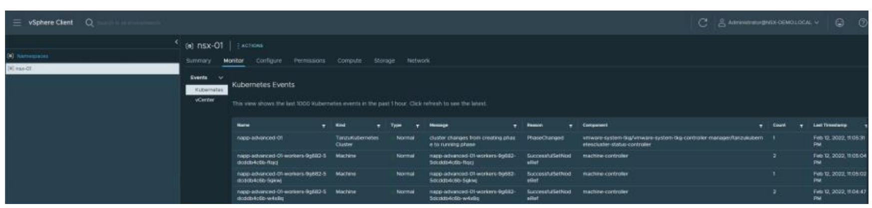
66. Return to the vCenter Server UI and under your **Namespace** in Workload Management you should be able to see VMs for the TKC are in a Phase of **Creating:**



You can also observe VMs for the TKC being provisioned under the **VMs and Templates** view, starting with the Control Plane node:



67. During this process you can monitor the status from either the UI, under `Workload Management -> Namespaces -> <namespace name> -> Monitor -> Kubernetes & vCenter`



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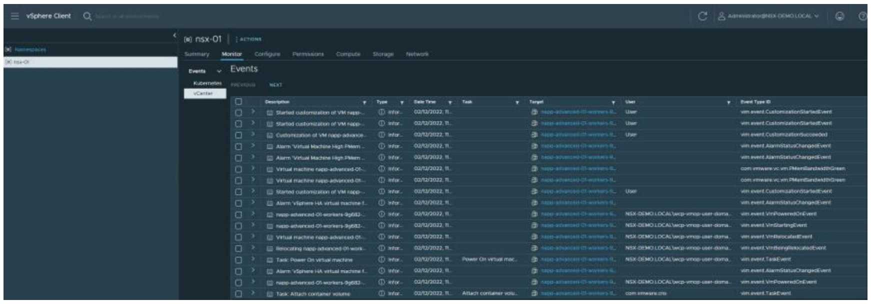
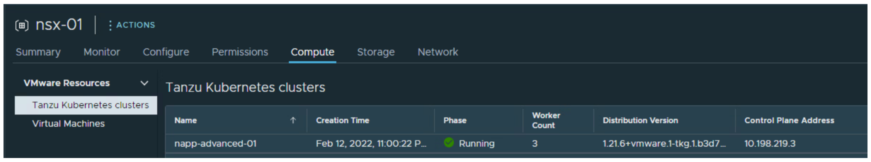


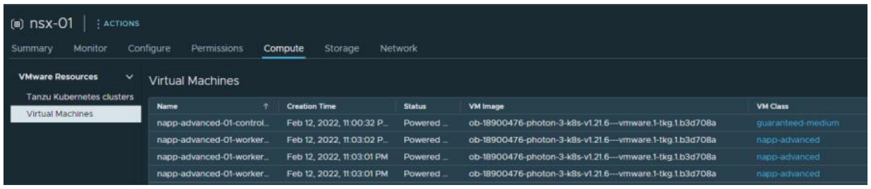
Or CLI using: `kubectl get svc -n <namespace name>`

68. The guest cluster deployment will take a couple of minutes, so be patient. Once you can see an Event message of: **Cluster changes from creating phase to running phase**

    You can click on the Compute tab of your namespace and verify the Tanzu Kubernetes Cluster is now in a Phase of: **Running**

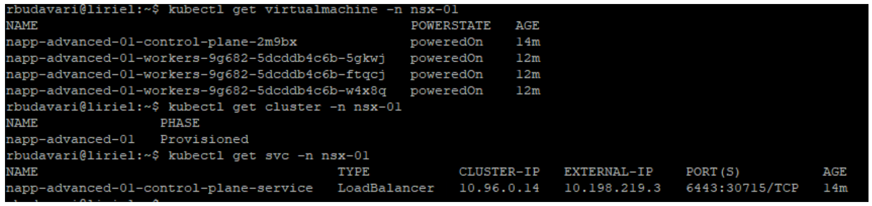


69. Under Virtual Machines you should see **1 control plane** and **3 worker nodes** using the **VM Class** defined in the `yaml` file:

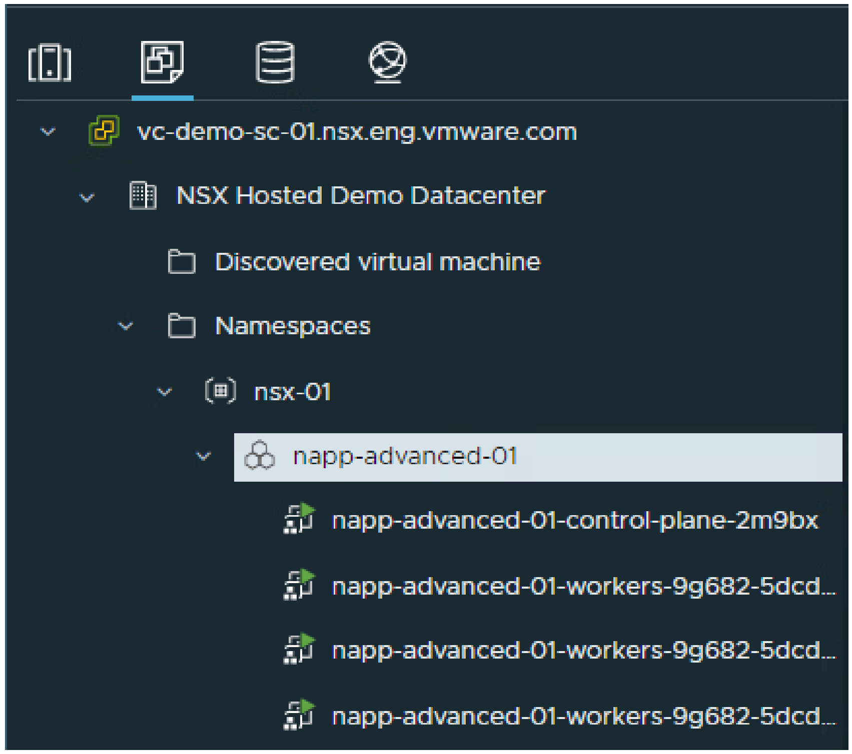


You can also use the following `kubectl` CLIs for verification:

```
kubectl get virtualmachine -n <namespace name>
kubectl get cluster -n <namespace name>
kubectl get svc -n <namespace name>
```



And under the **Virtual Machines** view, confirm all nodes are powered on:

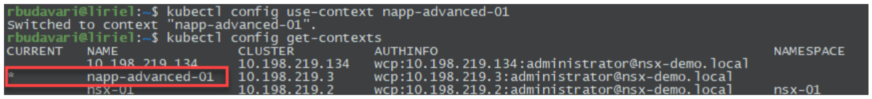


70. The final step of your TKGs setup is to retrieve the `kubeconfig` file that will be used by NSX Manager to connect to your TKC cluster for the NSX Application Platform installation.

    From the admin host you have installed the `kubectl` CLI utilities on, first verify you are in the correct context for your new guest cluster by running:

    ```
    kubectl config use-context <cluster name>

    kubectl config get-contexts
    ```



This will ensure the configuration file used for the NAPP deployment will install into the correct environment. If you do not see your guest cluster listed as a context, login again using the following syntax:

```
kubectl-vsphere login --server=https://<control-plane-ip> --vsphere-username <vcenter-admin> --insecure-skip-tls-verify --tanzu-kubernetes-cluster-namespace <namespace-name> --tanzu-kubernetes-cluster-name <cluster-name>
```

Once you have authenticated and confirmed the context you can retrieve the configuration file at:

```
~username/.kube/config
```

The example here is from a Linux host although other operating systems will use the same location:

```
$ cat .kube/config
```

```
rbudavari@liriel:~$ cat .kube/config
apiversion: v1
clusters:
- cluster:
    certificate-authority-data: [illegible]
    server: https://10.198.219.13:6443
  name: 10.198.219.13
- cluster:
    insecure-skip-tls-verify: true
    server: https://10.198.219.134:6443
  name: 10.198.219.134
- cluster:
    insecure-skip-tls-verify: true
    server: https://10.198.219.2:6443
  name: 10.198.219.2
- cluster:
    certificate-authority-data: [illegible]
    server: https://10.198.219.3:6443
  name: 10.198.219.3
- cluster:
    insecure-skip-tls-verify: true
    server: https://10.198.219.134:6443
  name: supervisor-10.198.219.134
contexts:
- context:
    cluster: 10.198.219.134
    user: wcp:10.198.219.134:administrator@nsx-demo.local
  name: 10.198.219.134
- context:
    cluster: 10.198.219.3
    user: wcp:10.198.219.3:administrator@nsx-demo.local
  name: napp-advanced-01
- context:
    cluster: 10.198.219.2
    namespace: nsx-01
    user: wcp:10.198.219.2:administrator@nsx-demo.local
  name: nsx-01
current-context: napp-advanced-01
kind: Config
preferences: {}
users:
- name: wcp:10.198.219.134:administrator@nsx-demo.local
  user:
    token: [illegible]
- name: wcp:10.198.219.13:administrator@nsx-demo.local
  user:
    token: [illegible]
- name: wcp:10.198.219.2:administrator@nsx-demo.local
  user:
    token: [illegible]
- name: wcp:10.198.219.3:administrator@nsx-demo.local
  user:
    token: [illegible]
rbudavari@liriel:~$
rbudavari@liriel:~$
```

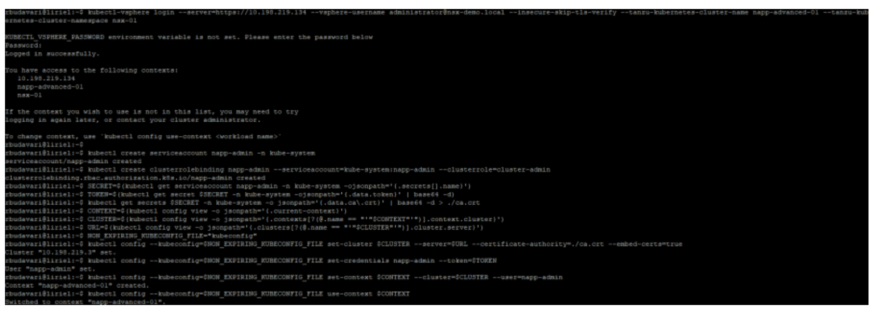
Note: By default, the `kubeconfig` token with TKGs expires after 10 hours. This is not technically an issue, as the `kubeconfig` details are only required for certain operations like upgrade or scale-out which also provide the opportunity to refresh your config file. But currently the NSX UI will display a warning that the `kubeconfig` credentials are invalid once it expires (this behavior will be changed in a future release to remove the warning).

71. To avoid any UI warnings once the token expires, you can create a non-expiring token and kubeconfig file for NAPP authentication.

    As mentioned, this is optional as the warning does not impact functionality and will be removed shortly.

For reference here are the steps used to generate a non-expiring file:

```
kubectl create serviceaccount napp-admin -n kube-system
kubectl create clusterrolebinding napp-admin --serviceaccount=kube-system:napp-admin --clusterrole=cluster-admin
SECRET=$(kubectl get serviceaccount napp-admin -n kube-system -ojsonpath='{.secrets[].name}')
TOKEN=$(kubectl get secret $SECRET -n kube-system -ojsonpath='{.data.token}' | base64 -d)
kubectl get secrets $SECRET -n kube-system -o jsonpath='{.data.ca\.crt}' | base64 -d > ./ca.crt
CONTEXT=$(kubectl config view -o jsonpath='{.current-context}')
CLUSTER=$(kubectl config view -o jsonpath='{.contexts[?(@.name == "'"$CONTEXT"'")].context.cluster}')
URL=$(kubectl config view -o jsonpath='{.clusters[?(@.name == "'"$CLUSTER"'")].cluster.server}')
NON_EXPIRING_KUBECONFIG_FILE="kubeconfig"
kubectl config --kubeconfig=$NON_EXPIRING_KUBECONFIG_FILE set-cluster $CLUSTER --server=$URL --certificate-authority=./ca.crt --embed-certs=true
kubectl config --kubeconfig=$NON_EXPIRING_KUBECONFIG_FILE set-credentials napp-admin --token=$TOKEN
kubectl config --kubeconfig=$NON_EXPIRING_KUBECONFIG_FILE set-context $CONTEXT --cluster=$CLUSTER --user=napp-admin
kubectl config --kubeconfig=$NON_EXPIRING_KUBECONFIG_FILE use-context $CONTEXT
```



And an example of the created `kubeconfig` file:

72. If the host you will be performing the NSX Application Platform deployment from (by connecting to NSX Manager via a browser) is different than the admin host, please copy the `kubeconfig` file to this host.

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NSX Application Platform Deployment – vSphere with Tanzu and NSX-T/Avi Networking

Note: If your management environment is already licensed and prepared for NSX-T and you also have ALB (Avi), then this is a great option to run NAPP. It provides the visibility, flexibility and security of NSX-based networking and consistency with other workloads in your environment. However, in most cases management environments will not have these components and the simplest TKGs solution of VDS networking with HAProxy is recommended.

NSX-T Configuration Steps

This section of the deployment guide outlines the NSX-T configuration required to support an NSX Application Platform deployment. It will not cover the process of installing NSX-T, preparing the environment or deploying gateways. Please refer to the product documentation for details on these steps if required.

In this guide the logical segments created for NAPP are attached to a Tier-0 gateway. These can also be attached to a Tier-1 gateway, although you must ensure you have enabled route advertisement from the Tier-1 so the networks are advertised to the Tier-0 gateway and the physical network.

Gateway Configuration

This section assumes the following:

- An Edge Cluster with Edge nodes has been deployed
- A Tier-0 Gateway has been configured with the following:
    - Uplink interfaces
    - Dynamic or Static routing configuration for external connectivity
    - Route re-distribution has been configured

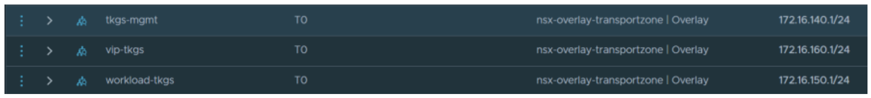
Segment Configuration

Three overlay segments are required for this solution to be deployed:

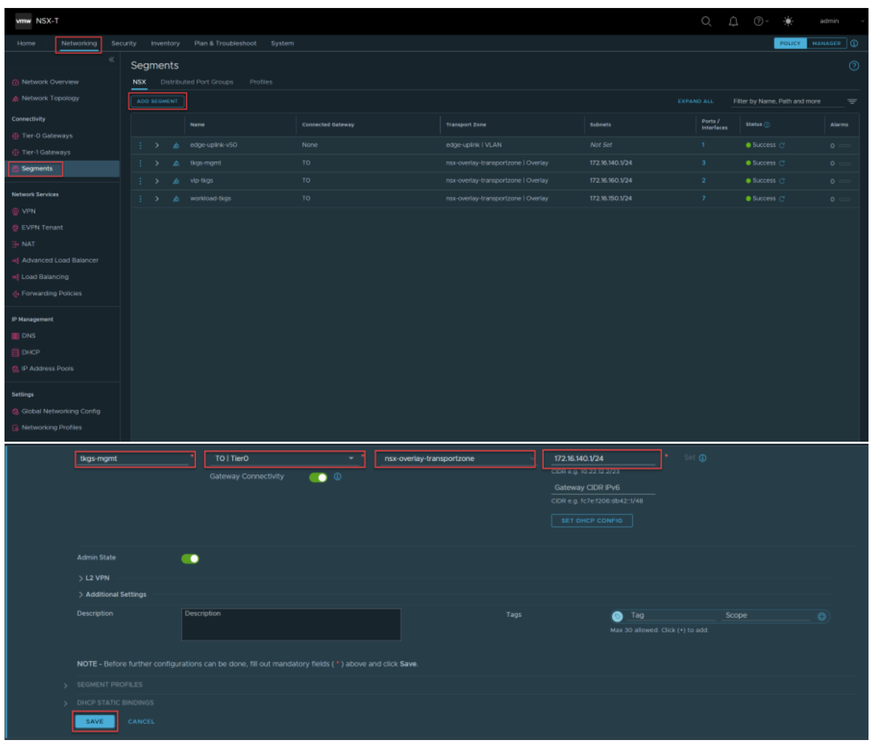
1. **Management**: This network will be attached to the Supervisor Control Plane Virtual Machines
2. **Workload**: This network will be utilized by the Supervisor Control Plane, and workload cluster virtual machines.

3. **VIP**: The VIPs will be configured in NSX ALB and used by the service engines for provide load balancer services to the NAPP environment.

The networks used in this guide are shown in the image below:



4. Login to the NSX-T Manager UI
5. Navigate to `Networking -> Segments -> Add Segment` and populate the details as required in your environment. Ensure the segments are Overlay segments and attached to the correct transport zone, are linked to the appropriate NSX Gateway:



6. Repeat for the additional networks required.

Deploying NSX Advanced Load Balancer Configuration

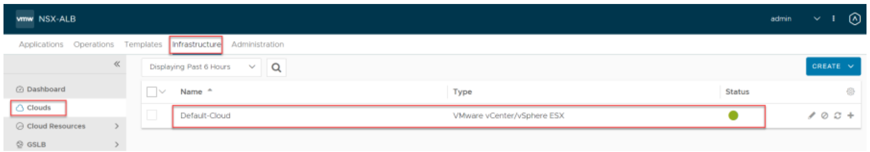
VMware also provides NSX Advanced Load Balancer which is a feature-rich, software-based Load Balancing solution that is supported for use with Tanzu. NSX ALB has several Editions, a full featured Enterprise Edition, a Basic Edition which is equivalent to the native NSX-T LB and also an Essentials Edition that is available for use with Tanzu for L4 Load Balancing services. For use with TKGs any of these Editions is supported.

https://avinetworks.com/docs/21.1/nsx-essentials-for-tanzu/
https://avinetworks.com/docs/21.1/avi-nsx-t-integration/

Similar to NSX-T, we will only document the NSX ALB configuration specific to NAPP. Refer to the Avi documentation for more information on setting up NSX ALB

This guide will demonstrate the process of using a self-signed certificate for NSX ALB.

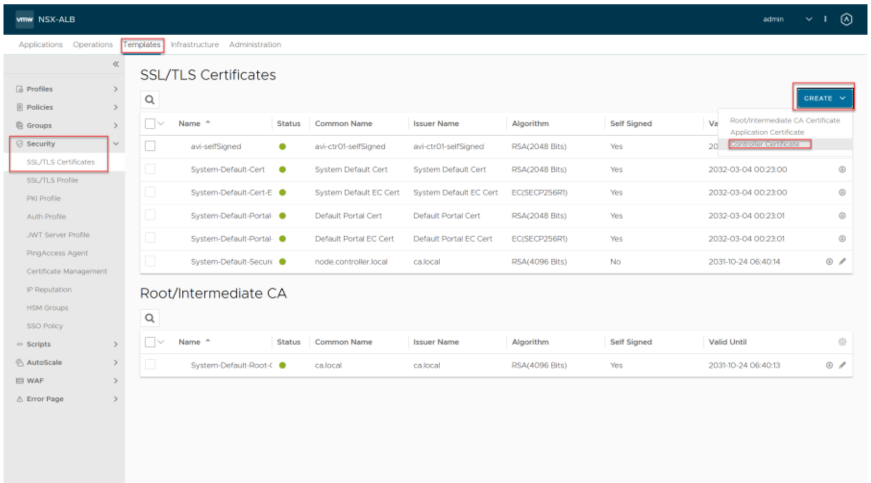
7. Login to the NSX ALB UI
8. Navigate to `Infrastructure -> Clouds`, and ensure your **Default-Cloud** is configured with VMware vCenter/vSphere ESX.



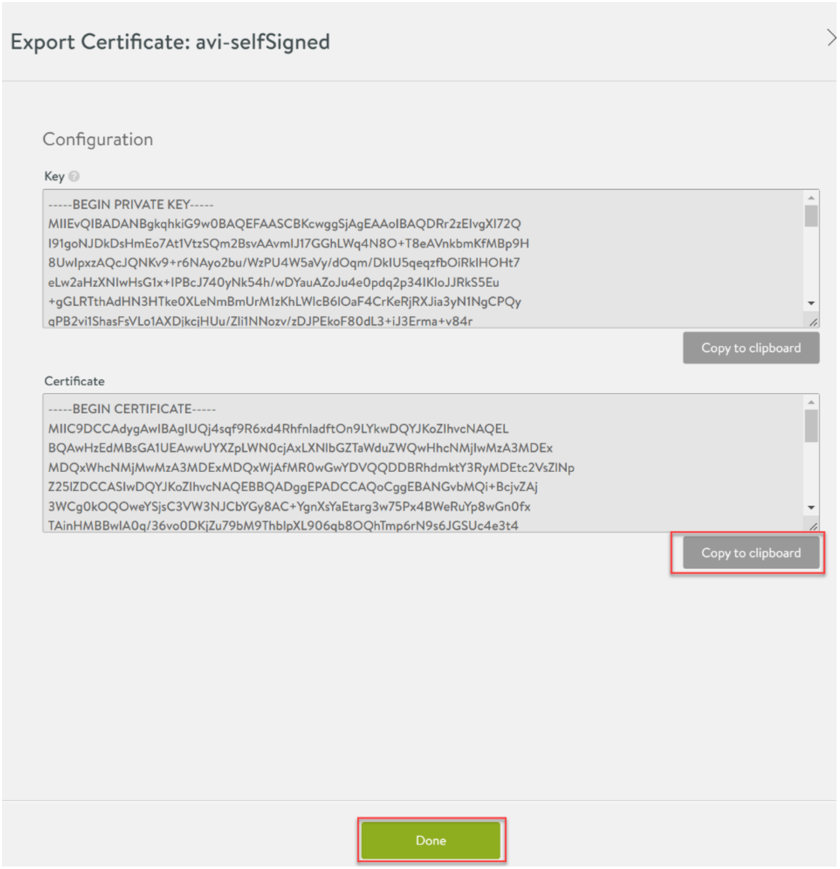
Self-Signed Certificate Creation

9. Next you will need to create the certificate. Navigate to `Templates -> Security -> SSL/TLS Certificates -> Create -> Controller Certificate.`

Populate the details relevant to your environment, ensure the Subject Alternate Name (SAN) includes and entry with the FQDN of the NSX ALB controller cluster, and a separate one for the VIP. Click save once complete.
Once it is created you should see output similar to the below

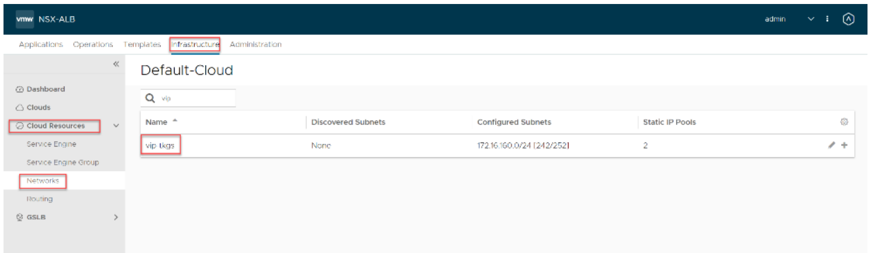


Click on the arrow pointing down, on the next screen, copy the certificate to clipboard, and paste it into notepad or any other text editor, click done.

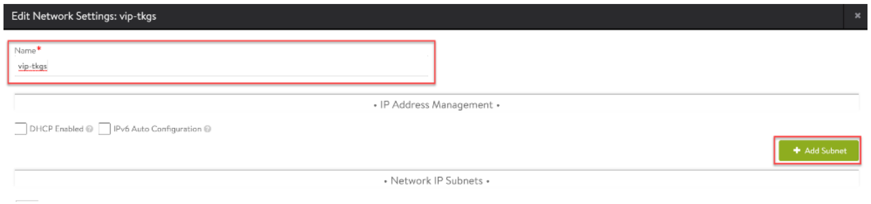


IPAM Profile

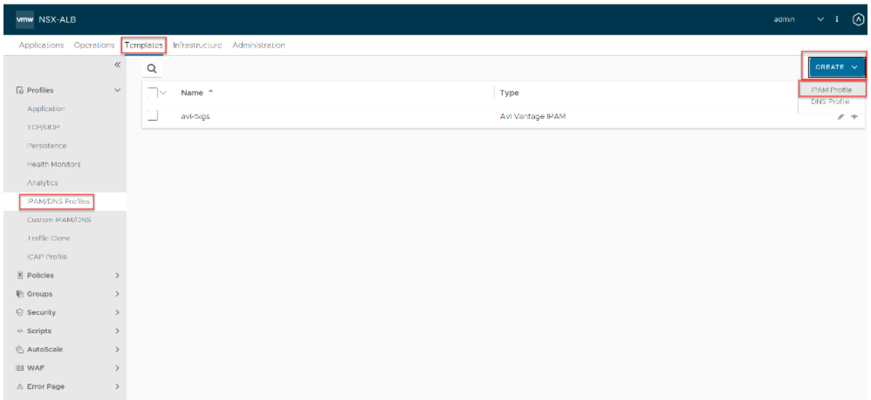
10. Navigate to `Infrastructure -> Cloud Resources -> Networks` and find the network you created in NSX-T for VIPs. Click on the pencil icon to edit it.



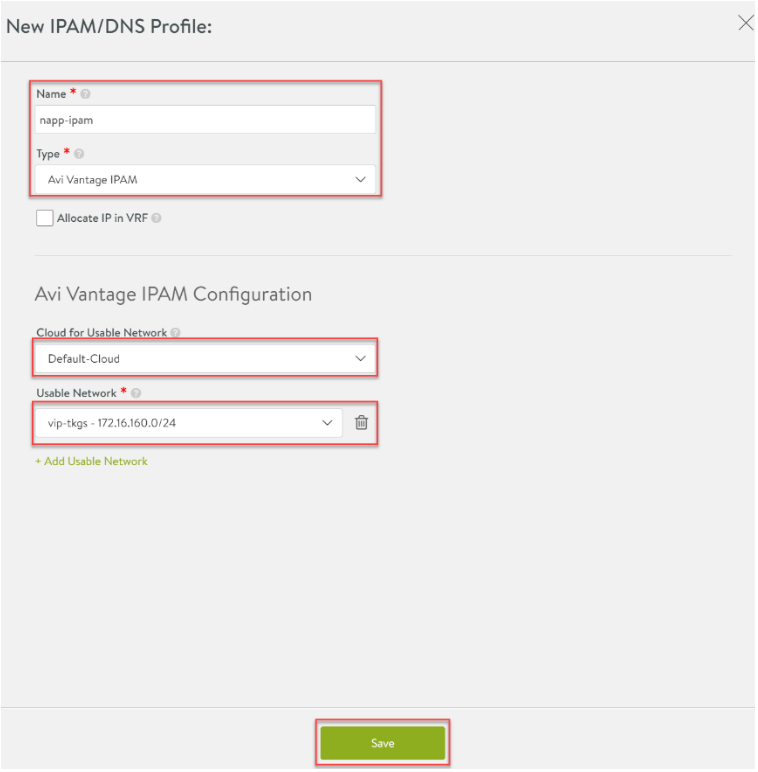
11. Click on Add subnet and define the subnet details for the network, and add an IP address pool, the IPAM profile will use this pool to assign VIP addresses to the various services for NAPP.



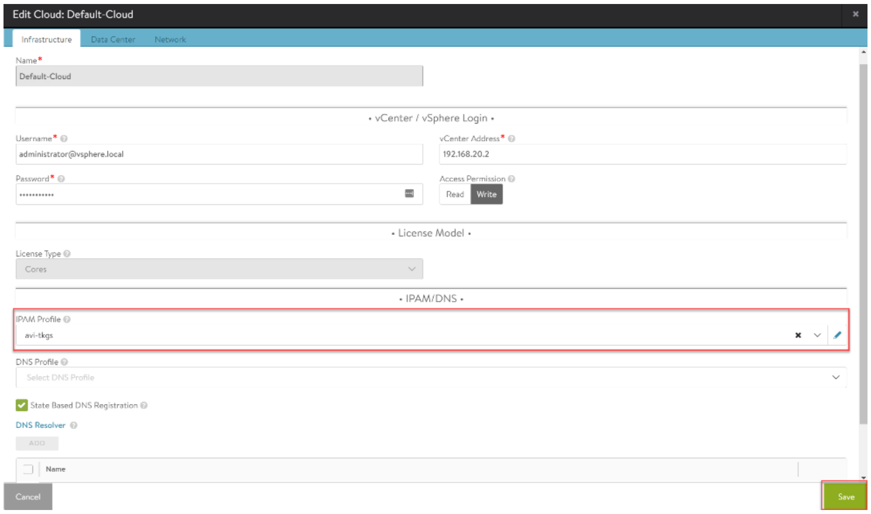
12. Navigate to `Templates -> IPAM/DNS Profiles -> Create`



13. Enter a name, ensure the type is Avi Vantage IPAM, Default-Cloud is selected, and the usable network is the VIP network that was created earlier.



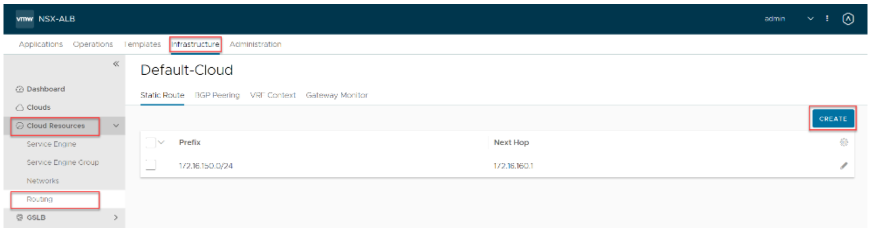
14. Now you will need to add the IPAM profile to the cloud profile, navigate to `Infrastructure -> Clouds -> Edit Default-Cloud` and add the IPAM profile created, then save



Static Route Configuration

A static route is required (unless you are using dynamic routing) to allow connectivity from the VIP network to the workload network. In this case, it will create a routing table entry on the service engines to tell them how to reach the workload-tkgs (172.16.150.0/24) network.

16. To create the static route, navigate to `Infrastructure -> Cloud Resources -> Routing` and click Create



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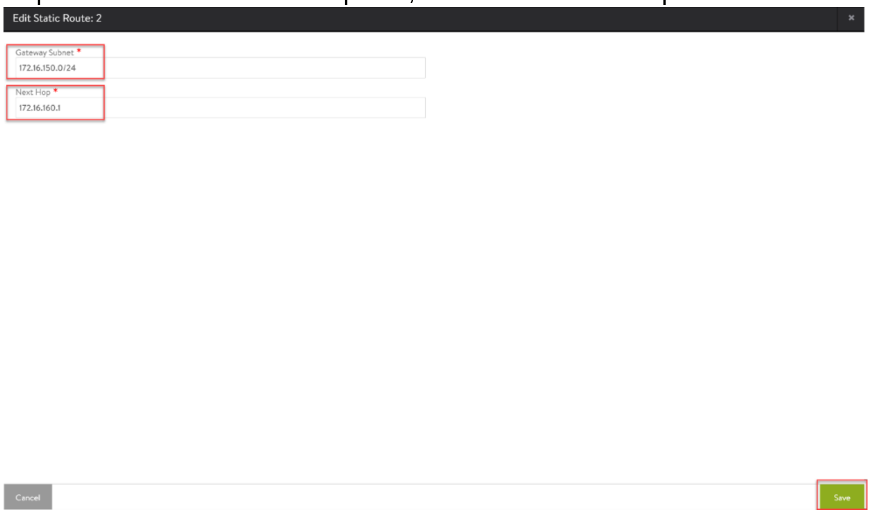
17. Populate the networks as required, click save once complete.



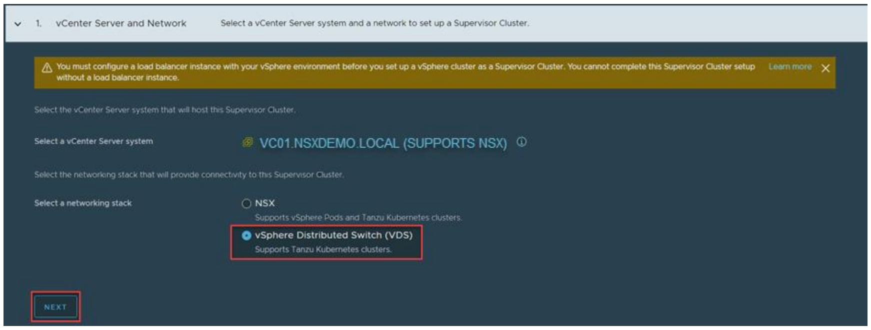
You are now ready to proceed with the TKGs configuration steps in the section below.

vSphere with Tanzu using NSX-T and ALB

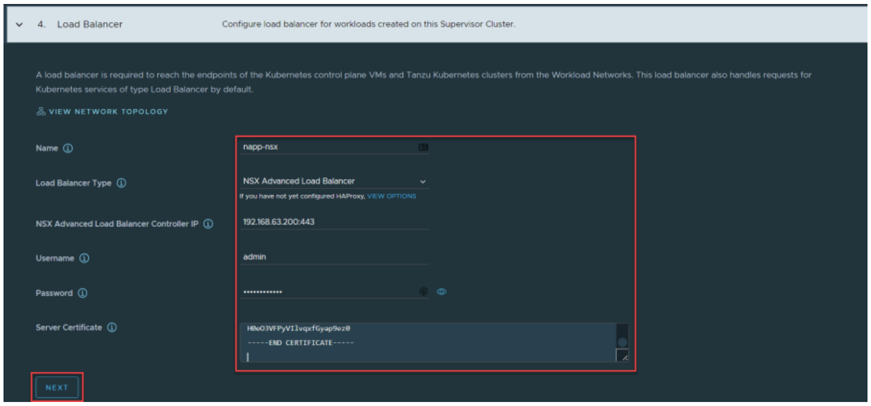
Regarding the deployment, setting up the TKGs environment is very similar to the HAProxy-based approach. However instead of creating dvPortgroups in vCenter you will use the overlays created earlier to support your NAPP environment.

Therefore, I am only highlighting the differences. Refer to the previous section for step-by-step details on deploying your TKGs Supervisor and Guest Clusters.

While you are enabling Workload Management and at the **Select a networking stack** screen (Step 1), even though you are using NSX-T Overlay segments it is important to select **vSphere Distributed Switch (VDS)**. This is not necessarily intuitive, but to use NSX Advanced Load Balancer instead of the native NSX Load balancer, you must select vSphere Distributed Switch. Click next when you have selected NSX.



Step 4 is where you configure the Load Balancer. As mentioned earlier, this guide will configure NSX Advanced Load Balancer. All the details shown in the image must be populated. The name needs to be a DNS-compliant format but does not require a DNS entry. Provide the NSX Advanced Load Balancer cluster IP if you have clustered control nodes. Paste the certificate contents from your text editor into the Server Certificate field. Click next when complete.



Step 5 is where the management network for the Supervisor Cluster nodes is defined. There are two options for address allocation. They are:

a. DHCP
b. Static

This guide will use static IP addresses for the Supervisor Cluster nodes.

Note: the network “**infrastructure-tkg**” is an Overlay segment created in NSX-T and not a dvPg (portgroup on a vSphere Distributed Switch)

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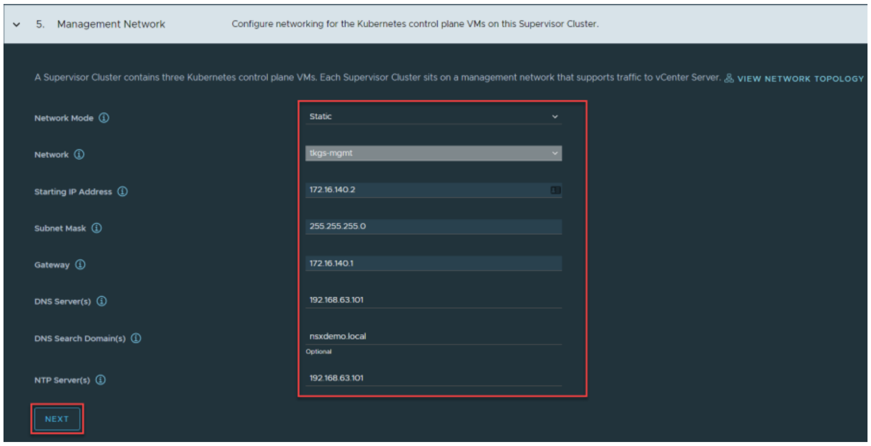
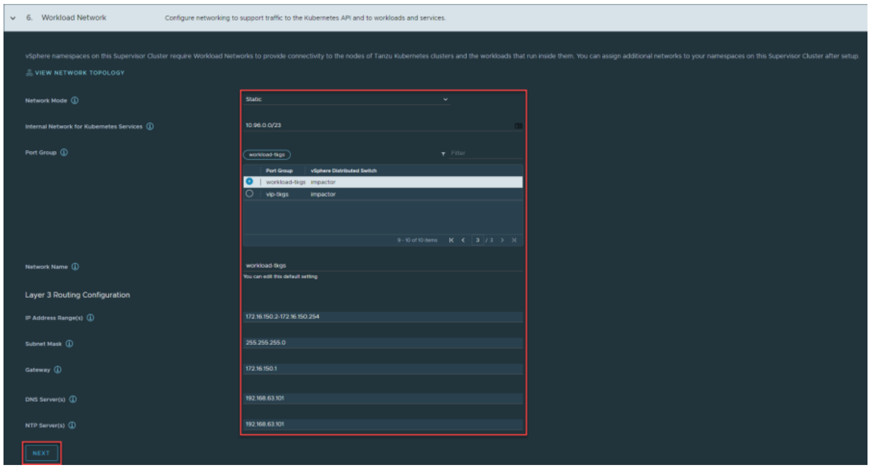
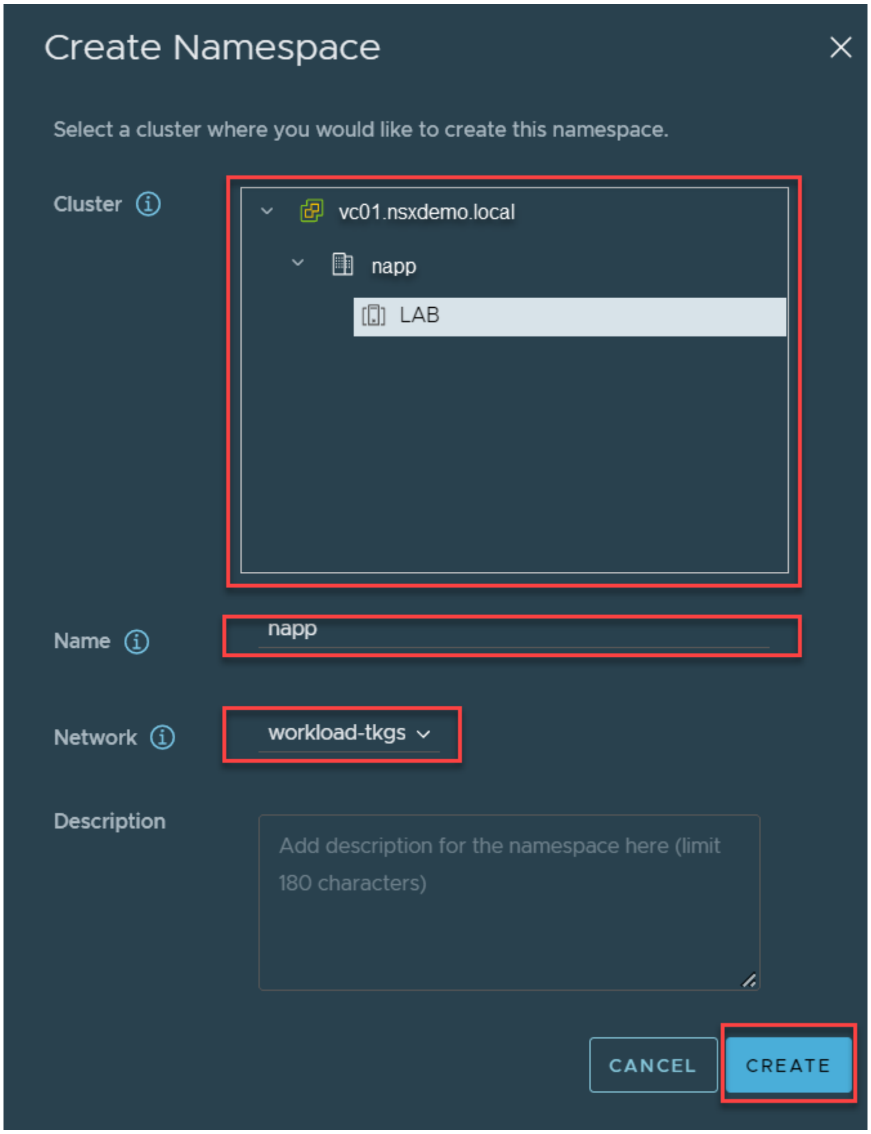


Next, the Workload Network must be defined. Similar to the management network, “**workload-tkg**” is an NSX Overlay segment. Ensure the address ranges used in your configuration exist and do not overlap with other subnets in the environment



Apart from the Create Namespace step (where you will select an NSX-T overlay as the workload network), all other steps are the same as the VDS with HAProxy, so refer to that section to setup TKGs and generate your kubeconfig file to proceed with deployment.

Here is a screenshot of the workload network selection for reference:



When you have TKGs deployed, proceed to the NAPP section of the document.

NSX Application Platform Deployment –

Upstream K8s or Tanzu CE

Note: Upstream Kubernetes or Tanzu Community Edition (TCE) are options for NSX Application Platform. However, these solutions do not come with official VMware support. Therefore, they should be used only in cases where it is not possible to deploy vSphere 7 or if you already have kubernetes expertise as support of the underlying microservices infrastructure would not be provided by VMware. As vSphere 6.x releases are starting to approach End of General Support in 2022, this will increasingly be an exception. As TCE is being retired we are **strongly** recommending the use of TKGs or Upstream/Vanilla K8s for NAPP deployments.

Tanzu Community Edition does benefit from using consistent VM images and vSphere integration so the steps are provided as a reference. TCE supports installation to multiple target platforms and for our NAPP deployment we will use **vSphere**.

First ensure you have met the following requirements:

Table 7. Tanzu Community Edition on vSphere Requirements

	Requirements	Comments
☐	A bootstrap system (Linux, MacOS or Windows operating systems are all supported), that meets the following prerequisites: https://tanzucommunityedition.io/docs/latest/cli-installation/ Which include the latest stable version of Docker and Kubernetes Tools	
☐	DHCP server on the management network to provide connectivity to all cluster node VMs that Tanzu Community Edition deploys. Sufficient IP addresses (12) in the DHCP range are required for all TCE node VMs. Also, this network must be able to connect to the vCenter Server and ESXi Management VMkernel interfaces in the vSphere environment.	
☐	A set of available static virtual IP (VIP) addresses for the clusters that you create, one for each management and workload cluster.	

☐	Traffic permitted between port 443 of all TCE cluster VMs and vCenter Server. Port 443 is where the vCenter Server API is exposed.	
☐	Traffic permitted between your local bootstrap machine and port 6443 of all TCE cluster VMs in the clusters you create. Port 6443 is where the Kubernetes API is exposed.	
☐	The Network Time Protocol (NTP) service running on all hosts	

This guide will use a Debian Linux system for the bootstrap host, although the steps are very similar across operating systems.

For the Docker requirement, you should always install an up-to-date release rather than relying on any distribution packages. Refer to the following website for download links and installation steps for Docker on the host you will be using to bootstrap your TCE deployment:

https://docs.docker.com/engine/install/

For Linux ensure you also follow the steps at:

https://docs.docker.com/engine/install/linux-postinstall/#manage-docker-as-a-non-root-user

```
sudo usermod -aG docker $USER
```

```
rbudavari@tce-mgt-sc-01:~/Downloads/tce-linux-amd64-v0.10.0$ sudo usermod -aG docker $USER
```

```
newgrp docker
```

```
groups
```

```
rbudavari@tce-mgt-sc-01:~/Downloads/tce-linux-amd64-v0.10.0$ newgrp docker
rbudavari@tce-mgt-sc-01:~/Downloads/tce-linux-amd64-v0.10.0$ groups
docker cdrom floppy sudo audio dip video plugdev netdev bluetooth scanner rbudavari
```

Also, you must install the kubernetes tools for your OS. Similarly, a version of the tools that is compatible with your cluster is required. In this guide we are installing a 1.21 K8s cluster and you must use a kubernetes tools version that is within one minor version difference of your cluster (so 1.20, 1.21 or 1.22 could be used):

https://kubernetes.io/docs/tasks/tools/

Before continuing with the TCE Installation, verify your software versions:

```
docker --version

kubectl version --client --short
```

```
rbudavari@tce-mgt-sc-01:~/.config/tanzu/tkg/clusterconfigs$ docker --version
Docker version 20.10.12, build e91ed57
rbudavari@tce-mgt-sc-01:~/.config/tanzu/tkg/clusterconfigs$ kubectl version --client --short
Client Version: v1.21.5
```

Another important step in the current versions of TCE is to ensure that you are using cgroups v1, otherwise the Docker runtime will not work. You can check this using the following command:

```
docker info | grep -I cgroup
```

```
rbudavari@tce-mgt-sc-01:~/Downloads/tce-linux-amd64-v0.10.0$ docker info | grep -i cgroup
 Cgroup Driver: systemd
 Cgroup Version: 2
  cgroupns
```

If it returns `Cgroup Version: 2` you will need to revert to v1. This can be done on Debian Linux by editing `/etc/default/grub` and adding the following line:

```
GRUB_CMDLINE_LINUX=" systemd.unified_cgroup_hierarchy=0"
```

```
GRUB_DEFAULT=0
GRUB_TIMEOUT=5
GRUB_DISTRIBUTOR=`lsb_release -i -s 2> /dev/null || echo Debian`
GRUB_CMDLINE_LINUX_DEFAULT="quiet"
GRUB_CMDLINE_LINUX="systemd.unified_cgroup_hierarchy=0"
```

Then run the following command to refresh your boot loader:

```
update-grub
```

```
rbudavari@tce-mgt-sc-01:~/Downloads/tce-linux-amd64-v0.10.0$ sudo update-grub
Generating grub configuration file ...
Found background image: /usr/share/images/desktop-base/desktop-grub.png
Found linux image: /boot/vmlinuz-5.10.0-11-amd64
Found initrd image: /boot/initrd.img-5.10.0-11-amd64
Found linux image: /boot/vmlinuz-5.10.0-9-amd64
Found initrd image: /boot/initrd.img-5.10.0-9-amd64
done
```

Then reboot the host and ensure you are now running with `Cgroup Version: v1`

```
rbudavari@tce-mgt-sc-01:~/Downloads/tce-linux-amd64-v0.10.0$ docker info | grep -i cgroup
 Cgroup Driver: cgroupfs
 Cgroup Version: 1
```

Note: A fix to enable cgroup v2 is targeted for TCE v0.11

Installing Tanzu Community Edition

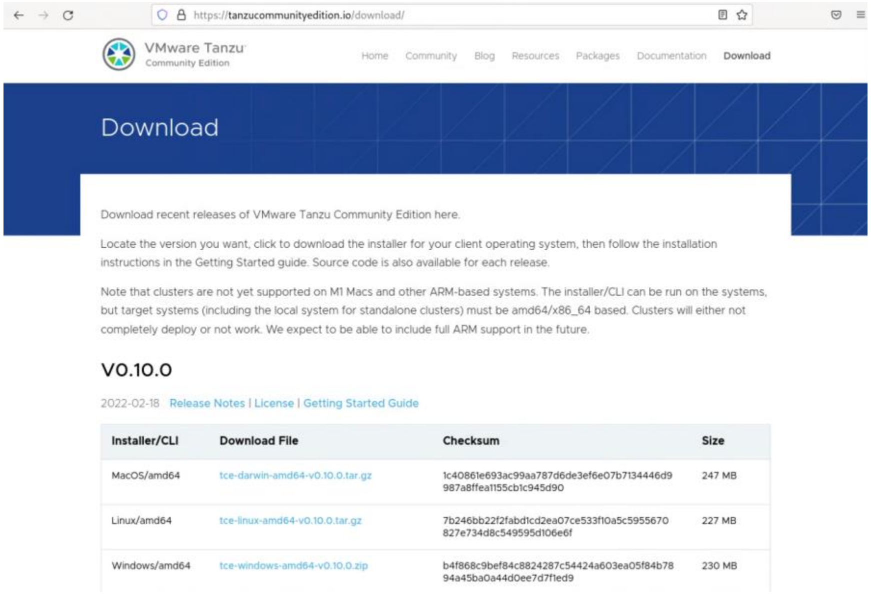
If you are installing on a Linux or Mac system where homebrew is available, that is the simplest method to install Tanzu CE:

```
brew install vmware-tanzu/tanzu/tanzu-community-edition
```

```
{HOMEBREW-INSTALL-LOCATION}/configure-tce.sh
```

As my system is Debian-based and I don't want to add another package manager, I am using the tarball-based CLI Installer. For a Windows-based installation, you will download the zip file from the same site:

https://tanzucommunityedition.io/docs/latest/cli-installation/



Once downloaded, extract the TCE tarball:

```
tar xvfz tce-linux-amd64-v0.10.0.tar.gz
```

```
rbudavari@tce-mgt-sc-01:~/Downloads$ tar xvfz tce-linux-amd64-v0.10.0.tar.gz
tce-linux-amd64-v0.10.0/
tce-linux-amd64-v0.10.0/bin/
tce-linux-amd64-v0.10.0/bin/tanzu
tce-linux-amd64-v0.10.0/bin/tanzu-plugin-cluster
tce-linux-amd64-v0.10.0/bin/tanzu-plugin-kubernetes-release
tce-linux-amd64-v0.10.0/bin/tanzu-plugin-login
tce-linux-amd64-v0.10.0/bin/tanzu-plugin-secret
tce-linux-amd64-v0.10.0/bin/tanzu-plugin-package
tce-linux-amd64-v0.10.0/bin/tanzu-plugin-pinniped-auth
tce-linux-amd64-v0.10.0/bin/tanzu-plugin-management-cluster
tce-linux-amd64-v0.10.0/bin/tanzu-plugin-builder
tce-linux-amd64-v0.10.0/bin/tanzu-plugin-codegen
tce-linux-amd64-v0.10.0/bin/tanzu-plugin-standalone-cluster
tce-linux-amd64-v0.10.0/bin/tanzu-plugin-conformance
tce-linux-amd64-v0.10.0/bin/tanzu-plugin-diagnostics
tce-linux-amd64-v0.10.0/bin/tanzu-plugin-unmanaged-cluster
tce-linux-amd64-v0.10.0/install.sh
tce-linux-amd64-v0.10.0/uninstall.sh
```

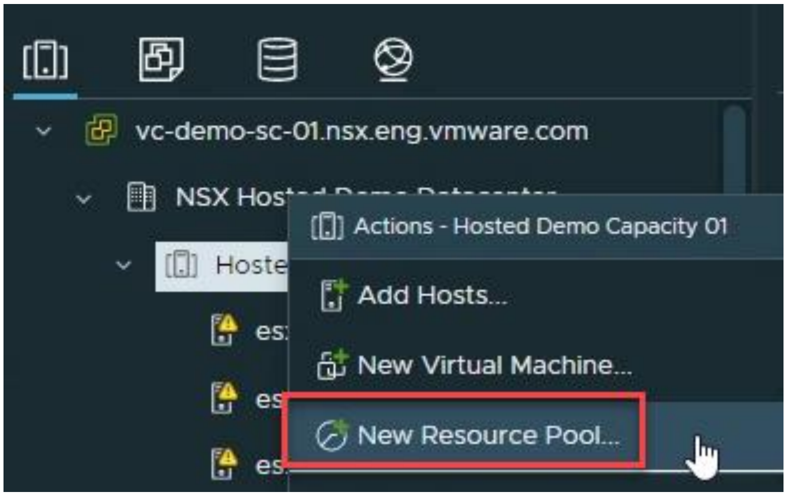
The run the installer (as a regular user and not as root):

`./install.sh`

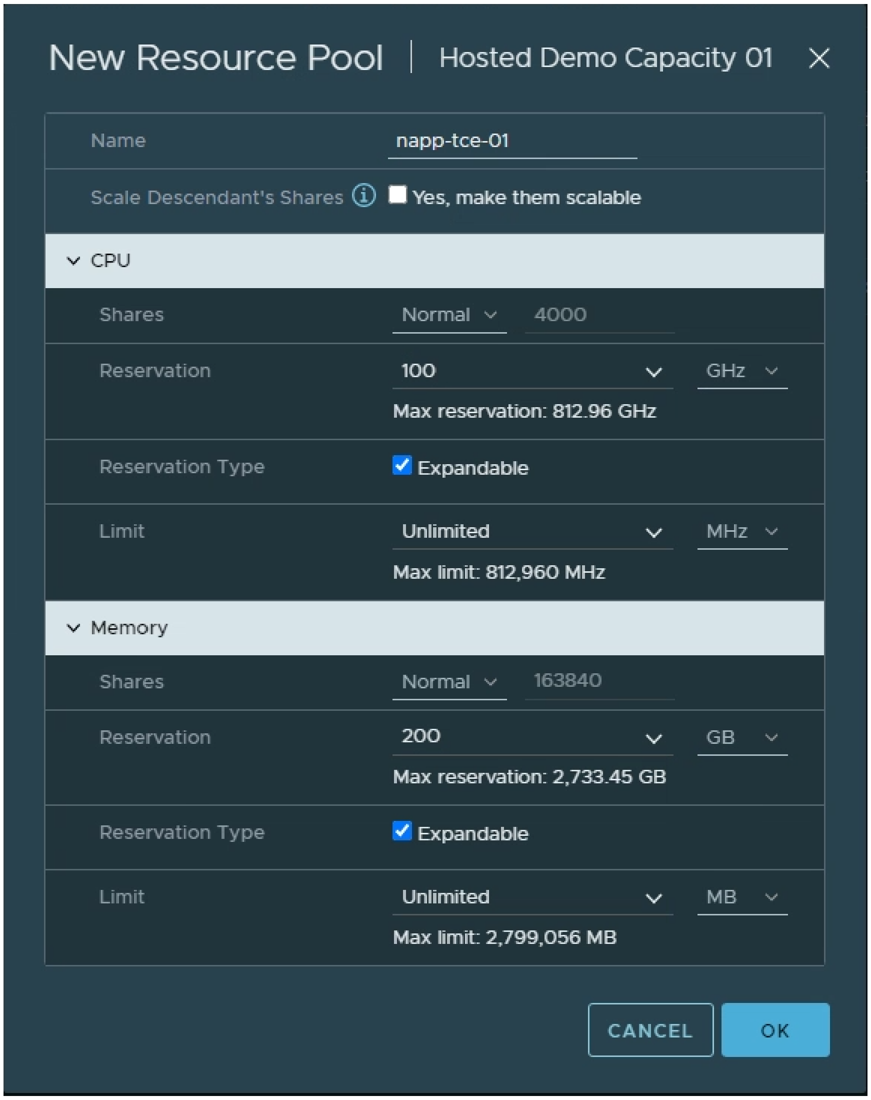
```
rbudavari@tce-mgt-sc-01:~/Downloads/tce-linux-amd64-v0.10.0$ ./install.sh
+ ALLOW_INSTALL_AS_ROOT=
+ [[ 1000 -eq 0 ]]
+++ dirname ./install.sh
++ cd .
++ pwd
+ MY_DIR=/home/rbudavari/Downloads/tce-linux-amd64-v0.10.0
++ uname
+ BUILD_OS=Linux
+ case "${BUILD_OS}" in
+ XDG_DATA_HOME=/home/rbudavari/.local/share
+ echo /home/rbudavari/.local/share
/home/rbudavari/.local/share
++ command -v tanzu
+ TANZU_BIN_PATH=
+ [[ -n '' ]]
+ TANZU_BIN_PATH=/usr/local/bin
+ [[ :/usr/local/bin:/usr/bin:/bin:/usr/local/games:/usr/games: == *\:\/\h\o\m\e\/\r\b\u\d\a\v\a\r\i\/\b\i\n\:* ]]
+ echo Installing tanzu cli to /usr/local/bin
Installing tanzu cli to /usr/local/bin
+ sudo install /home/rbudavari/Downloads/tce-linux-amd64-v0.10.0/bin/tanzu /usr/local/bin
+ mkdir -p /home/rbudavari/.local/share/tanzu-cli
+ for plugin in "${MY_DIR}"/bin/tanzu-plugin*
+ install /home/rbudavari/Downloads/tce-linux-amd64-v0.10.0/bin/tanzu-plugin-builder /home/rbudavari/.local/share/tanzu-cli
+ for plugin in "${MY_DIR}"/bin/tanzu-plugin*
+ install /home/rbudavari/Downloads/tce-linux-amd64-v0.10.0/bin/tanzu-plugin-cluster /home/rbudavari/.local/share/tanzu-cli
+ for plugin in "${MY_DIR}"/bin/tanzu-plugin*
+ install /home/rbudavari/Downloads/tce-linux-amd64-v0.10.0/bin/tanzu-plugin-codegen /home/rbudavari/.local/share/tanzu-cli
+ for plugin in "${MY_DIR}"/bin/tanzu-plugin*
+ install /home/rbudavari/Downloads/tce-linux-amd64-v0.10.0/bin/tanzu-plugin-conformance /home/rbudavari/.local/share/tanzu-cli
+ for plugin in "${MY_DIR}"/bin/tanzu-plugin*
+ install /home/rbudavari/Downloads/tce-linux-amd64-v0.10.0/bin/tanzu-plugin-diagnostics /home/rbudavari/.local/share/tanzu-cli
+ for plugin in "${MY_DIR}"/bin/tanzu-plugin*
+ install /home/rbudavari/Downloads/tce-linux-amd64-v0.10.0/bin/tanzu-plugin-kubernetes-release /home/rbudavari/.local/share/tanzu-cli
+ for plugin in "${MY_DIR}"/bin/tanzu-plugin*
+ install /home/rbudavari/Downloads/tce-linux-amd64-v0.10.0/bin/tanzu-plugin-login /home/rbudavari/.local/share/tanzu-cli
+ for plugin in "${MY_DIR}"/bin/tanzu-plugin*
+ install /home/rbudavari/Downloads/tce-linux-amd64-v0.10.0/bin/tanzu-plugin-management-cluster /home/rbudavari/.local/share/tanzu-cli
+ for plugin in "${MY_DIR}"/bin/tanzu-plugin*
+ mkdir -p /home/rbudavari/.local/share/tce
+ install /home/rbudavari/Downloads/tce-linux-amd64-v0.10.0/uninstall.sh /home/rbudavari/.local/share/tce
+ TANZU_PLUGIN_CACHE=/home/rbudavari/.cache/tanzu/catalog.yaml
+ [[ -n /home/rbudavari/.cache/tanzu/catalog.yaml ]]
+ echo 'Removing old plugin cache from /home/rbudavari/.cache/tanzu/catalog.yaml'
Removing old plugin cache from /home/rbudavari/.cache/tanzu/catalog.yaml
+ rm -f /home/rbudavari/.cache/tanzu/catalog.yaml
+ tanzu init
| initializing ✔  successfully initialized CLI
++ tanzu plugin repo list
++ grep tce
+ TCE_REPO=
+ [[ -z '' ]]
+ tanzu plugin repo add --name tce --gcp-bucket-name tce-tanzu-cli-plugins --gcp-root-path artifacts
++ grep core-admin
++ tanzu plugin repo list
+ TCE_REPO=
+ [[ -z '' ]]
+ tanzu plugin repo add --name core-admin --gcp-bucket-name tce-tanzu-cli-framework-admin --gcp-root-path artifacts-admin
+ echo 'Installation complete!'
Installation complete!
```

Now you will start preparing your vSphere environment before deploying a Management Cluster. Start by navigating to the `Hosts and`

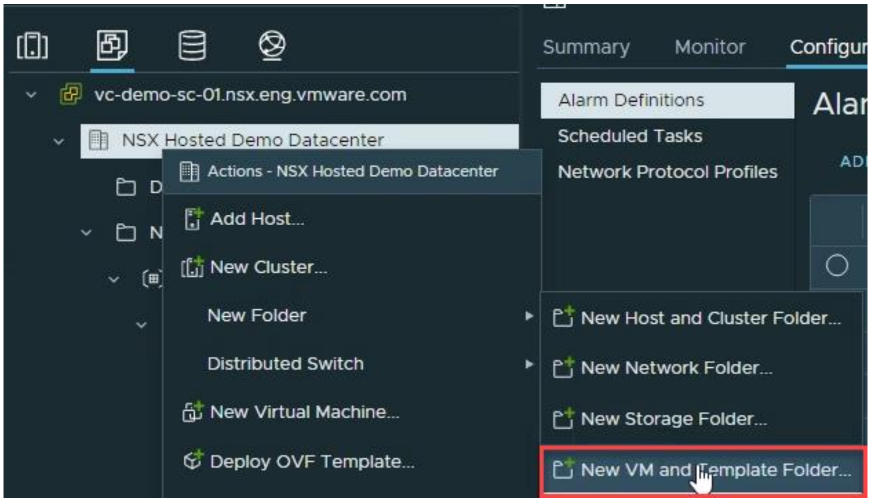
`Clusters` view in the vSphere UI, right-click on your cluster and create a **New Resource Pool**:



If you have sufficient resources a CPU and Memory reservation is recommended, but isn't a requirement:



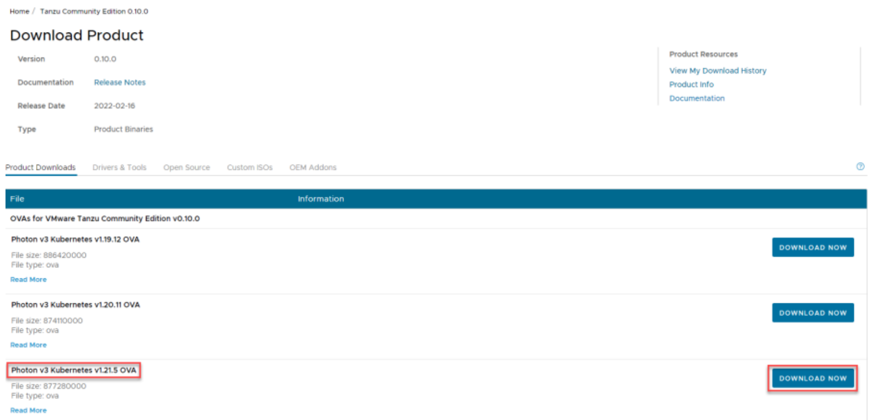
Then from the `VMs and Templates` view, create a new **VM and Template Folder**



It is recommended to use a consistent naming scheme for TCE objects:

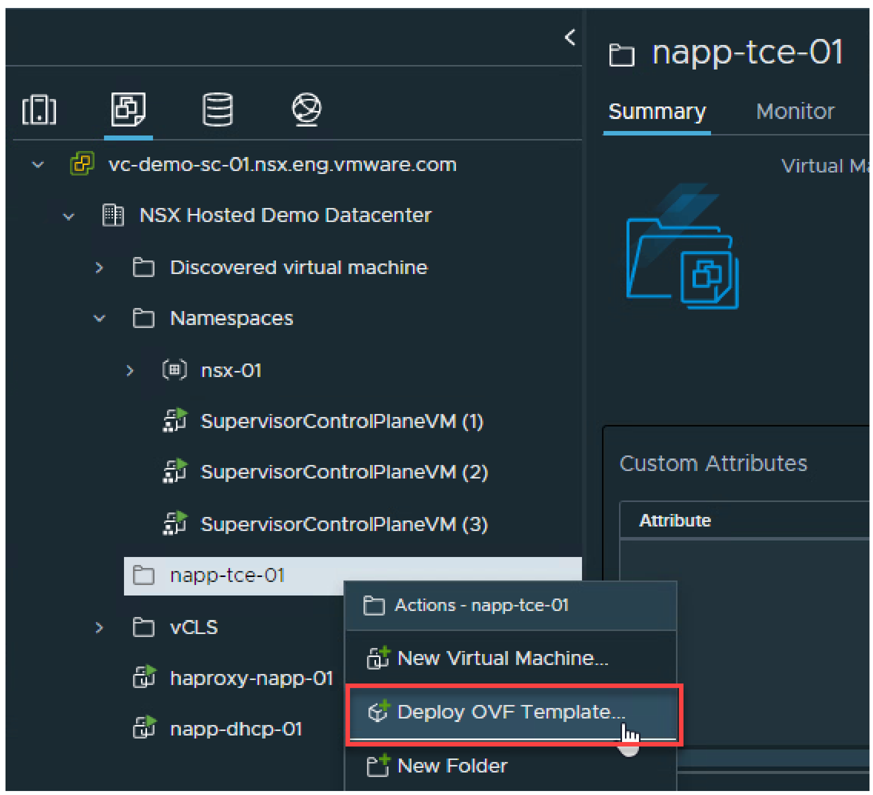
In the next step you need to download the required Photon-based Kubernetes OVA from the Tanzu Community Edition 0.10.0 page (you will need a myvmware.com account):

https://customerconnect.vmware.com/downloads/get-download?downloadGroup=TCE-0100

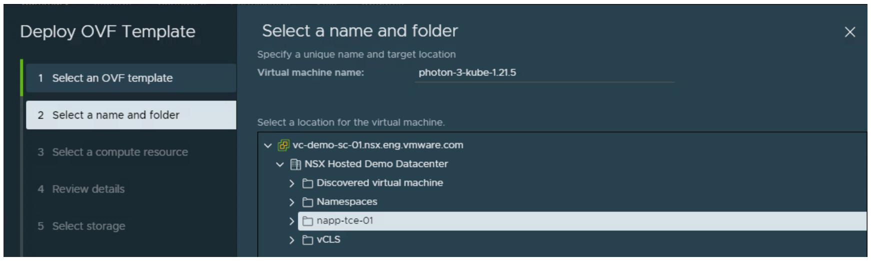


Photon v3 Kubernetes v1.12.5 is the latest at the time of this guide and is the recommended version.

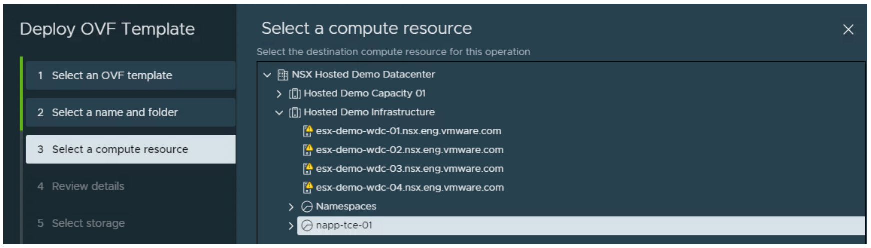
Now select the Folder we created earlier for TCE and right-click **Deploy OVF Template** and choose the Photon OVA you just downloaded:



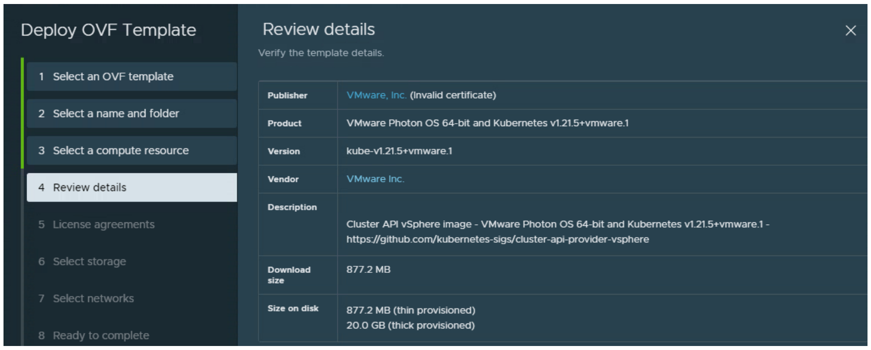
It is recommended to use a shorter/more user-friendly **name** for the Virtual machine:



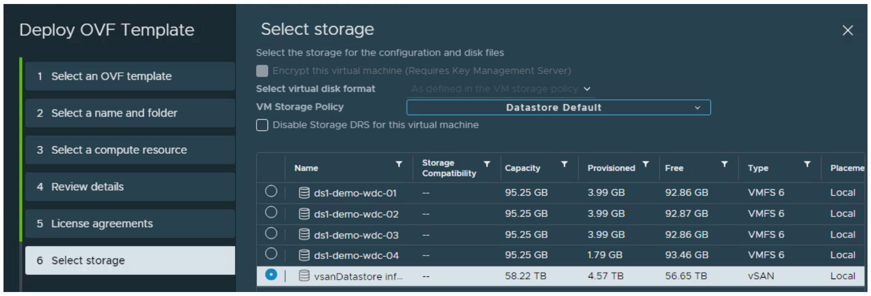
Then for the **compute resource**, select the Resource pool you just created for TCE:



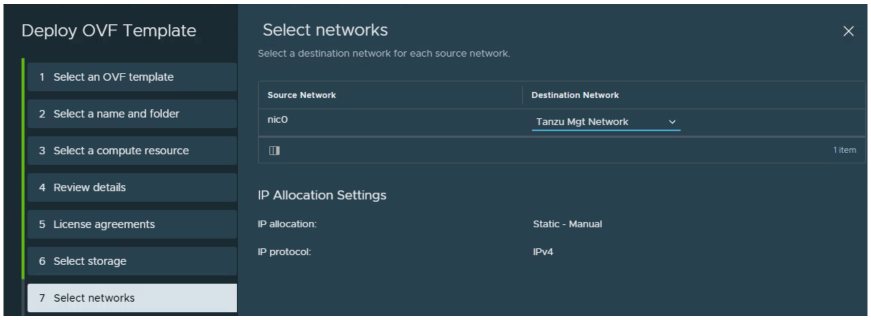
On the **Review** screen make sure your version is the same as below:



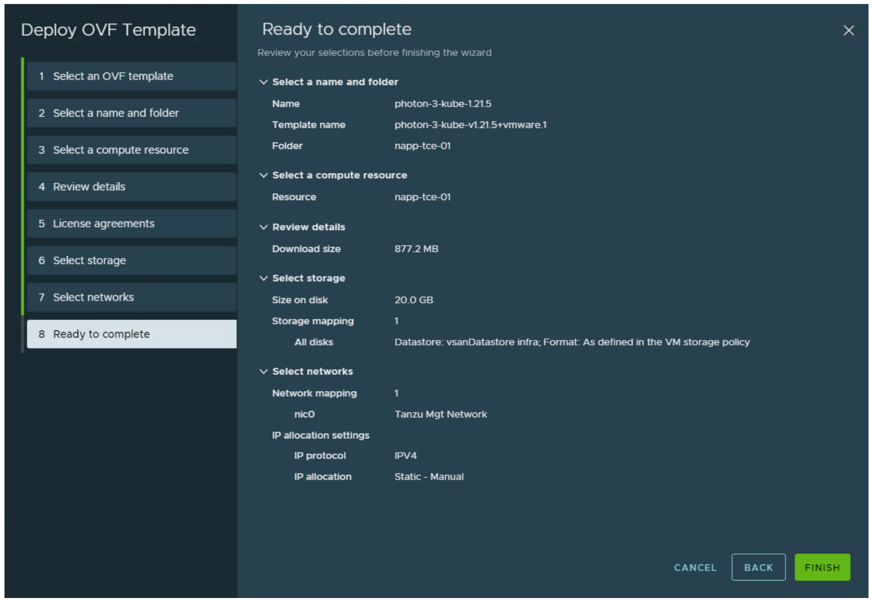
Accept the **License agreement**, then **Select storage** and choose a shared datastore (VSAN in my case):



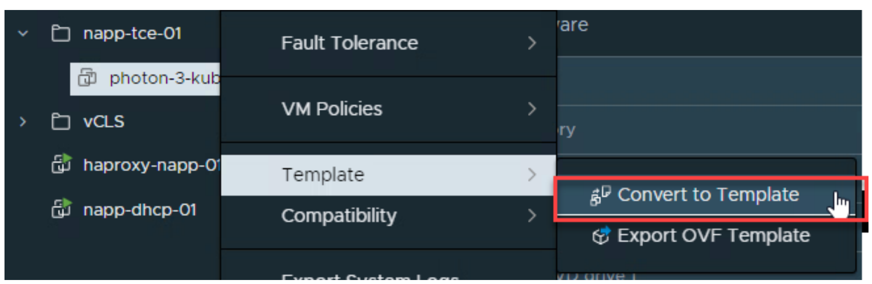
On the **Select networks** screen choose a Management dvPortgroup which has connectivity to vCenter, ESXi Management and your bootstrap machine. You must also ensure an active DHCP server connected to this dvPortgroup:



On the final **Ready to complete** screen, review your inputs. When you are satisfied click **FINISH**:



When the OVA deployment finishes, right-click on the Photon VM and select `Template > Convert to Template`:



Now we will return to the bootstrap machine where we just installed the Tanzu CLI, run the following `ssh-keygen` command and enter the same password twice:

```
ssh-keygen -t rsa -b 4096 -C "your@email.address"
```

```
rbudavari@tce-mgt-sc-01:~/Downloads/tce-linux-amd64-v0.10.0$ ssh-keygen -t rsa -b 4096 -C "rbudavari@vmware.com"
Generating public/private rsa key pair.
Enter file in which to save the key (/home/rbudavari/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/rbudavari/.ssh/id_rsa
Your public key has been saved in /home/rbudavari/.ssh/id_rsa.pub
The key fingerprint is:
SHA256: [redacted] rbudavari@vmware.com
The key's randomart image is:
+---[RSA 4096]----+
| .  . o          |
|.oo. o +         |
|o =oo o .        |
|oB ==o.o         |
|=.==+o= S        |
|.Eoo o .         |
|=o+.   .         |
|Boo. .o          |
|XB  o+           |
+----[SHA256]-----+
```

And add the RSA identity to your SSH authentication agent by running:

```
ssh-add ~/.ssh/id_rsa
```

```
rbudavari@tce-mgt-sc-01:~/Downloads/tce-linux-amd64-v0.10.0$ ssh-add ~/.ssh/id_rsa
Enter passphrase for /home/rbudavari/.ssh/id_rsa:
Identity added: /home/rbudavari/.ssh/id_rsa (rbudavari@vmware.com)
```

Deploy TCE Management Cluster

You are now ready to start deploying your Tanzu Community Edition Management Cluster.

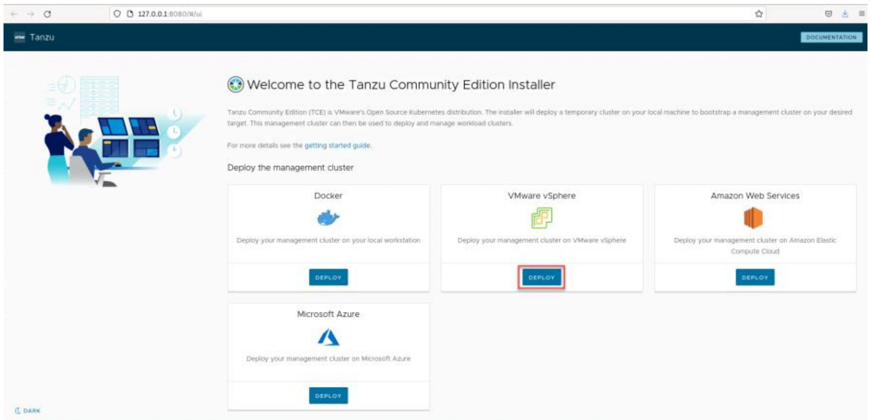
Begin by running the following command on your bootstrap host:

```
tanzu management-cluster create --ui
```

```
rbudavari@tce-mgt-sc-01:~/Downloads/tce-linux-amd64-v0.10.0$ tanzu management-cluster create --ui

Validating the pre-requisites...
Serving kickstart UI at http://127.0.0.1:8080
```

This should immediately launch a browser and display the following page, where you will click on the **DEPLOY** link in the VMware vSphere option:



And proceed to a guided wizard.

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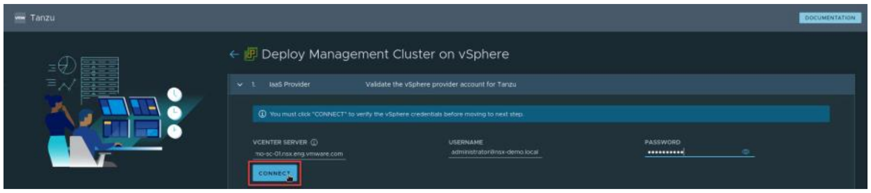
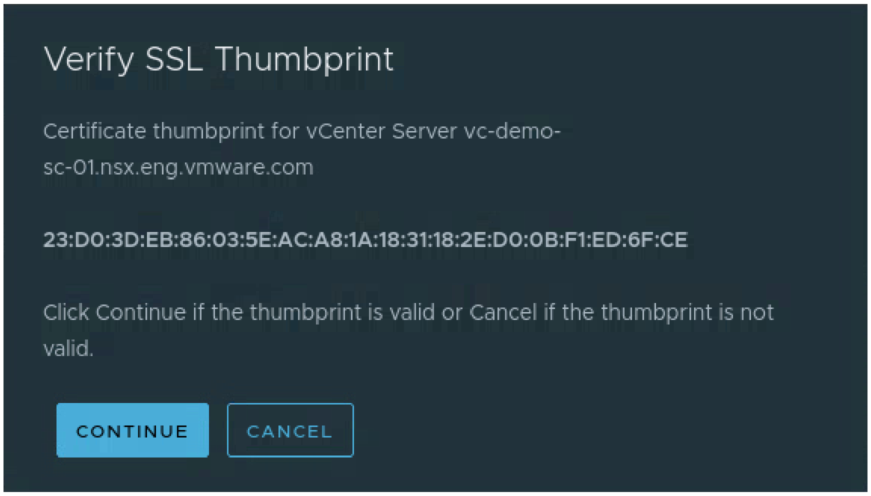
Step 1 is **IaaS Provider** which is where you will add your **VCENTER SERVER and Credentials**, then click on **CONNECT**

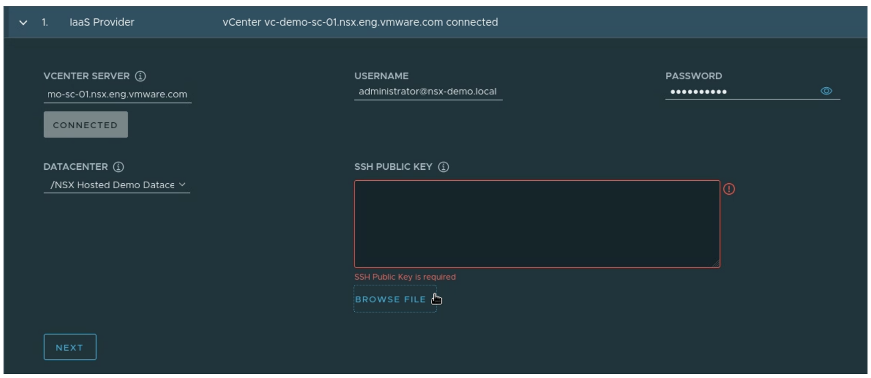


Accept the SSL thumbprint for your vCenter Server by clicking **CONTINUE**:

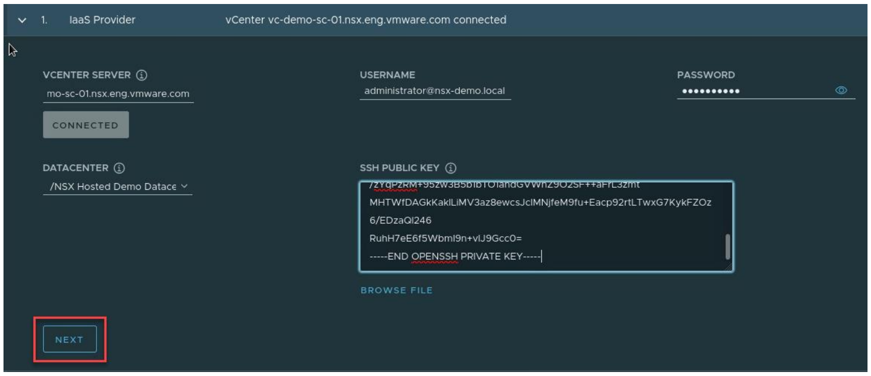


And once you are connected, select the appropriate target **DATACENTER** (if you have multiple), and add the **SSH Public Key** you generated before.

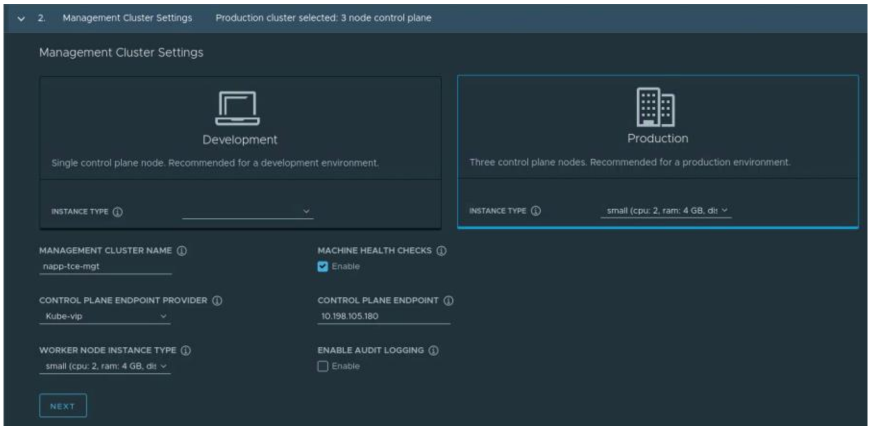
It is recommended to browse to the `id_rsa` file rather than copy and paste to avoid any formatting issues:



Once your SSH Public Key has been added, click **NEXT**:



Step 2 is where you will configure your **Management Cluster Settings**

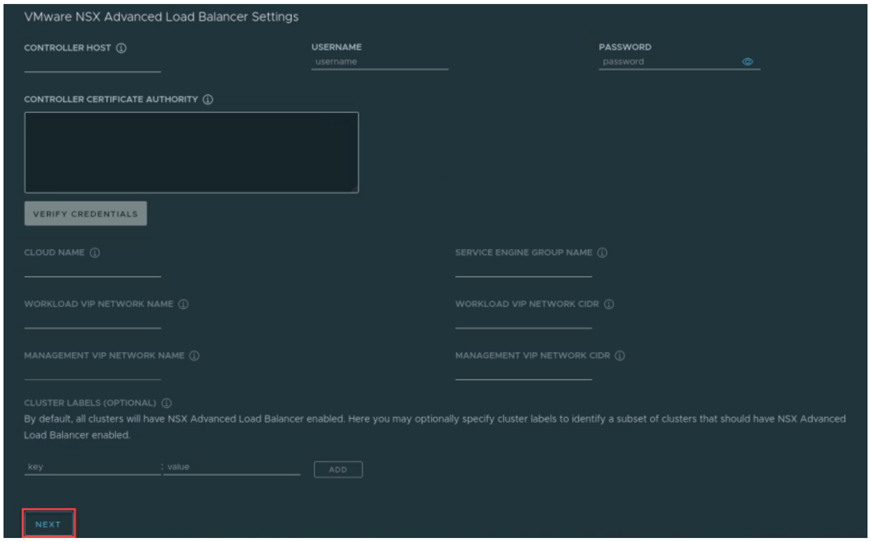


Here we will select:

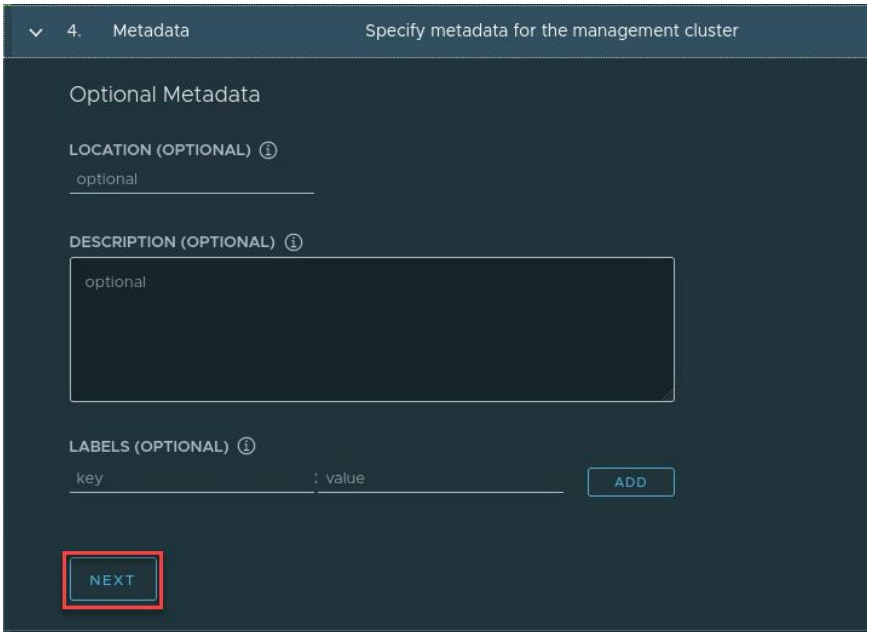
- **Production** Plan with **INSTANCE TYPE** of **small**. If this is an evaluation or non-production environment you can select the **Dev** Plan to reduce resource utilization which will deploy a single node for control plane and workers vs 3 nodes for Production.
- Enter a **MANAGEMENT CLUSTER NAME**
- Under **WORKER NODE INSTANCE TYPE** select **small**
- For **CONTROL PLANE ENDPOINT PROVIDER,** you can use the default of **kube-vip** as the requirements for control plane load balancing are basic and this has the lowest footprint. If you do already have NSX ALB in the environment you can use it as the CP Provider.

For kube-vip the only configuration required is a **CONTROL PLANE ENDPOINT** IP address on the same network as your DHCP scope (Tanzu Mgt in my case). This address should not be part of the range of DHCP addresses, as it is statically assigned to kube-vip.

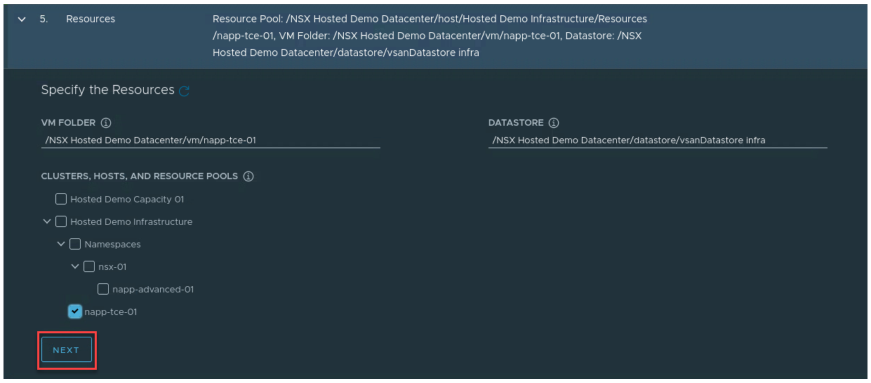
If you choose to use NSX ALB you will need to provide the Controller Host, Credentials, Certificate and Configuration details:



On the **Metadata** step, feel free to just click **NEXT**:

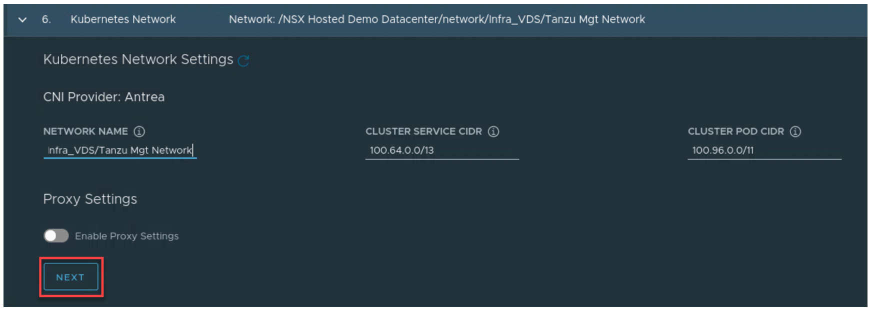


At the **Resources** step you will map your TCE cluster to vSphere Resources. Click on the **VM Folder**, **Datastore** links to select the appropriate folder for TCE and shared datastore from the drop-down menus. Also under **Clusters, Hosts and Resource Pools** select the TCE Resource Pool created earlier:



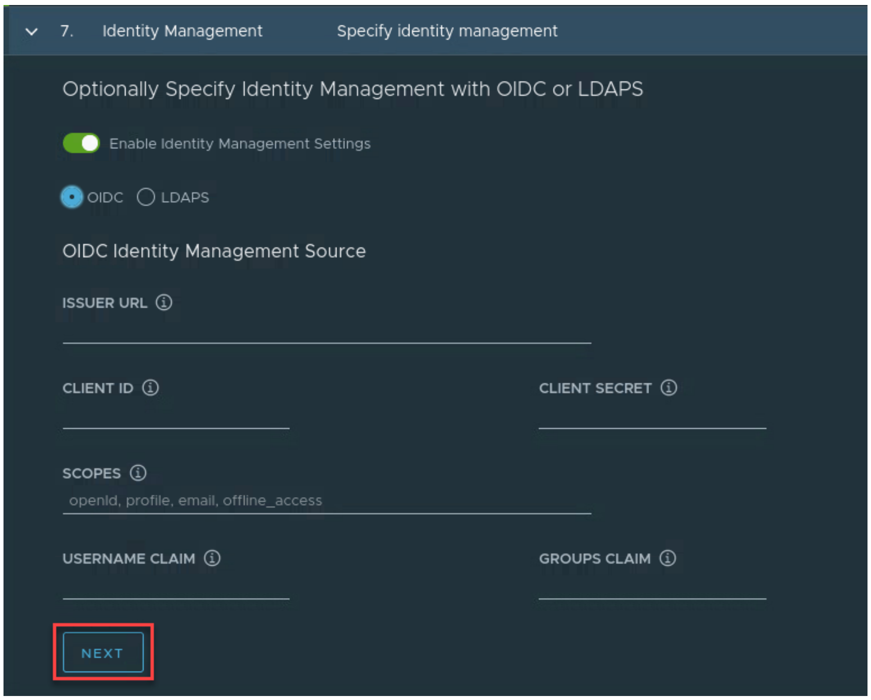
Step 6 is **Kubernetes Network**

Choose the dvPortgroup for your TCE node VMs (management network with DHCP enabled) from the **NETWORK NAME** drop-down. Unless you know these will conflict, it is recommended to leave the default internal-only subnets for **CLUSTER SERVICE CIDR** and **CLUSTER POD CIDR**. Define your proxy if one is needed to reach the external networks and click **NEXT**.



Step 7 **Identity Management**. In the Identity Management section, optionally uncheck Enable Identity Management Settings.

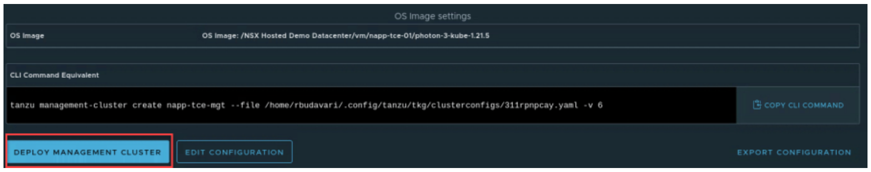
You can deactivate identity management for proof-of-concept deployments, but it is recommended to implement identity management (OIDC or LDAPS) in production. If you deactivate identity management here, you can always activate it later if required. Click **NEXT** when ready:



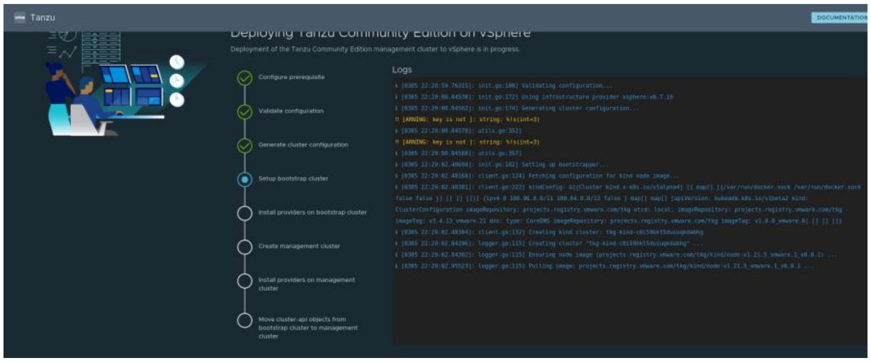
The final step is **OS Image** where you will select the Photon Kubernetes template you uploaded in the vSphere UI, then click **NEXT:**

Then Review Configuration to ensure all your inputs are correct:

Once you are happy, at the bottom of the screen click the **DEPLOY MANAGEMENT CLUSTER** link



Your TCE Management cluster is now provisioning and will take several minutes to complete. You can monitor the step-by-step progress of the deployment from the UI, starting with **Setup bootstrap cluster**:



Followed by **Install providers on bootstrap cluster**:

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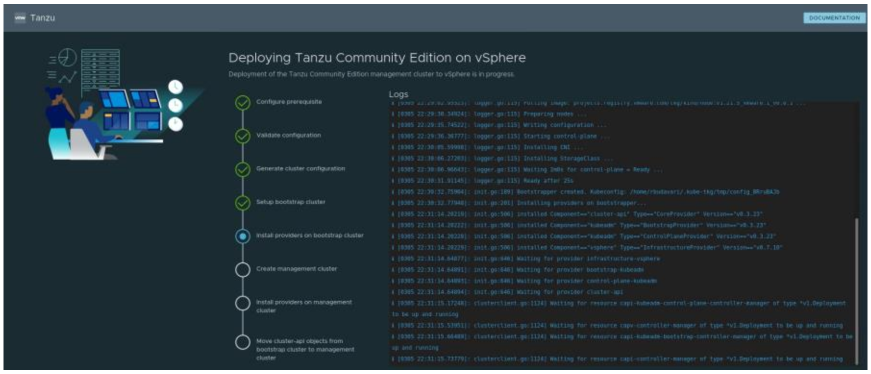
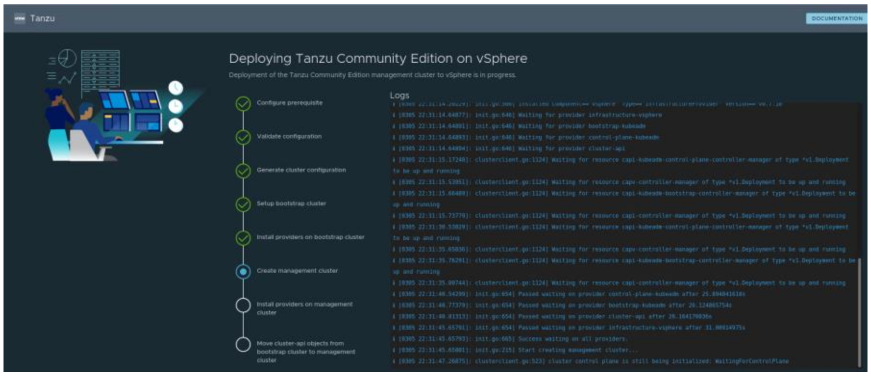
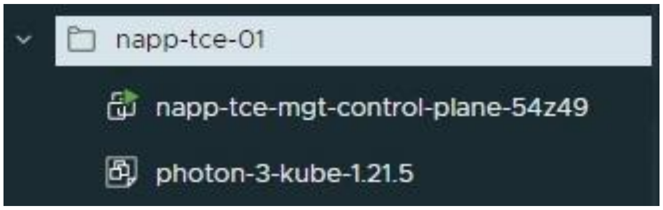
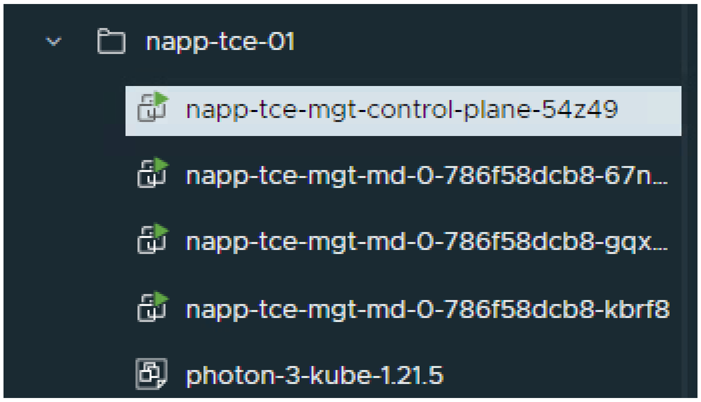


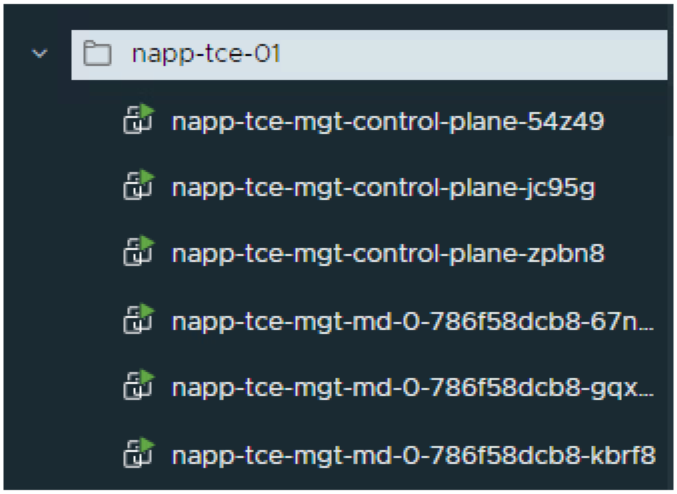
Once the local bootstrap environment is provisioned in Docker, TCE will move on to **Create management cluster:**



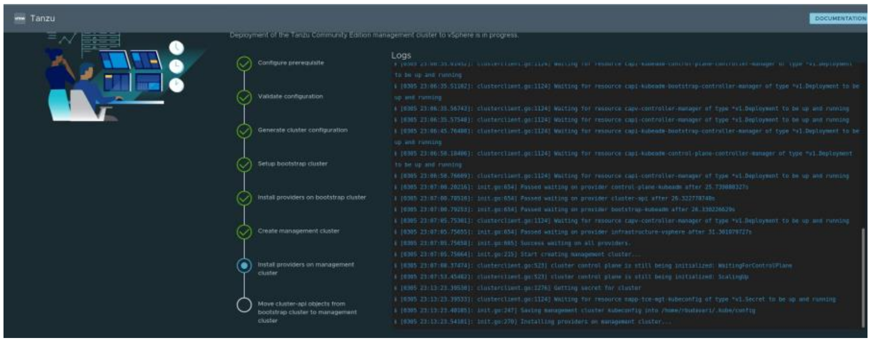
Here you can also observe the management control and worker nodes being provisioned in vSphere, where you should ultimately have 3 VMs for each node type:



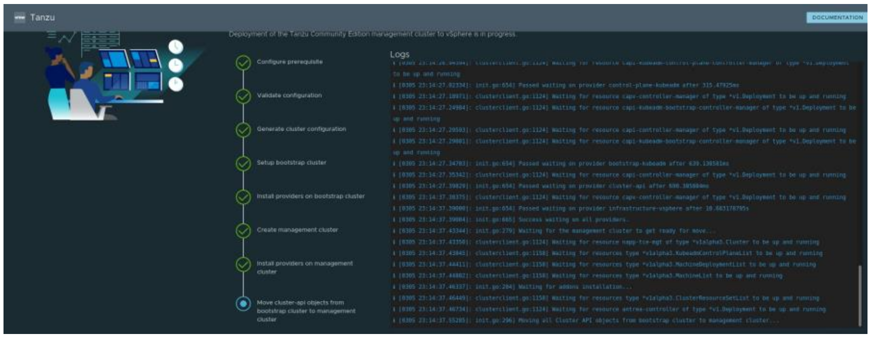




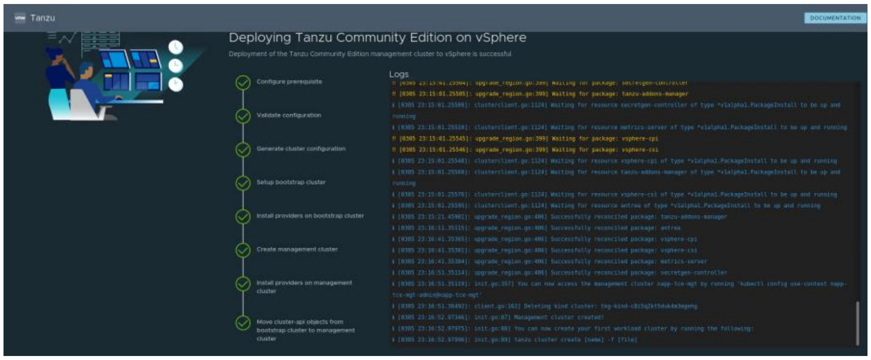
Once the management cluster has been provisioned, the deployment will move on to **Install providers on management cluster**:



Before the last step of **Move cluster-api objects from bootstrap cluster to management cluster**:



And if all goes well your management cluster deployment will complete successfully:

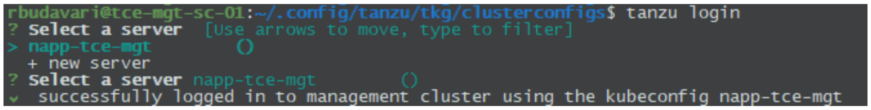


At this point you can close the browser as your remaining interaction to deploy the workload cluster will be via the kubernetes tools.

Deploy TCE Workload Cluster

Authenticate to your management cluster to generate your kubeconfig credentials:

```
tanzu login
```



Then verify your K8s version to make sure it is supported for NAPP as per the System Requirements table:

```
tanzu management-cluster get
```

```
rbudavari@tce-mgt-sc-01:~/.config/tanzu/tkg/clusterconfigs$ tanzu management-cluster get
  NAME           NAMESPACE   STATUS   CONTROLPLANE  WORKERS  KUBERNETES        ROLES
  napp-tce-mgt   tkg-system  running  3/3           3/3      v1.21.5+vmware.1  management

Details:

NAME                                                                READY  SEVERITY  REASON  SINCE  MESSAGE
/napp-tce-mgt                                                       True                     16h
├─ClusterInfrastructure - VSphereCluster/napp-tce-mgt               True                     16h
├─ControlPlane - KubeadmControlPlane/napp-tce-mgt-control-plane     True                     16h
│ └─3 Machines...                                                   True                     16h    See napp-tce-mgt-control-plane-54z49, napp-tce-mgt-control-plane-jc95g, ...
└─Workers
  └─MachineDeployment/napp-tce-mgt-md-0
    └─3 Machines...                                                 True                     16h    See napp-tce-mgt-md-0-786f58dcb8-67ng8, napp-tce-mgt-md-0-786f58dcb8-gqx42, ...

Providers:

  NAMESPACE                          NAME                    TYPE                    PROVIDERNAME  VERSION  WATCHNAMESPACE
  capi-kubeadm-bootstrap-system      bootstrap-kubeadm       BootstrapProvider       kubeadm       v0.3.23
  capi-kubeadm-control-plane-system  control-plane-kubeadm   ControlPlaneProvider    kubeadm       v0.3.23
  capi-system                        cluster-api             CoreProvider            cluster-api   v0.3.23
  capv-system                        infrastructure-vsphere  InfrastructureProvider  vsphere       v0.7.10
```

Now that we have logged in to the management cluster, we can use the kubernetes tools (`kubectl`) commands to verify the management cluster status:

```
kubectl get nodes
```

```
rbudavari@tce-mgt-sc-01:~/.config/tanzu/tkg/clusterconfigs$ kubectl get nodes
NAME                                  STATUS   ROLES                  AGE   VERSION
napp-tce-mgt-control-plane-54z49      Ready    control-plane,master   17h   v1.21.5+vmware.1
napp-tce-mgt-control-plane-jc95g      Ready    control-plane,master   17h   v1.21.5+vmware.1
napp-tce-mgt-control-plane-zpbn8      Ready    control-plane,master   17h   v1.21.5+vmware.1
napp-tce-mgt-md-0-786f58dcb8-67ng8    Ready    <none>                 17h   v1.21.5+vmware.1
napp-tce-mgt-md-0-786f58dcb8-gqx42    Ready    <none>                 17h   v1.21.5+vmware.1
napp-tce-mgt-md-0-786f58dcb8-kbrf8    Ready    <none>                 17h   v1.21.5+vmware.1
```

And check that the namepsaces are all reporting a status of **Active**:

```
kubectl get ns
```

```
rbudavari@tce-mgt-sc-01:~/.config/tanzu/tkg/clusterconfigs$ kubectl get ns
NAME                                STATUS   AGE
capi-kubeadm-bootstrap-system       Active   17h
capi-kubeadm-control-plane-system   Active   17h
capi-system                         Active   17h
capi-webhook-system                 Active   17h
capv-system                         Active   17h
cert-manager                        Active   17h
default                             Active   17h
kube-node-lease                     Active   17h
kube-public                         Active   17h
kube-system                         Active   17h
tanzu-package-repo-global           Active   17h
tanzu-system                        Active   17h
tkg-system                          Active   17h
tkg-system-public                   Active   17h
tkr-system                          Active   17h
```

Once you have verified the management cluster status is healthy, we will move on to deploying a Workload Cluster to run NAPP. For best practice of resource isolation, we will create a namespace for this workload cluster. I have used the name **nsx-01** in my case:

```
kubectl create ns nsx-01
```

```
rbudavari@tce-mgt-sc-01:~/.config/tanzu/tkg/clusterconfigs$ kubectl create ns nsx-01
namespace/nsx-01 created
```

To create the workload cluster with the correct settings, we will modify the configuration file that was generated when the management cluster was deployed. Change to the `~username/.config/tanzu/tkg/clusterconfigs` directory

```
cd .config/tanzu/tkg/clusterconfigs
```

and check that you have a yaml file for the management cluster

```
rbudavari@tce-mgt-sc-01:~/.config/tanzu/tkg/clusterconfigs$ pwd
/home/rbudavari/.config/tanzu/tkg/clusterconfigs
rbudavari@tce-mgt-sc-01:~/.config/tanzu/tkg/clusterconfigs$ ls -l napp-tce-mgt.yaml
-rw------- 1 rbudavari rbudavari 5835 Mar  5 23:23 napp-tce-mgt.yaml
```

Note: The yaml file name will be different based on the name of your management cluster or may be auto generated. This is not important.

Start by creating a copy of the management cluster yaml file for the workload cluster:

```
cp <management cluster name.yaml> <guest cluster
name.yaml>
```

```
rbudavari@tce-mgt-sc-01:~/.config/tanzu/tkg/clusterconfigs$ cp napp-tce-mgt.yaml napp-advanced-01.yaml
```

It is important to be accurate in the next step, since to meet the specific requirements of NAPP we will be using a different workflow for the workload cluster deployment.

Then open the guest cluster yaml file in an editor and modify the following fields. You will configure the **CLUSTER_NAME** and **VSPHERE_CONTROL_PLANE_ENDPOINT** with appropriate values based on your environment and resources for **CPUS/MEM/DISK**, based on your NAPP form factor:

Advanced:

```
CLUSTER_NAME: napp-advanced-01
CLUSTER_PLAN: prod
NAMESPACE: nsx-01
CNI: antrea
ENABLE_AUTOSCALER: false
VSPHERE_CONTROL_PLANE_ENDPOINT: 10.198.105.178
VSPHERE_WORKER_DISK_GIB: "64"
VSPHERE_WORKER_MEM_MIB: "65536"
VSPHERE_WORKER_NUM_CPUS: "16"
```

Standard:

```
CLUSTER_NAME: napp-standard-01
CLUSTER_PLAN: prod
NAMESPACE: nsx-01
CNI: antrea
ENABLE_AUTOSCALER: false
VSPHERE_CONTROL_PLANE_ENDPOINT: 10.198.105.178
VSPHERE_WORKER_DISK_GIB: "64"
VSPHERE_WORKER_MEM_MIB: "16384"
VSPHERE_WORKER_NUM_CPUS: "4"
```

Note: We have increased the node disk to account for the additional ephemeral storage required by NAPP so an additional volume is not required. You also need to use an available IP address from the Management network you are using for TCE for the Control Plane Endpoint. This address must be in the same network but not in in the DHCP scope.

Your options for Cluster Plan are **Prod** (which will deploy 3 control plane nodes and 3 worker nodes by default) and **Dev** (1 of each node).

I have also made these files available via github, although generally it is better to edit your own template from the management cluster because your vSphere environment details will be different:

TCE Guest Cluster Advanced Configuration

TCE Guest Cluster Standard Configuration

Here is the complete **YAML** file from my environment for reference:

VMware, Inc. 3401 Hillview Avenue Palo Alto CA 94304 USA Tel 877-486-9273 Fax 650-427-5001 www.vmware.com

```yaml
AVI_CA_DATA_B64: ""
AVI_CLOUD_NAME: ""
AVI_CONTROL_PLANE_HA_PROVIDER: ""
AVI_CONTROLLER: ""
AVI_DATA_NETWORK: ""
AVI_DATA_NETWORK_CIDR: ""
AVI_ENABLE: "false"
AVI_LABELS: ""
AVI_MANAGEMENT_CLUSTER_VIP_NETWORK_CIDR: ""
AVI_MANAGEMENT_CLUSTER_VIP_NETWORK_NAME: ""
AVI_PASSWORD: ""
AVI_SERVICE_ENGINE_GROUP: ""
AVI_USERNAME: ""
CLUSTER_CIDR: 100.96.0.0/11
CLUSTER_NAME: napp-advanced-01
CLUSTER_PLAN: prod
NAMESPACE: nsx-01
CNI: antrea
ENABLE_AUTOSCALER: false
ENABLE_AUDIT_LOGGING: "false"
ENABLE_CEIP_PARTICIPATION: "false"
ENABLE_MHC: "true"
IDENTITY_MANAGEMENT_TYPE: none
INFRASTRUCTURE_PROVIDER: vsphere
LDAP_BIND_DN: ""
LDAP_BIND_PASSWORD: ""
LDAP_GROUP_SEARCH_BASE_DN: ""
LDAP_GROUP_SEARCH_FILTER: ""
LDAP_GROUP_SEARCH_GROUP_ATTRIBUTE: ""
LDAP_GROUP_SEARCH_NAME_ATTRIBUTE: cn
LDAP_GROUP_SEARCH_USER_ATTRIBUTE: DN
LDAP_HOST: ""
LDAP_ROOT_CA_DATA_B64: ""
LDAP_USER_SEARCH_BASE_DN: ""
LDAP_USER_SEARCH_FILTER: ""
LDAP_USER_SEARCH_NAME_ATTRIBUTE: ""
LDAP_USER_SEARCH_USERNAME: userPrincipalName
OIDC_IDENTITY_PROVIDER_CLIENT_ID: ""
OIDC_IDENTITY_PROVIDER_CLIENT_SECRET: ""
OIDC_IDENTITY_PROVIDER_GROUPS_CLAIM: ""
OIDC_IDENTITY_PROVIDER_ISSUER_URL: ""
OIDC_IDENTITY_PROVIDER_NAME: ""
OIDC_IDENTITY_PROVIDER_SCOPES: ""
OIDC_IDENTITY_PROVIDER_USERNAME_CLAIM: ""
OS_ARCH: amd64
OS_NAME: photon
OS_VERSION: "3"
SERVICE_CIDR: 100.64.0.0/13
TKG_HTTP_PROXY_ENABLED: "false"
TKG_IP_FAMILY: ipv4
VSPHERE_CONTROL_PLANE_DISK_GIB: "20"
VSPHERE_CONTROL_PLANE_ENDPOINT: 10.198.105.178
VSPHERE_CONTROL_PLANE_MEM_MIB: "4096"
VSPHERE_CONTROL_PLANE_NUM_CPUS: "2"
VSPHERE_DATACENTER: /NSX Hosted Demo Datacenter
VSPHERE_DATASTORE: /NSX Hosted Demo Datacenter/datastore/vsanDatastore infra
VSPHERE_FOLDER: /NSX Hosted Demo Datacenter/vm/napp-tce-01
VSPHERE_NETWORK: /NSX Hosted Demo Datacenter/network/Infra_VDS/Tanzu Mgt Network
VSPHERE_PASSWORD: <encoded:Vk13YXJlYDEyMw==>
VSPHERE_RESOURCE_POOL: /NSX Hosted Demo Datacenter/host/Hosted Demo Infrastructure/Resources/napp-tce-01
VSPHERE_SERVER: vc-demo-sc-01.nsx.eng.vmware.com
VSPHERE_SSH_AUTHORIZED_KEY: |-
    -----BEGIN OPENSSH PRIVATE KEY-----
    -----END OPENSSH PRIVATE KEY-----
VSPHERE_TLS_THUMBPRINT: 23:D0:3D:EB:86:03:5E:AC:A8:1A:18:31:18:2E:D0:0B:F1:ED:6F:CE
VSPHERE_USERNAME: administrator@nsx-demo.local
VSPHERE_WORKER_DISK_GIB: "1024"
VSPHERE_WORKER_MEM_MIB: "65536"
VSPHERE_WORKER_NUM_CPUS: "16"
```


Now that you have configured the workload cluster YAML, you will have to create a manifest by providing the `--dry-run` option so you can modify the standard TCE deployment configuration. Run the following command replacing the example with your guest cluster name:

```
tanzu cluster create <guest-cluster-name> --file <guest-cluster-name>.yaml --dry-run > <guest-cluster-name>-manifest.yaml
```

```
rbudavari@tce-mgt-sc-01:~/.config/tanzu/tkg/clusterconfigs$ tanzu cluster create napp-advanced-01 --file napp-advanced-01.yaml --dry-run > napp-advanced-01-manifest.yaml
```

This will create a large manifest file:

```
rbudavari@tce-mgt-sc-01:~/.config/tanzu/tkg/clusterconfigs$ ls -l napp-advanced-01-manifest.yaml
-rw-r--r-- 1 rbudavari rbudavari 64344 Mar  8 16:07 napp-advanced-01-manifest.yaml
```

But fortunately, we only need to change one value in this file. Open it with an editor:

```
rbudavari@tce-mgt-sc-01:~/.config/tanzu/tkg/clusterconfigs$ vi napp-advanced-01-manifest.yaml
```

And search for the section of the yaml file that contains the following value (this should be approximately around line 95 of your file):

```
kind: KubeadmControlPlane
metadata:
  name: <guest-cluster-name>-control-plane
  namespace: <namespace-name>
```

then approximately 100 lines down search for the following section:

```
replicas: 3
```

and change the value to:

```
replicas: 1
```

This will deploy 1 control-plane node as required by NAPP instead of 3, reducing the footprint of your deployment.

Technically we could also create a custom plan to achieve the same outcome or provide a different `CONTROL_PLANE_MACHINE_COUNT`, but this option supports additional customizations of the nodes as needed in future (such as extra volumes, which at one point in the NAPP deployment process was required)

Now you will run the following command to create the workload cluster:

```
kubectl apply -f < guest-cluster-name-manifest.yaml>
```

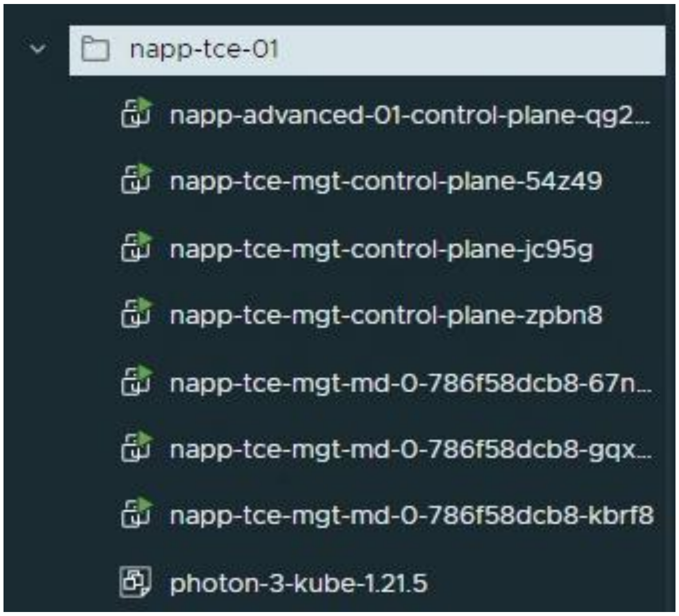
```
rbudavari@tce-mgt-sc-01:~/.config/tanzu/tkg/clusterconfigs$ kubectl apply -f napp-advanced-01-manifest.yaml
cluster.cluster.x-k8s.io/napp-advanced-01 created
vspherecluster.infrastructure.cluster.x-k8s.io/napp-advanced-01 created
vspheremachinetemplate.infrastructure.cluster.x-k8s.io/napp-advanced-01-control-plane unchanged
vspheremachinetemplate.infrastructure.cluster.x-k8s.io/napp-advanced-01-worker unchanged
kubeadmcontrolplane.controlplane.cluster.x-k8s.io/napp-advanced-01-control-plane created
kubeadmconfigtemplate.bootstrap.cluster.x-k8s.io/napp-advanced-01-md-0 created
machinedeployment.cluster.x-k8s.io/napp-advanced-01-md-0 created
secret/napp-advanced-01 created
secret/napp-advanced-01-antrea-addon created
secret/napp-advanced-01-vsphere-cpi-addon created
secret/napp-advanced-01-vsphere-csi-addon created
secret/napp-advanced-01-kapp-controller-addon created
secret/napp-advanced-01-tkg-metadata-namespace-role created
secret/napp-advanced-01-tkg-metadata-configmap created
secret/napp-advanced-01-tkg-metadata-bom-configmap created
clusterresourceset.addons.cluster.x-k8s.io/napp-advanced-01-tkg-metadata created
secret/napp-advanced-01-metrics-server-addon created
clusterresourceset.addons.cluster.x-k8s.io/napp-advanced-01-core-package-repository created
secret/napp-advanced-01-core-package-repository-crs created
secret/napp-advanced-01-secretgen-controller-addon created
machinehealthcheck.cluster.x-k8s.io/napp-advanced-01 created
machinehealthcheck.cluster.x-k8s.io/napp-advanced-01-control-plane created
clusterresourceset.addons.cluster.x-k8s.io/napp-advanced-01-default-storage-class created
secret/napp-advanced-01-default-storage-class created
clusterresourceset.addons.cluster.x-k8s.io/napp-advanced-01-capabilities created
secret/napp-advanced-01-capabilities created
secret/napp-advanced-01-config-values created
```

and assuming you see a similar output to the above screenshot you can monitor the progress of the guest cluster deployment at the CLI:

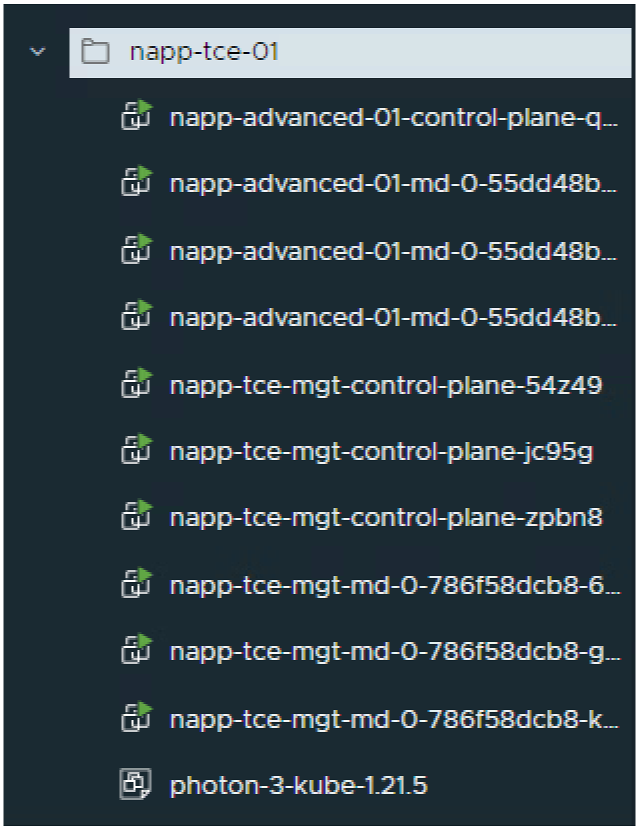
```
tanzu cluster list
```

```
rbudavari@tce-mgt-sc-01:~/.config/tanzu/tkg/clusterconfigs$ tanzu cluster list
  NAME               NAMESPACE  STATUS    CONTROLPLANE  WORKERS  KUBERNETES         ROLES   PLAN
  napp-advanced-01   nsx-01     creating  0/1           0/3      v1.21.5+vmware.1   <none>  prod
```

Or within the vSphere UI, where you can see the control plane node VM being provisioned:

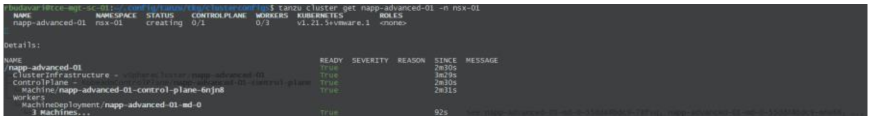


Followed by the 3 worker nodes:



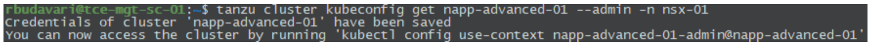
After a couple of minutes, your guest cluster should return a status of READY=True. You can verify this with:

```
tanzu <guest-cluster-name> -n <namespace-name>
```



Next you will refresh your credentials for workload cluster by running:

```
tanzu cluster kubeconfig get <guest-cluster-name> --
admin -n <namespace-name>
```



Then switch your context from management to workload cluster:

```
kubectl config use-context <guest-cluster-name>-admin@
<guest-cluster-name>
```

```
rbudavari@tce-mgt-sc-01:~$ kubectl config use-context napp-advanced-01-admin@napp-advanced-01
Switched to context "napp-advanced-01-admin@napp-advanced-01".
```

And verify all cluster nodes are both **Ready** and running a **Version** of K8s supported by NAPP:

```
kubectl get nodes
```

```
rbudavari@tce-mgt-sc-01:~$ kubectl get nodes
NAME                                  STATUS   ROLES                  AGE   VERSION
napp-advanced-01-control-plane-fcq8j  Ready    control-plane,master   9h    v1.21.5+vmware.1
napp-advanced-01-md-0-55dd48bdc9-4qj2c  Ready  <none>                 9h    v1.21.5+vmware.1
napp-advanced-01-md-0-55dd48bdc9-68tjn  Ready  <none>                 9h    v1.21.5+vmware.1
napp-advanced-01-md-0-55dd48bdc9-z77r2  Ready  <none>                 9h    v1.21.5+vmware.1
```

We are almost done and just need to configure the container networking and storage dependencies for NAPP before we proceed.

Tanzu CE uses **kube-vip** by default for load balancing access to the K8s cluster API server, but not for workload clusters. When using a vSphere endpoint to load balance workloads you have the option to use NSX Advanced Load Balancer (Essentials Edition or greater) if it is available in your environment, however given NAPP is a single cluster the load balancing requirements are minimal. So, unless you already have an NSX ALB deployment, rather than provisioning ALB Controller and Service Engine VMs, NAPP has also been validated with **MetalLB**. This is a very simple Load Balancing service with straightforward configuration and runs as a native pod in your Tanzu environment, so it helps keep the footprint down and is the recommended approach for TCE. Additional details on MetalLB are available from:

https://metallb.universe.tf/

Installing MetalLB in your workload cluster

Note: Make sure you are in the correct workload cluster context before proceeding. If you have followed the steps above in sequence this will be the case.

Start the MetalLB installation by running the following commands to apply manifests which will create the namespace and deploy pods for Load Balancing:

```
kubectl apply -f
https://raw.githubusercontent.com/metallb/metallb/v0.12.1/manifests/namespace.yaml
```

```
kubectl apply -f
https://raw.githubusercontent.com/metallb/metallb/v0.12.1/manifests/metallb.yaml
```

v0.12.1 is the latest MetalLB release at the time of creating this document, and is known to work reliably with NAPP

```
rbudavari@tce-mgt-sc-01:~$ kubectl apply -f https://raw.githubusercontent.com/metallb/metallb/v0.12.1/manifests/namespace.yaml
namespace/metallb-system created
rbudavari@tce-mgt-sc-01:~$
rbudavari@tce-mgt-sc-01:~$ kubectl apply -f https://raw.githubusercontent.com/metallb/metallb/v0.12.1/manifests/metallb.yaml
Warning: policy/v1beta1 PodSecurityPolicy is deprecated in v1.21+, unavailable in v1.25+
podsecuritypolicy.policy/controller created
podsecuritypolicy.policy/speaker created
serviceaccount/controller created
serviceaccount/speaker created
clusterrole.rbac.authorization.k8s.io/metallb-system:controller created
clusterrole.rbac.authorization.k8s.io/metallb-system:speaker created
role.rbac.authorization.k8s.io/config-watcher created
role.rbac.authorization.k8s.io/pod-lister created
role.rbac.authorization.k8s.io/controller created
clusterrolebinding.rbac.authorization.k8s.io/metallb-system:controller created
clusterrolebinding.rbac.authorization.k8s.io/metallb-system:speaker created
rolebinding.rbac.authorization.k8s.io/config-watcher created
rolebinding.rbac.authorization.k8s.io/pod-lister created
rolebinding.rbac.authorization.k8s.io/controller created
daemonset.apps/speaker created
deployment.apps/controller created
```

Now you need to configure MetalLB.

This is very simple and involves creating a yaml file (`lb-config.yaml` used in the guide):

```
rbudavari@tce-mgt-sc-01:~$ vi lb-config.yaml
```

Copy the following text into your yaml file and ensure the `addresses` field has a range with a minimum of 5 and ideally 8 IPs per NAPP deployment:

```
apiVersion: v1
kind: ConfigMap
metadata:
  namespace: metallb-system
  name: config
data:
  config: |
    address-pools:
    - name: default
      protocol: layer2
      addresses:
      - <your-management-ip-range>
```

Here is my reference file:

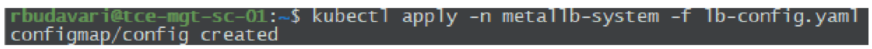
```
apiVersion: v1
kind: ConfigMap
metadata:
  namespace: metallb-system
  name: config
data:
  config: |
    address-pools:
    - name: default
      protocol: layer2
      addresses:
      - 10.198.105.150-10.198.105.159
```

Note: The first IP in your range will be allocated for NAPP interface access and will need to be mapped to an FQDN in DNS for the NAPP Service Name

MetalLB Configuration

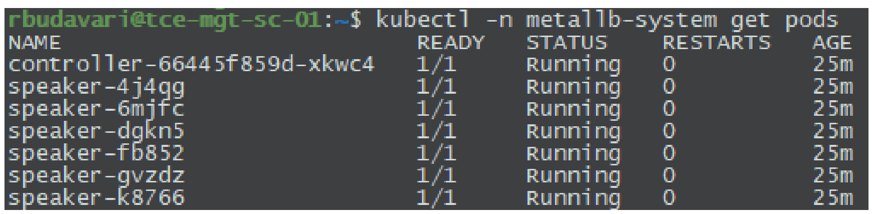
Next apply the LB configuration:

```
kubectl apply -n metallb-system -f <lb-config.yaml>
```



And verify the MetalLB pods have a **STATUS** of Running (until you apply a valid configuration all pods will not be running):

```
kubectl -n metallb-system get pods
```



You can also use kubectl to view MetalLB logs in the unlikely event troubleshooting is required:

```
kubectl logs -n metallb-system <pod-name>
```

This completes the MetalLB configuration for TCE.

Create a storage class and retrieve your kubeconfig file

Finally, you need to create a storage class which will be used for all persistent storage volumes for NAPP. This will be backed by a vSphere datastore that needs sufficient space for your form factor.

Note: Refer to the **Preparing your vSphere environment for Tanzu** section of this guide to create a Storage Policy if needed. Having a Storage Policy assigned to a vSphere datastore is a required input for this operation.

Open an editor and create a yaml file for the storage class:

```
rbudavari@tce-mgt-sc-01:~$ vi sc-config.yaml
```

Copy the following text and update the **storagePolicyName** for your environment:

```
kind: StorageClass
apiVersion: storage.k8s.io/v1
metadata:
  name: vsan-default-storage-policy
  annotations:
    storageclass.kubernetes.io/is-default-class:
"true"
provisioner: csi.vsphere.vmware.com
parameters:
  storagePolicyName: "<your-storage-policy-name>"
```

As always here is my reference:

```
kind: StorageClass
apiVersion: storage.k8s.io/v1
metadata:
  name: vsan-default-storage-policy
  annotations:
    storageclass.kubernetes.io/is-default-class: "true"
provisioner: csi.vsphere.vmware.com
parameters:
  storagePolicyName: "vSAN Default Storage Policy"
```

Storage Class Configuration

Now create the Storage Class with the following command:

```
kubectl create -f <your-sc-file.yaml>
```

```
rbudavari@tce-mgt-sc-01:~$ kubectl create -f sc-config.yaml
storageclass.storage.k8s.io/vsan-default-storage-policy created
```

Then verify the storage class was correctly configured and is the default option using:

```
kubectl get sc
```

```
rbudavari@tce-mgt-sc-01:~$ kubectl get sc
NAME                                  PROVISIONER              RECLAIMPOLICY   VOLUMEBINDINGMODE   ALLOWVOLUMEEXPANSION   AGE
vsan-default-storage-policy (default)   csi.vsphere.vmware.com   Delete          Immediate           false                  34s
rbudavari@tce-mgt-sc-01:~$ kubectl describe sc vsan-default-storage-policy
Name:                  vsan-default-storage-policy
IsDefaultClass:        Yes
Annotations:           storageclass.kubernetes.io/is-default-class=true
Provisioner:           csi.vsphere.vmware.com
Parameters:            storagePolicyName=vSAN Default Storage Policy
AllowVolumeExpansion:  <unset>
MountOptions:          <none>
ReclaimPolicy:         Delete
VolumeBindingMode:     Immediate
Events:                <none>
```

Note: In my environment I created the storage class in the management cluster to provide more flexibility for multiple workload clusters. If you also do so, it will automatically be available to the workload cluster as a default and your output will look like:

```
rbudavari@tce-mgt-sc-01:~$ kubectl get sc
NAME                PROVISIONER              RECLAIMPOLICY   VOLUMEBINDINGMODE   ALLOWVOLUMEEXPANSION   AGE
default (default)   csi.vsphere.vmware.com   Delete          Immediate           true                   9h
```

At this point you can retrieve your `kubeconfig` file. For the NAPP deployment it is essential to make sure this file has the context of the workload cluster and not your management cluster. As you have potentially been running commands against both, it is recommended to either confirm the context now, or delete the existing config file and regenerate it for the workload cluster to be sure you have the correct target cluster.

First run:

```
kubectl config get-contexts
```

```
rbudavari@tce-mgt-sc-01:~$ kubectl config get-contexts
CURRENT   NAME                                        CLUSTER             AUTHINFO                  NAMESPACE
*         napp-advanced-01-admin@napp-advanced-01     napp-advanced-01    napp-advanced-01-admin
```

And if the **CLUSTER** returned matches your workload cluster you can continue without refreshing your kubeconfig file.

Otherwise perform the following steps:

```
rm ~username/.kube/config
```

```
rbudavari@tce-mgt-sc-01:~/.config/tanzu/tkg/clusterconfigs$ rm ~rbudavari/.kube/config
```

```
tanzu cluster kubeconfig get <guest-cluster-name> --
admin -n <namespace-name>
```

```
rbudavari@tce-mgt-sc-01:~$ tanzu cluster kubeconfig get napp-advanced-01 --admin -n nsx-01
Credentials of cluster 'napp-advanced-01' have been saved
You can now access the cluster by running 'kubectl config use-context napp-advanced-01-admin@napp-advanced-01'
```

```
kubectl config use-context <guest-cluster-name>-admin@
<guest-cluster-name>
```

```
rbudavari@tce-mgt-sc-01:~$ kubectl config use-context napp-advanced-01-admin@napp-advanced-01
Switched to context "napp-advanced-01-admin@napp-advanced-01".
```

You can also follow the sequence in Step 71 of the TKGs deployment to generate a non-expiring token if you choose.

Finally, if you are using a different system to access the NSX-T Manager UI and deploy NAPP, copy the kubeconfig file (`~username/.kube/config`) to this system for the next operation.

NSX Application Platform Deployment – NSX-T Data Center

1. Verify you have satisfied all requirements listed in the **NSX Application Platform System Requirements** section of this Deployment Guide.

   Also ensure you have a valid `kubeconfig` file for the correct context.

   Note: If you are referring to the public documentation on vmware.com, you can ignore any references to a Private Harbor registry (if they are listed) as we will use the VMware Public Distribution Harbor instance for our deployment. The documentation should be updated by the time you are viewing this guide.

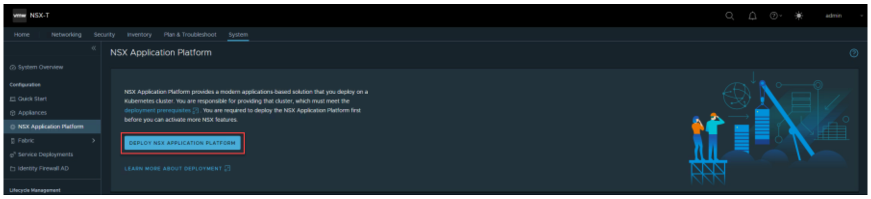
   Note: You will also need the ability to add a **DNS** record for the **NAPP Service Name and Messaging Name**

2. Deploy an **NSX-T Data Center 3.2.0.1** or newer unified appliance and ensure there is IP connectivity between the Manager and your K8s environment. Your NSX-T Manager can be either a standalone or clustered deployment, although as mentioned in the prerequisites, clustered is recommended for production.

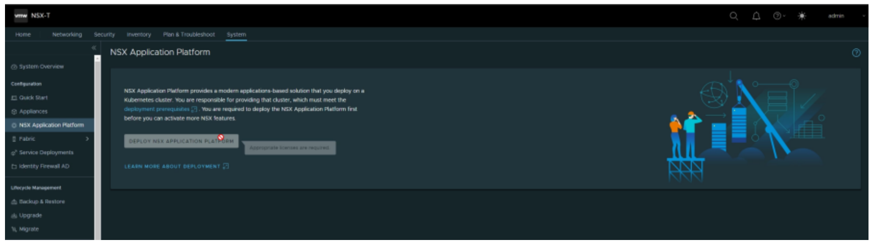
   It is important to note that all NAPP Deployment, Lifecycle Management and Monitoring steps are performed via the NSX-T Manager. After the underlying cluster infrastructure is provided, no knowledge of microservices application provisioning or operations is required to run NAPP.

3. Once your **NSX Manager** has been provisioned, powered on, and an appropriate license applied, you are now ready to start the NAPP deployment. Login to NSX-T Manager and navigate to `System -> NSX Application Platform`

   If you have a valid NSX license you will be able to click on the **DEPLOY NSX APPLICATION PLATFORM** link:

   Otherwise, this action will be greyed out until you apply a license:



Note: NSX Evaluation licenses are able to deploy NAPP, otherwise a valid NSX Data Center Plus, Advanced with ATP or NSX Security license is required

4. The first screen in the NAPP deployment workflow is **Prepare to Deploy.**

   Here you are prompted to provide the **Helm Repository** and **Docker Registry** hosting NAPP charts and images.

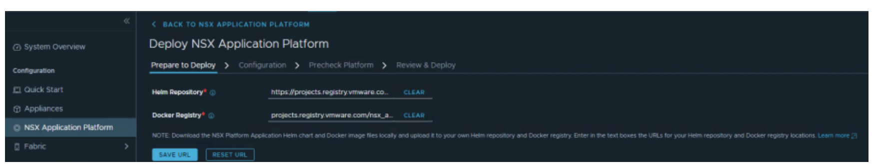
   Note: It is **strongly** recommended to use the VMware distribution Helm Repository/Docker Registry wherever possible to simplify the installation. No data is exchanged or stored during the deployment process and it is simply a pull of the required charts and images.

   If your environment has external access, you can use the following URLs for your installation:

   Helm Repository
   https://projects.registry.vmware.com/chartrepo/nsx_application_platform

   Docker Registry
   projects.registry.vmware.com/nsx_application_platform/clustering

   Note: It is also important that you use **https** for the **Repository** and **http** for the **Registry**



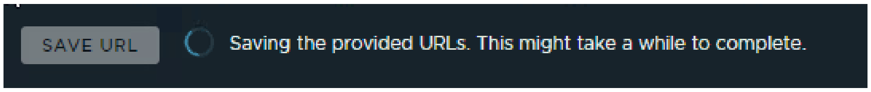
   Alternatively, if this is an air-gapped deployment without external access, you can set up a private registry. This process is documented at:

   https://docs.vmware.com/en/VMware-NSX-T-Data-Center/3.2/nsx-application-platform/GUID-FAC9DBE3-A8EE-4891-A723-942D0AB679F6.html#GUID-FAC9DBE3-A8EE-4891-A723-942D0AB679F6

and VMware has also provided a supported script for populating your registry with the NSX Application Platform installation files.

Note: You will currently need to provide a **valid CA-Signed Certificate** for the registry access. Self-signed certificates are not currently supported although this is planned as a future enhancement. Therefore, the public registry should be used wherever possible.

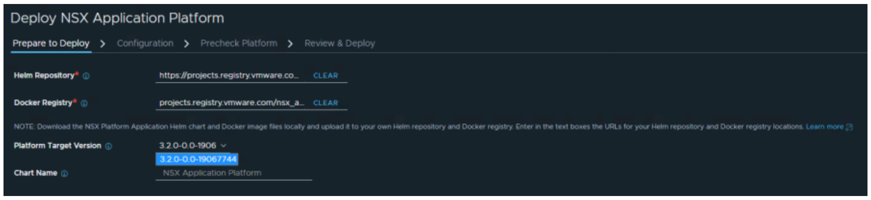
5. Once you click on **SAVE URL** this can take NSX Manager a short period of time to access and save all Docker images/Helm Charts:



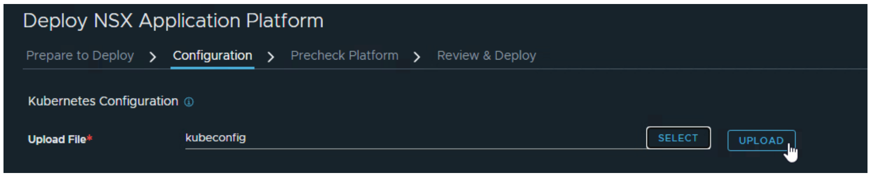
6. When this operation completes you must select a version number from the **Platform Target Version** dropdown. For **3.2.0 GA** the required build and version number is:

```
3.2.0-0.0.19067744
```

At the time of this document, **19067744** is the only available version, but check for the latest NSX Application Platform Release Notes at the time you are deploying NAPP to see if a newer version is generally available



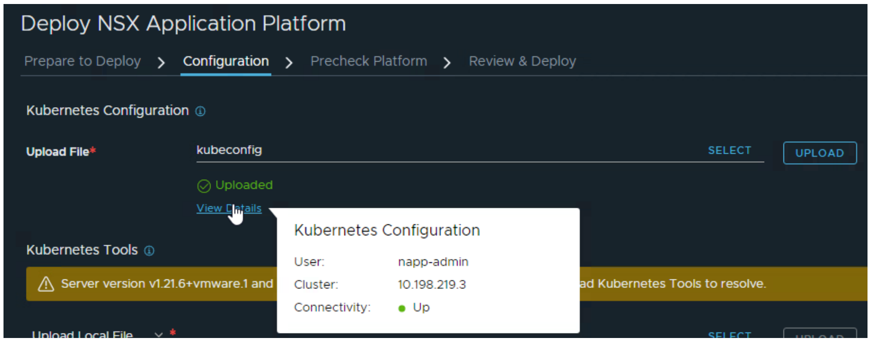
7. When you advance to the **Configuration** screen click **SELECT** and provide the **kubeconfig** file generated earlier during your guest/workload cluster deployment. Then click **UPLOAD** to store the file on the NSX Manager and enable access to your Kubernetes Guest Cluster:



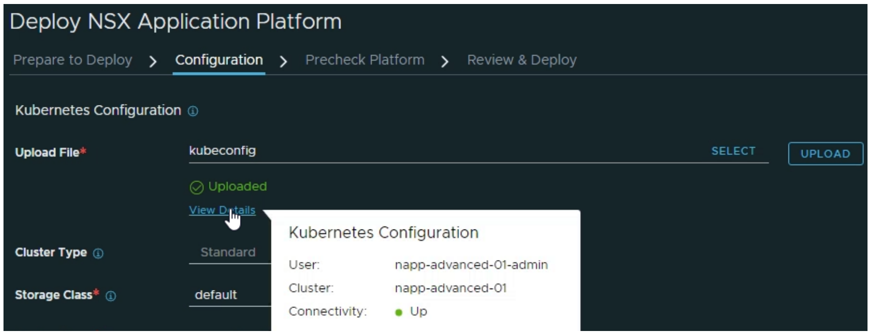
Once the **kubeconfig** file has been uploaded you can click on **View Details** to ensure your Connectivity is reporting a status of **Up**

Example from TKGs:

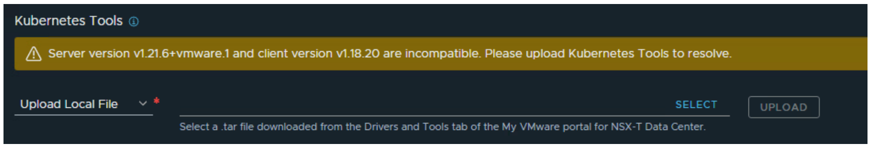
VMware, Inc. 3401 Hillview Avenue Palo Alto CA 94304 USA Tel 877-486-9273 Fax 650-427-5001 www.vmware.com



Example from TCE:

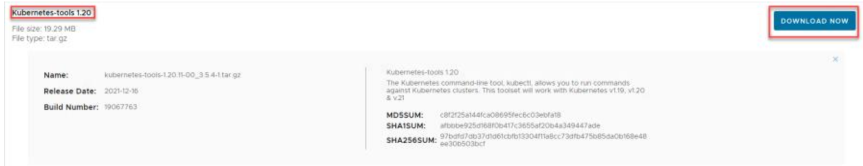


8. Dependent on the version of NSX-T and your Kubernetes cluster, you may be prompted to update your kubernetes tools if the server is two or more releases newer than **1.18.**

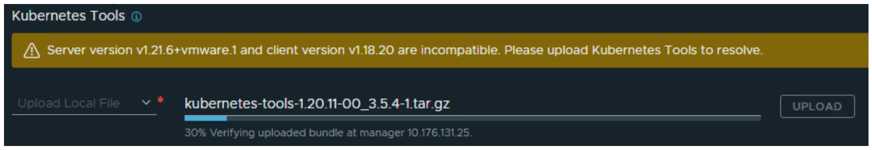


This message means you need to update the k8s tools available on NSX Manager to connect to your cluster. This is a simple process which can be performed independently of any NSX Manager upgrades. After logging in to the VMware Customer Connect portal you can download a newer **Kubernetes Tools package** from here:

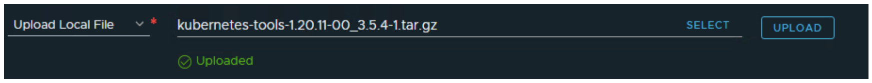
https://customerconnect.vmware.com/downloads/details?downloadGroup=NSX-T-320&productId=982&rPId=81133



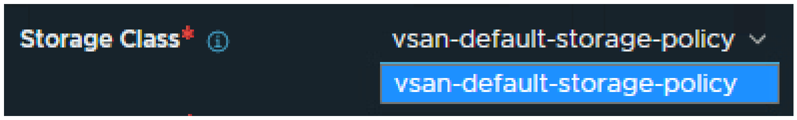
We will select the **1.20** tools to connect to our version **1.21** cluster. Once you have downloaded the tarball, click **SELECT** and upload the tarball using the same process as your kubeconfig file. Then **UPLOAD** which will handle the process of verifying the checksum and validity of the tools and upgrading across all NSX Manager nodes in your cluster**:**



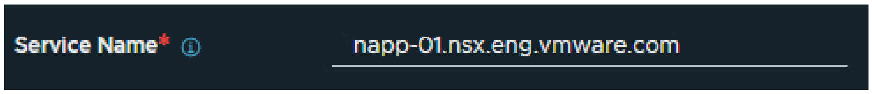
Once this process has completed you should receive confirmation of the **Upload** and the warning banner will disappear:



9. Then select your **Storage Class**. In almost all cases there will only be one auto-populated option available based on the values provided in the yaml file used to create the guest cluster:



10. Next enter a **Service Name and Messaging Name** for your NSX Application Platform environment (this is the endpoint NSX clients will use to connect to NAPP):



These service names **must** be a Fully Qualified Domain Name (FQDN) that resolves to the External IPs allocated to the load balancer VIP for the NAPP external services

To create these FQDNs there are several options:

a. **Recommended Approach:** Create DNS entries for the next 2 available VIP address **before** proceeding with the installation. LB IPs are always assigned in ascending order, which means for a NAPP deployment the IP Allocations for the VIPs are predictable and deterministic as the sequence will be:

i. vSphere CSI Driver
ii. Supervisor Cluster Control Plane Address
iii. Guest Cluster Control Plane Address
iv. Contour (used by NAPP for Interface Service Access)
v. Kafka External (used by NAPP for Messaging Service Access)

and will not change while the platform is deployed, making this is a safe option. You just need to create a DNS record for the 4th (**Service Name**) and 5th (**Messaging Name**) IP addresses

You can retrieve the currently-allocated external IPs using (where namespace is your vSphere with Tanzu namespace specifically): `kubectl get svc -n <namespace>`

```
rbudavari@liriel:~$ kubectl get svc -n nsx-01
NAME                                  TYPE           CLUSTER-IP    EXTERNAL-IP     PORT(S)            AGE
napp-advanced-01-control-plane-service   LoadBalancer   10.96.0.14    10.198.219.3    6443:30715/TCP     12d
```

b. Using NSX-T networking you can statically allocate these External-IPs using NSX ALB and pre-create DNS entries: https://docs.vmware.com/en/VMware-vSphere/7.0/vmware-vsphere-with-tanzu/GUID-83060EA7-991B-4E1E-BBE4-F53258A77A9C.html

c. Allow HAProxy to assign an IP address from a Pool then use External-DNS to dynamically create the FQDN: https://github.com/kubernetes-sigs/external-dns/

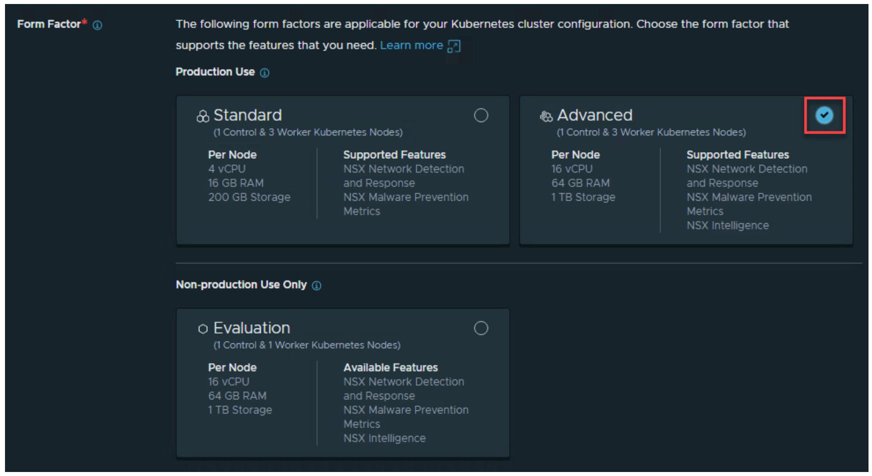
Here is a reference external-dns on vSphere with Tanzu:

https://docs.vmware.com/en/VMware-vSphere/7.0/vmware-vsphere-with-tanzu/GUID-6551DCCF-0ADF-4416-B475-7FF19C4A7636.html

While External-DNS is a good option for dynamic applications, a single NAPP deployment does not require this additional component.

11. One you have verified and entered your Service Name and Messaging Name are resolving correctly, select the appropriate Form Factor for

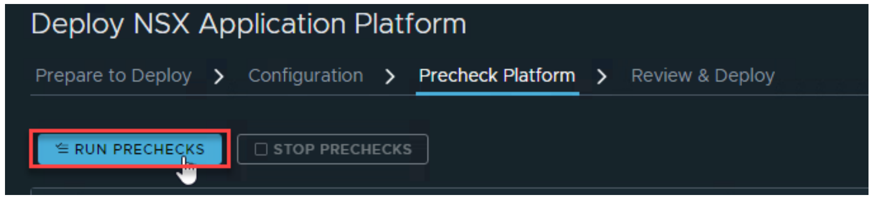
NAPP based on the features you will be running (Advanced in this Deployment Guide, because we are also covering NSX Intelligence):



Based on the node sizes in your environment, you will only be able to select a form factor if sufficient resources have been provisioned.

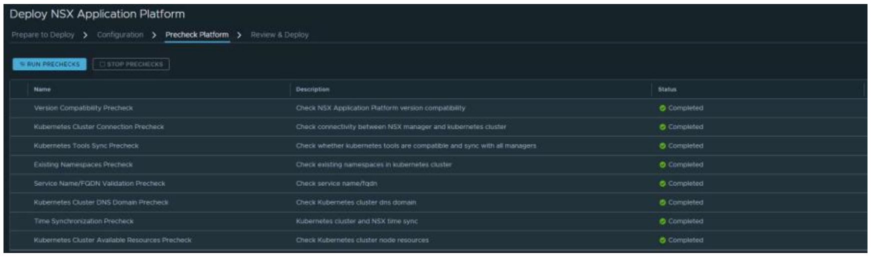
Once you are ready, click **NEXT** to continue the deployment

12. At the **Precheck Platform** screen click on **RUN PRECHECKS**



Your environment will now be validated to ensure you can proceed with NAPP deployment. This will confirm details such as your K8s version, no conflicting namespaces in the cluster, that your Service Name is resolvable and you have provided sufficient cluster resources. If you have closely followed the steps in this guide all pre-checks should return a Status of **Completed**

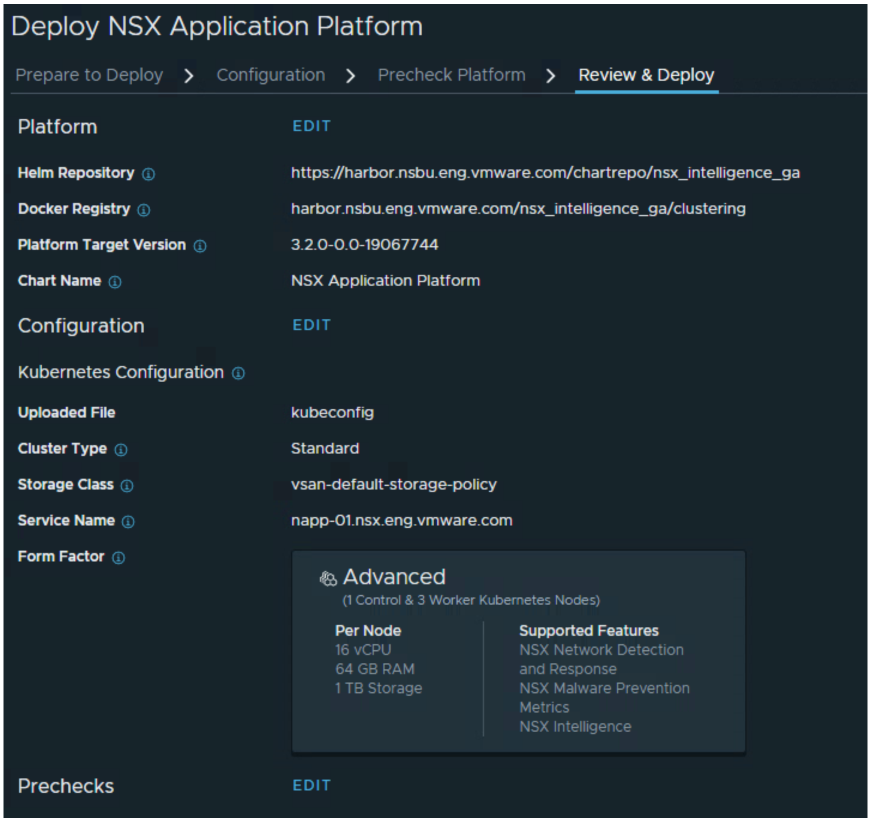
VMware, Inc. 3401 Hillview Avenue Palo Alto CA 94304 USA Tel 877-486-9273 Fax 650-427-5001 www.vmware.com



The prechecks include failure reporting on both **Warnings**, which are not deployment blockers, and critical **Issues** which will block deployment. Warnings can typically be ignored, but Issues must be fixed at this stage and the **PRECHECKS** run again before continuing. If you receive a warning for the **Time Synchronization Precheck** this can safely be ignored as it will not impact NAPP operations.

When your **PRECHECK**s are **Completed** and you are ready, click **NEXT**

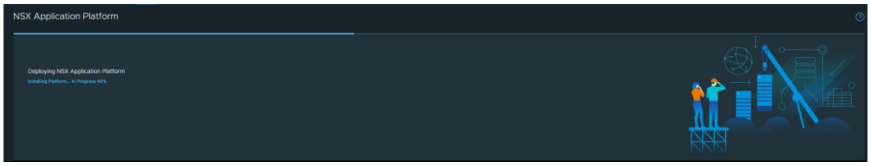
13. At the **Review & Deploy** screen you have a final chance to review all NAPP deployment inputs:

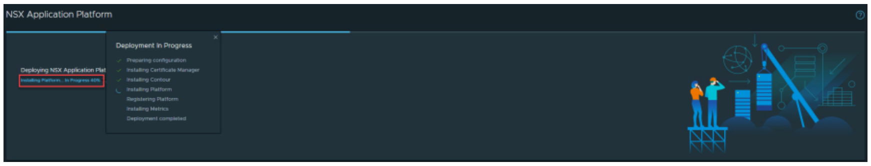


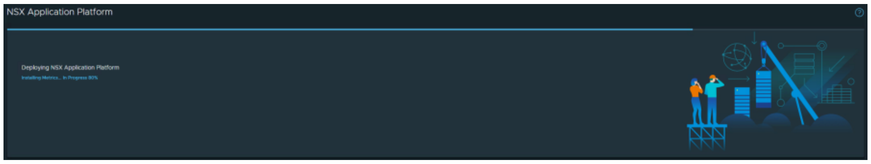
And can make any final changes if required. Once you have verified your settings, click **DEPLOY**

14. At this point the deployment of NAPP is fully orchestrated by NSX Manager, and all services and configuration will be automatically performed. You can track the progress of your deployment and click the highlighted link for step-by-step status information:

These steps include installing a Certificate Manager, Ingress services, all NAPP pods and registering your NAPP instance with NSX Manager:

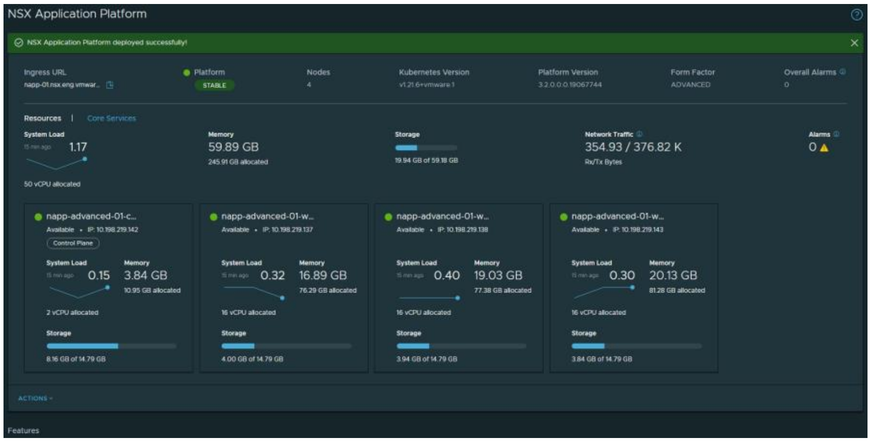




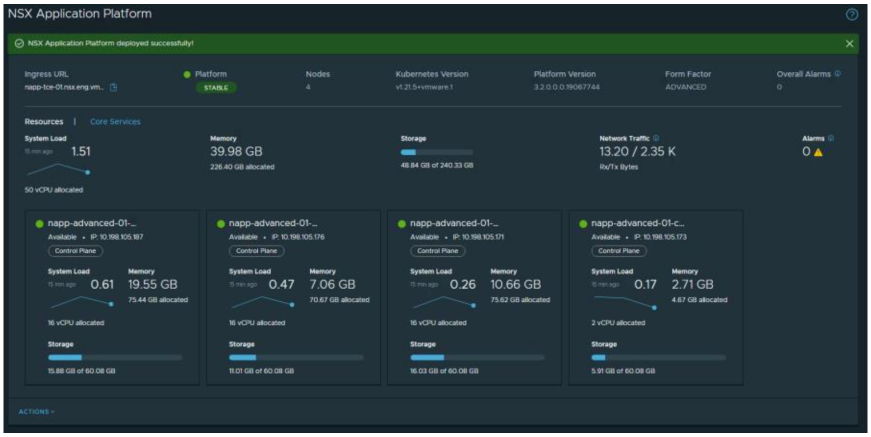


15. Once the install has completed successfully you will be presented with the **NSX Application Platform dashboard**

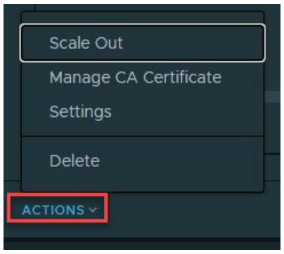
TKGs:



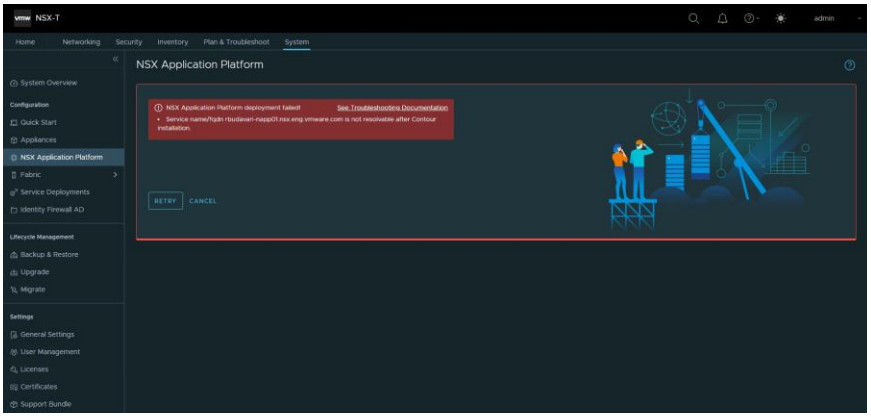
TCE:



From here you can monitor the health of the environment, utilization of resources and status of services, perform **ACTIONS** such as **Scale Out**, **Manage Certificates**, modify deployment **Settings** (updating Repository/Registry details, kubeconfig, etc) or **Delete** NAPP.



16. If for any reason a different VIP address is assigned to the NAPP external service than the one which you have mapped to the **Service Name** in DNS, the install will fail at **40%** while Installing Contour when NSX Manager attempts to access the Interface **Service Name** to connect to NAPP:

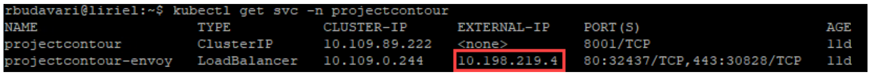


At this time, you can safely retrieve the allocated IP via kubectl then just update the required DNS entry and **RETRY** the deployment.

Use the following command:

```
kubectl get svc -n projectcontour
```

To retrieve the **External-IP** addresses allocated from the Load Balancer VIPs and look for the IP bound to **projectcontour-envoy** and ports **80** and **443**. Then create a DNS entry which maps this IP to your FQDN for the NAPP **Service Name**



Once you have corrected the DNS entry, click **RETRY** on the NSX UI

17. Optional Post-install verification.

If you would like to run additional CLI commands to understand what has been deployed for NSX Application Platform and verify the environment status, a number of reference commands have been provided.

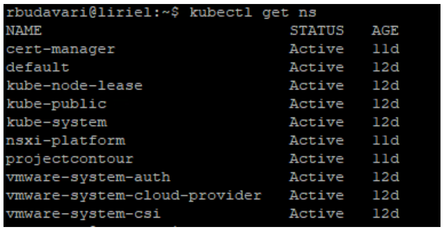
Note: These steps are entirely optional and issues with the underlying Tanzu with vSphere environment will be reported in the NSX UI.

18. If you need to refresh your kubernetes configuration, log in again, specifying the namespace and cluster name:

```
kubectl-vsphere login --server=https://<control-plane-ip> --vsphere-username <vcenter-admin> --insecure-skip-tls-verify --tanzu-kubernetes-cluster-namespace <namespace-name> --tanzu-kubernetes-cluster-name <cluster-name>
```

Start by listing all available namespaces to confirm they are **Active**:

```
kubectl get ns
```



```
rbudavari@liriel:~$ kubectl get ns
NAME                          STATUS   AGE
cert-manager                  Active   11d
default                       Active   12d
kube-node-lease               Active   12d
kube-public                   Active   12d
kube-system                   Active   12d
nsxi-platform                 Active   11d
projectcontour                Active   11d
vmware-system-auth            Active   12d
vmware-system-cloud-provider  Active   12d
vmware-system-csi             Active   12d
```

Verify all pods have a status of **Running** or **Completed**:

```
kubectl get pods -n <namespace>
```

```
rbudavari@liriel:~$ kubectl get pods -n nsxi-platform
NAME                                                READY   STATUS      RESTARTS   AGE
authserver-74f49d45f9-fxdw7                         1/1     Running     0          11d
cluster-api-7d79ffc5c9-qdr9x                        2/2     Running     2          11d
common-agent-7fffd9c7f5-69h4z                       1/1     Running     3          11d
common-agent-create-kafka-topic-jh6hh               0/1     Completed   0          11d
configure-druid-k6ncr                               0/1     Completed   0          11d
create-kubeapi-networkpolicy-job-874v2              0/1     Completed   0          11d
druid-broker-f545d98d5-spvz5                        1/1     Running     1          11d
druid-config-historical-0                           1/1     Running     1          11d
druid-coordinator-697ff5564c-8znfz                  1/1     Running     1          11d
druid-historical-0                                  1/1     Running     1          11d
druid-historical-1                                  1/1     Running     1          11d
druid-middle-manager-0                              1/1     Running     1          11d
druid-middle-manager-1                              1/1     Running     1          11d
druid-middle-manager-2                              1/1     Running     0          11d
druid-overlord-68867fdffd-hbvkz                     1/1     Running     1          11d
fluentd-0                                           1/1     Running     0          11d
kafka-0                                             1/1     Running     0          11d
kafka-1                                             1/1     Running     0          11d
kafka-2                                             1/1     Running     0          11d
metrics-app-server-6c7f5f77f9-sfffr                 1/1     Running     0          11d
metrics-db-helper-7dc8fb55b8-jl6md                  1/1     Running     1          11d
metrics-manager-5c5f5469b7-5trk8                    1/1     Running     0          11d
metrics-nsx-config-79db48d48-pr4x6                  1/1     Running     0          11d
metrics-nsx-config-create-kafka-topic-mr4xh         0/1     Completed   0          11d
metrics-postgresql-ha-pgpool-854f5d55c4-pclll       1/1     Running     2          11d
metrics-postgresql-ha-pgpool-854f5d55c4-tpzbl       1/1     Running     2          11d
metrics-postgresql-ha-postgresql-0                  1/1     Running     0          11d
metrics-postgresql-ha-postgresql-1                  1/1     Running     1          11d
metrics-postgresql-ha-postgresql-2                  1/1     Running     1          11d
metrics-query-server-866f9fcbcb-7qp78               1/1     Running     1          11d
minio-0                                             1/1     Running     0          11d
minio-1                                             1/1     Running     0          11d
minio-2                                             1/1     Running     0          11d
minio-3                                             1/1     Running     0          11d
minio-make-bucket-for-druid-8qd57                   0/1     Completed   0          11d
monitor-8648c78d7f-bw77b                            1/1     Running     0          11d
monitor-create-kafka-topic-hrl2h                    0/1     Completed   0          11d
nsxi-platform-fluent-bit-84fdl                      1/1     Running     0          11d
nsxi-platform-fluent-bit-nnwf8                      1/1     Running     0          11d
nsxi-platform-fluent-bit-vnqq2                      1/1     Running     0          11d
pod-cleaner-27430830-sdwxh                          0/1     Completed   0          11m
postgresql-ha-pgpool-69c4b78d5d-cmsq8               1/1     Running     0          11d
postgresql-ha-pgpool-69c4b78d5d-kqmj8               1/1     Running     0          11d
postgresql-ha-postgresql-0                          1/1     Running     0          11d
redis-master-0                                      1/1     Running     0          11d
redis-slave-0                                       1/1     Running     0          11d
redis-slave-1                                       1/1     Running     0          11d
routing-controller-6c584b4799-nd7dx                 1/1     Running     0          11d
spark-operator-7f7d6957d8-tmjgj                     1/1     Running     0          11d
spark-operator-webhook-init-xpg2c                   0/1     Completed   0          11d
telemetry-844f6446f5-qlv4d                          1/1     Running     0          11d
telemetry-create-kafka-topic-nnlbb                  0/1     Completed   0          11d
trust-manager-698555bdf8-2rj74                      1/1     Running     0          11d
zookeeper-0                                         1/1     Running     0          11d
zookeeper-1                                         1/1     Running     0          11d
zookeeper-2                                         1/1     Running     0          11d
```

View all events for a namespace pod:

```
kubectl get events -n nsxi-platform
```

```
rbudavari@liriel:~$ kubectl get events -n nsxi-platform
LAST SEEN   TYPE     REASON               OBJECT                               MESSAGE
46m         Normal   Scheduled            pod/pod-cleaner-27430800-vz9bx       Successfully assigned nsxi-platform/pod-cleaner-27430800-vz9bx to napp-advanced-01-workers-9g682-5dcddb4c6b-5gkwj
16m         Normal   Created              pod/pod-cleaner-27430830-sdwxh       Created container cleaner
16m         Normal   Started              pod/pod-cleaner-27430830-sdwxh       Started container cleaner
16m         Normal   SuccessfulCreate     job/pod-cleaner-27430830             Created pod: pod-cleaner-27430830-sdwxh
16m         Normal   Completed            job/pod-cleaner-27430830             Job completed
46m         Normal   SuccessfulCreate     cronjob/pod-cleaner                  Created job pod-cleaner-27430800
46m         Normal   SawCompletedJob      cronjob/pod-cleaner                  Saw completed job: pod-cleaner-27430800, status: Complete
46m         Normal   SuccessfulDelete     cronjob/pod-cleaner                  Deleted job pod-cleaner-27430770
16m         Normal   SuccessfulCreate     cronjob/pod-cleaner                  Created job pod-cleaner-27430830
16m         Normal   SawCompletedJob      cronjob/pod-cleaner                  Saw completed job: pod-cleaner-27430830, status: Complete
16m         Normal   SuccessfulDelete     cronjob/pod-cleaner                  Deleted job pod-cleaner-27430800
```

Retrieve more detailed information on a specific pod:

```
kubectl describe pod <pod name> -n <namespace>
```

```
rbudavari@liriel:~$ kubectl describe pod fluentd-0 -n nsxi-platform
Name:         fluentd-0
Namespace:    nsxi-platform
Priority:     0
Node:         napp-advanced-01-workers-9g682-5dcddb4c6b-5gkwj/10.198.219.137
Start Time:   Mon, 14 Feb 2022 16:36:53 -0800
Labels:       allow-traffic-to-dns=true
              app=aggregator
              app.kubernetes.io/component=aggregator
              app.kubernetes.io/instance=nsxi-platform
              app.kubernetes.io/managed-by=Helm
              app.kubernetes.io/name=fluentd
              controller-revision-hash=fluentd-5d99c754bd
              helm.sh/chart=fluentd-3.4.0
              logger=aggregator
              statefulset.kubernetes.io/pod-name=fluentd-0
              vmware/version=3.2.0-0.0-19067763
Annotations:  checksum/config: 06e342f1ccec940f9c085ba023bd72ad7bab769c9ac701a4bdf0352286e9a365
              kubernetes.io/psp: vmware-napp-platform-psp
Status:       Running
IP:           192.168.2.22
IPs:
  IP:           192.168.2.22
Controlled By:  StatefulSet/fluentd
Containers:
  fluentd:
    Container ID:   containerd://432515061d0dca0b135659ade9dd5390e9bd71e001136820bee78cb16c32643d
    Image:          harbor.nsbu.eng.vmware.com/nsx_intelligence_ga/clustering/third-party/fluentd:19067763
    Image ID:       harbor.nsbu.eng.vmware.com/nsx_intelligence_ga/clustering/third-party/fluentd@sha256:151ba8d89421ba8f64c6356ef045190f70671dadc6665919fab79dc2c056112b
    Ports:          24224/TCP, 9880/TCP
    Host Ports:     0/TCP, 0/TCP
    State:          Running
      Started:      Mon, 14 Feb 2022 16:37:44 -0800
    Ready:          True
    Restart Count:  0
    Liveness:       http-get http://:http/fluentd.healthcheck%3Fjson=%257B%2522ping%2522%253A+%2522pong%2522%257D delay=60s timeout=5s period=10s #success=1 #failure=6
    Readiness:      http-get http://:http/fluentd.healthcheck%3Fjson=%257B%2522ping%2522%253A+%2522pong%2522%257D delay=5s timeout=5s period=10s #success=1 #failure=6
    Environment:
      FLUENTD_CONF:  td-agent.conf
      FLUENTD_OPT:   --log-rotate-age 10 --log-rotate-size 31457280
    Mounts:
      /etc/td-agent from fluentd-config (rw)
      /opt/bitnami/fluentd/certs from certs (rw)
      /opt/bitnami/fluentd/logs/buffers from fluentd-buffer (rw)
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-8kvzk (ro)
Conditions:
  Type              Status
  Initialized       True
  Ready             True
  ContainersReady   True
  PodScheduled      True
Volumes:
  certs:
    Type:                  Projected (a volume that contains injected data from multiple sources)
    SecretName:            fluentd-tls-cert
    SecretOptionalName:    <nil>
    SecretName:            fluentd-truststore
    SecretOptionalName:    <nil>
  fluentd-config:
    Type:      ConfigMap (a volume populated by a ConfigMap)
    Name:      fluentd-aggregator-cm
    Optional:  false
  fluentd-buffer:
    Type:       PersistentVolumeClaim (a reference to a PersistentVolumeClaim in the same namespace)
    ClaimName:  shared-supportbundle-buffer
    ReadOnly:   false
  kube-api-access-8kvzk:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    ConfigMapOptional:       <nil>
    DownwardAPI:             true
QoS Class:                   BestEffort
Node-Selectors:              <none>
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:                      <none>
```

View the External IP (Load Balancer VIP) assigned to NAPP:

```
kubectl get svc -n projectcontour
```

```
rbudavari@liriel:~$ kubectl get svc -n projectcontour
NAME                    TYPE           CLUSTER-IP      EXTERNAL-IP     PORT(S)                       AGE
projectcontour          ClusterIP      10.109.89.222   <none>          8001/TCP                      11d
projectcontour-envoy    LoadBalancer   10.109.0.244    10.198.219.4    80:32437/TCP,443:30828/TCP    11d
```

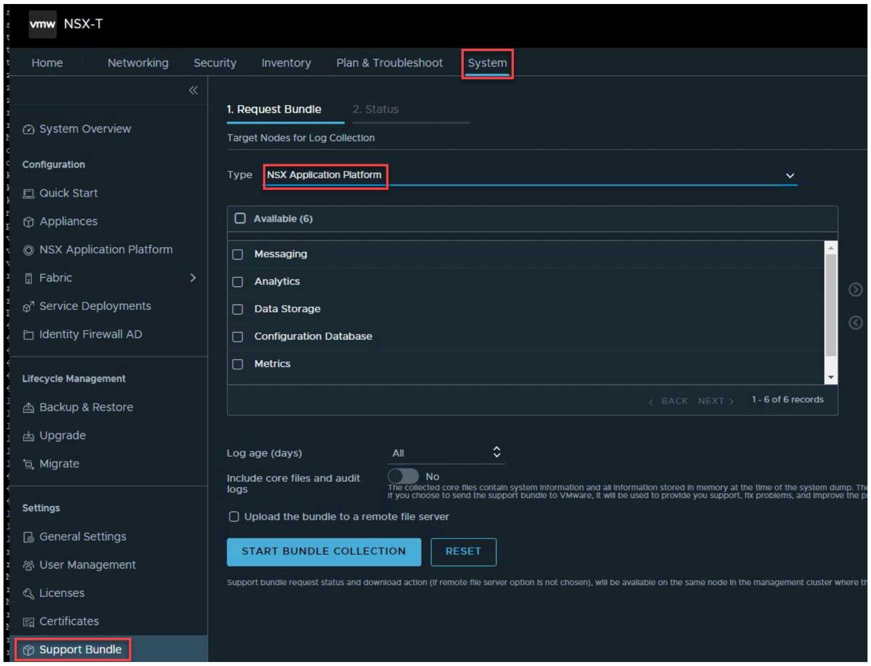
On the NSX Manager, events for registering, connecting and interacting with the NSX Application Platform are available in the manager.log file:

```
get log-file manager.log
```

NAPP errors are also surfaced in the NSX UI, however for reference here is an example of the logging **ERROR** when the Helm Repository is inaccessible and charts cannot be retrieved:

Note: Any further troubleshooting at this point would typically be performed under the direction and guidance of VMware support after collecting a Support Bundle. The Support Bundle can be generated in NSX Manager under:

```
System -> Support Bundle -> Type -> NSX
Application Platform
```



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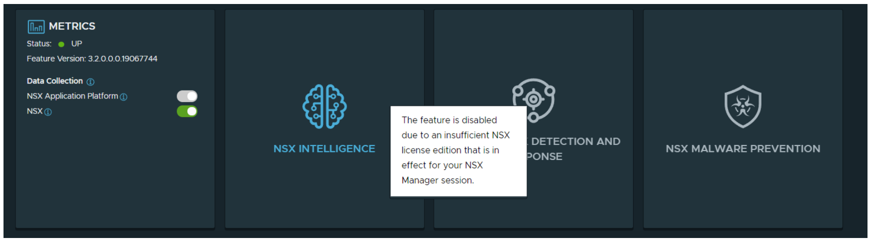
Enabling Features on NSX Application Platform

You are now ready to start enabling features that run on the NSX Application Platform.

Feature enablement is based on your license type and if you do not have the correct licenses assigned, features will remain greyed out and cannot be selected.

Licensing requirements for NAPP are documented at:
https://docs.vmware.com/en/VMware-NSX-T-Data-Center/3.2/nsx-application-platform/GUID-34EC60F7-E335-45AF-A237-727E72E0F1F6.html

Note that the Metrics feature is always enabled once NAPP is deployed (as it is used to monitor NAPP itself) and NSX Intelligence is available to all deployments with NSX Enterprise Plus and Firewall licenses. However, for the Network Detection and Response and Malware Prevention features you will see the message below if you do not have an appropriate Advanced Threat Prevention license:

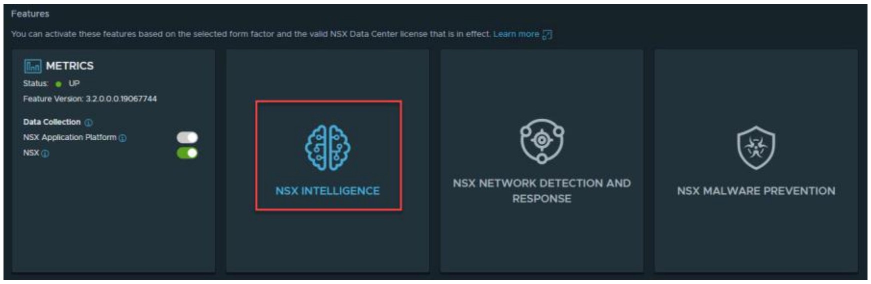


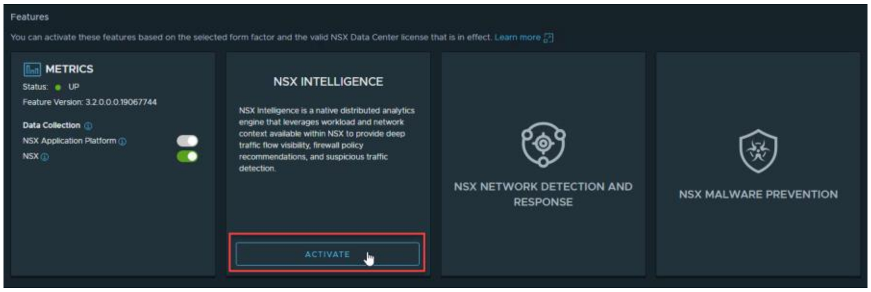
After applying a valid ATP license, you will see all remaining tiles available for enablement:

NSX Intelligence

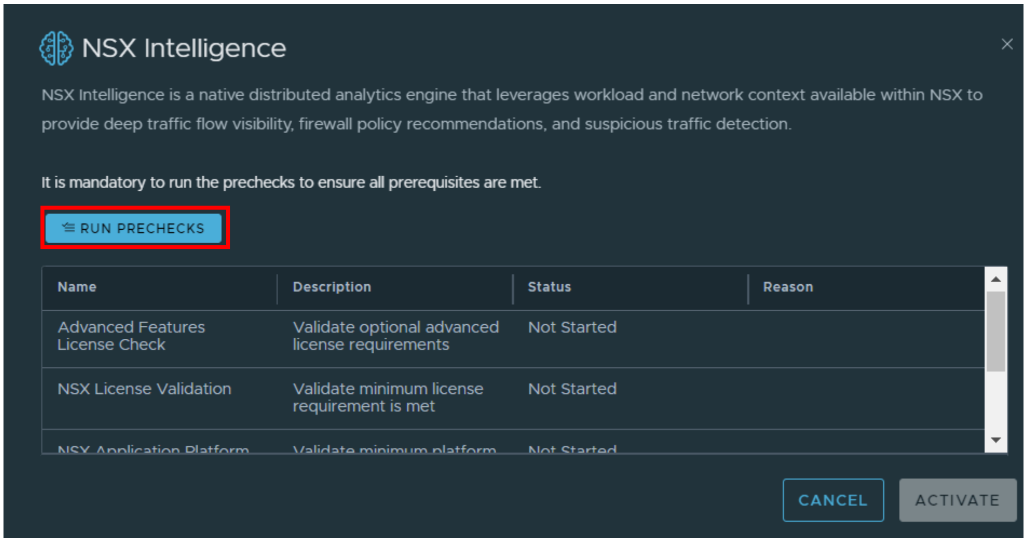
This section will highlight the process to activate NSX Intelligence.

Begin by clicking on **NSX Intelligence** in the NSX Application Platform tile, then followed by **ACTIVATE**:

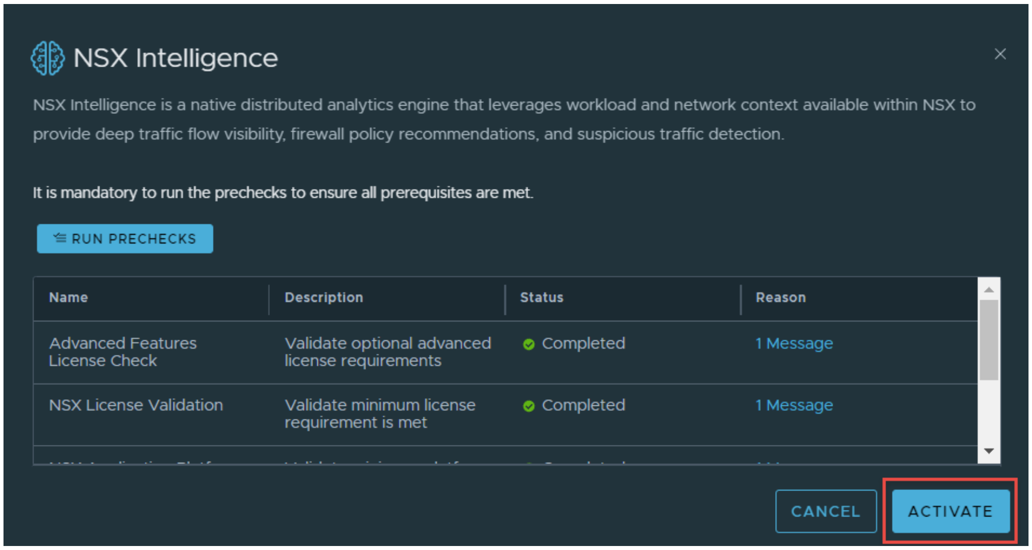




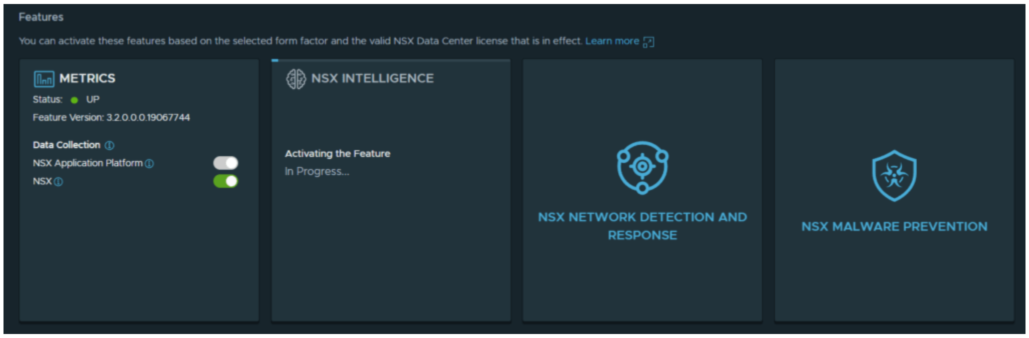
You will next see the precheck screen where you will click on **RUN PRECHECKS** and make sure all checks pass successfully:



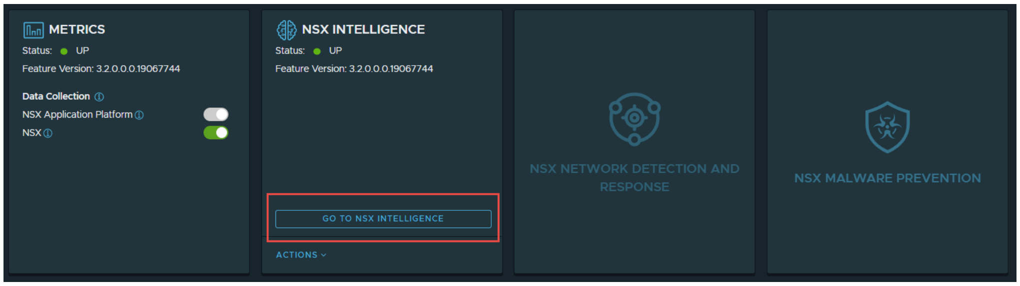
Once the prechecks have passed validation the **ACTIVATE** button will be available and you can install NSX Intelligence.



The time taken to complete activation will depend on the environment, however it can take several minutes to deploy and configure all the required services for the NSX Intelligence Analytics and Data Platform. During this time the display will show In Progress:



And once it has completed successfully the status will be reported as: **UP**

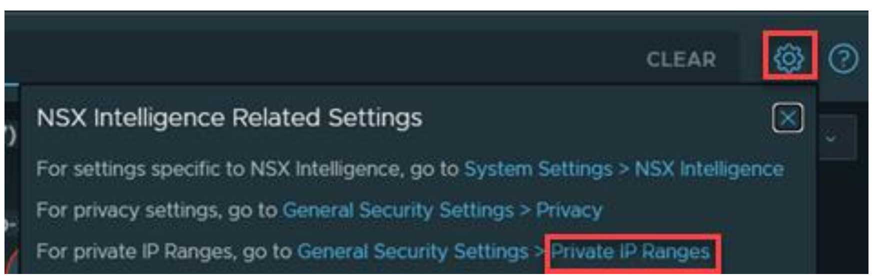


Clicking **GO TO NSX INTELLIGENCE** will redirect you to the **PLAN & TROUBLESHOOT** screen where you can use NSX Intelligence visualizations to view the security posture of your environment and create security policy recommendations to automate your DFW rules, groups and services:

This completes the activation of NSX Intelligence on NAPP.

After installation there are two key settings for NSX Intelligence configuration. The first is **Private IP Ranges**. The RFC1918 addresses are configured as Private IP Ranges by default. Any IP addresses outside of these CIDR ranges that do not resolve to an object in the NSX Inventory are categorized as Public IPs and aggregated. By adding additional CIDRs to Private IP Ranges this will allow you to see these SRC/DST IP addresses within NSX Intelligence visualizations.

In the Plan & Troubleshoot screen of NSX Intelligence, click on the settings cog in the top right and click on the third option for Private IP Ranges



You will then be presented with the following screen where you can add additional IPv4 or IPv6 IP Ranges that are used within your network environment:

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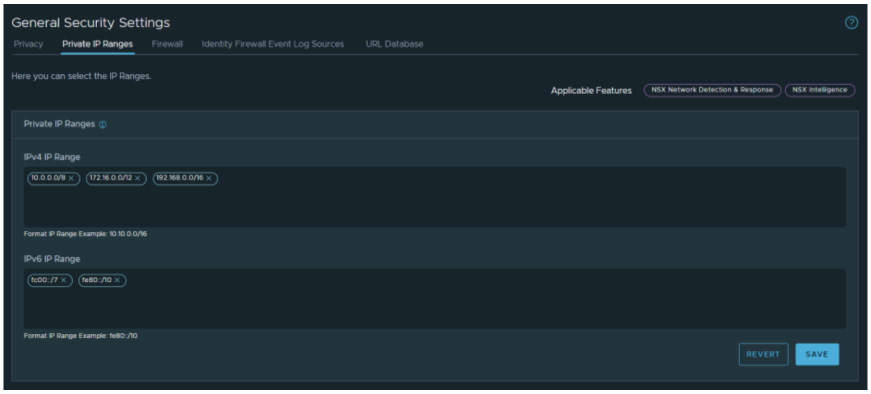
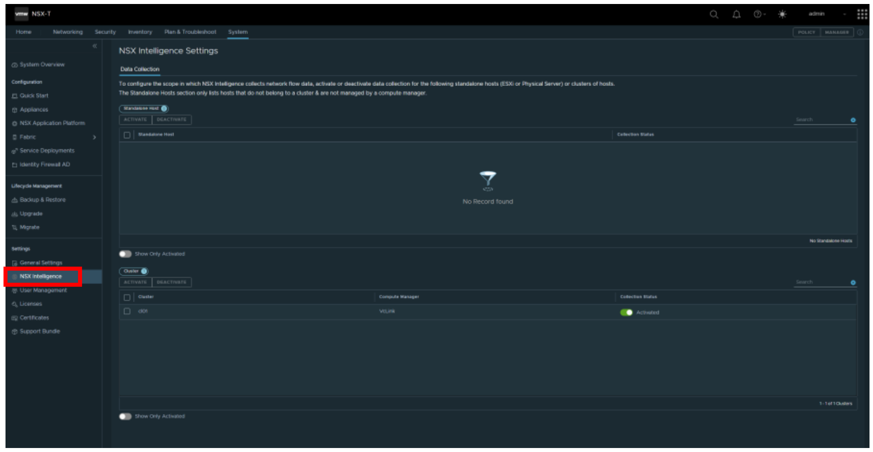


The second post installation configuration setting to note is **Data Collection**.

By default, NSX Intelligence Data Collection is enabled on the clusters/hosts in your environment. If for licensing or scalability reasons you wish to exclude a subset of clusters or hosts from the:

`System -> Settings -> NSX Intelligence` page.



Here you can disable or enable data collection either at the cluster level or an individual host level so these hypervisors will stop streaming data to the NSX Intelligence Data Platform.

Network Detection and Response

This section will highlight the process to activate **NSX Network Detection and Response.**

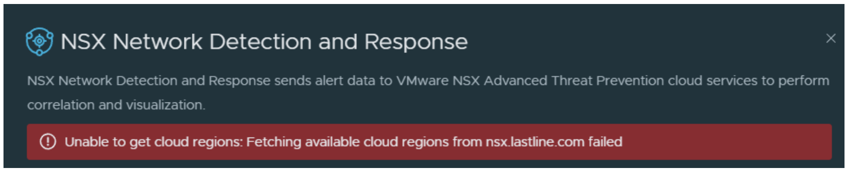
Navigate to: `System -> NSX Application Platform`

Then click on **ACTIVATE** in the NSX Network Detection and Response tile.

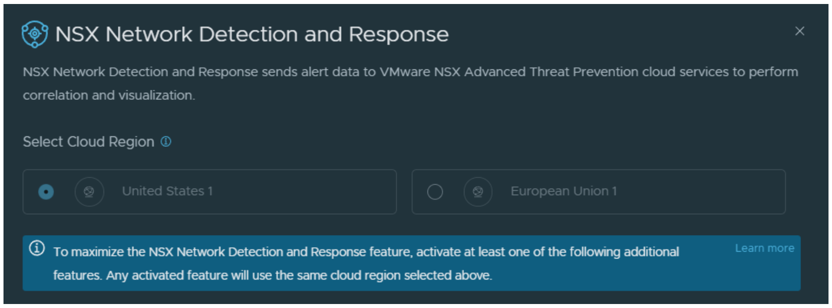
Note: As part of the activation workflow a GET request is sent to:

https://nsx.lastline.com/nsx/cloud-connector/api/v1/papi/accounting/nsx/get_cloud_regions.json

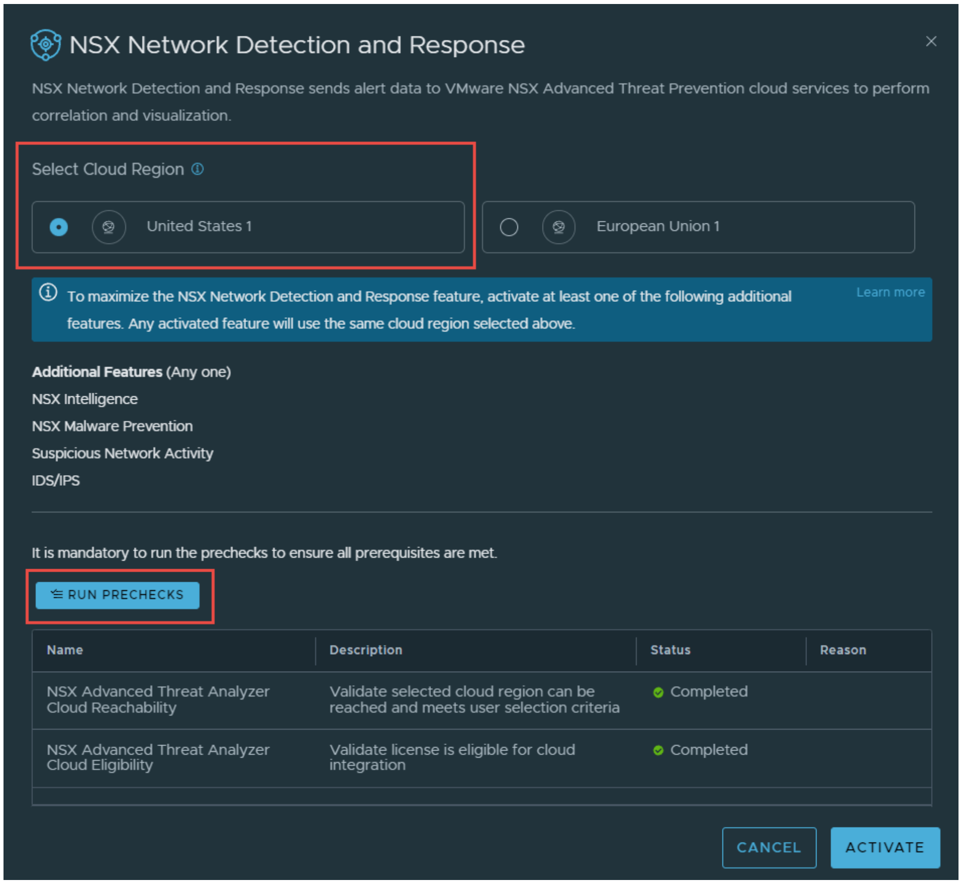
which is used to retrieves the list of available cloud regions to connect to. If your Tanzu guest cluster worker nodes do not have outbound internet access for this request or if it is blocked by a firewall, the following error will be returned:



Once the request is permitted and completes successfully, you will see the below screen:

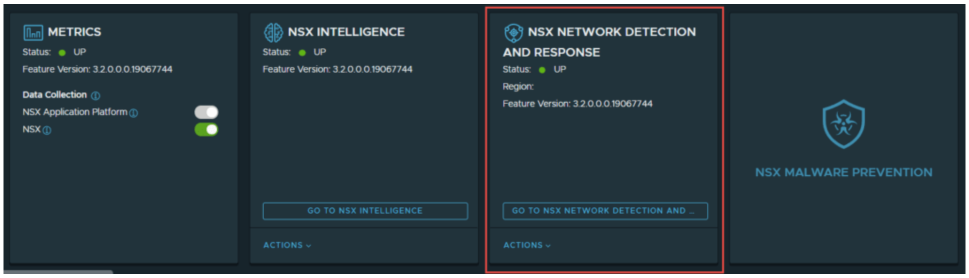


Here you will select your preferred **Cloud Region** based on your location, then click on **RUN PRECHECKS**



Once the prechecks have completed successfully you will be able to continue by clicking **ACTIVATE**.

The deployment of NSX NDR and all related components on NAPP will commence and may take a minute or two, dependent on your environment and network connectivity. Once the deployment has completed the tile will change to return a Status of: **UP**



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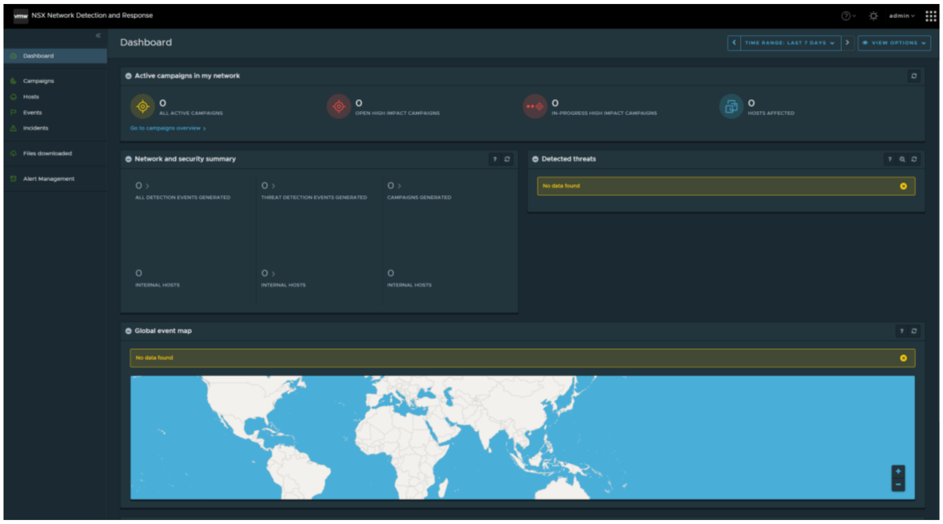
And you can click on **Go To NSX Network Detection and Response** link. This will redirect you to the following screen where you can begin using NSX NDR:

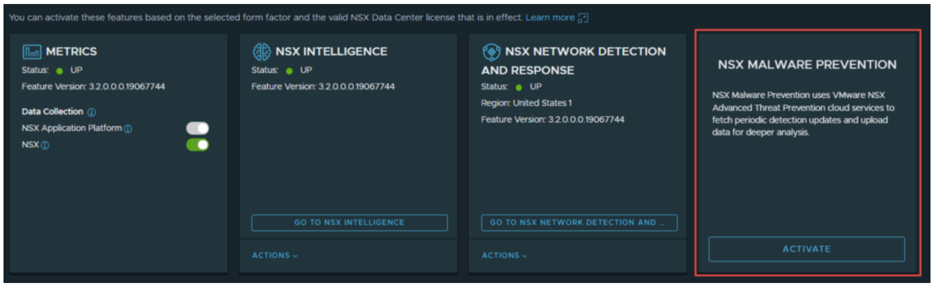


NSX Malware Prevention

Here we will cover the process to enable NSX Malware Prevention. The first stage is enabling the feature, which you begin by navigating to:

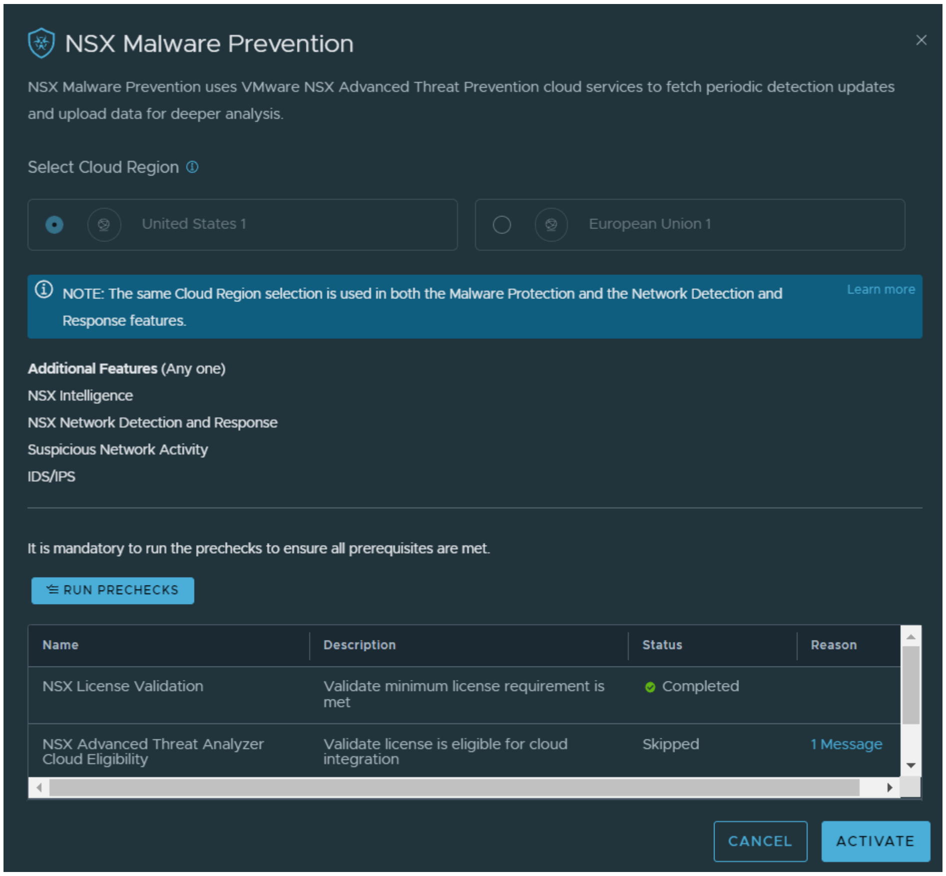
```
System -> NSX Application Platform
```

Then click on **ACTIVATE** in NSX Malware Prevention tile:



Assuming you have already enabled NSX Network Detection and Response the Cloud Region will already be configured, as the same one that was selected for NDR will be used for Malware prevention. Otherwise follow the steps in the NDR section to configure the Cloud Region.

As with other features click on the **RUN PRECHECKS** box and ensure all return a successful status of Completed.



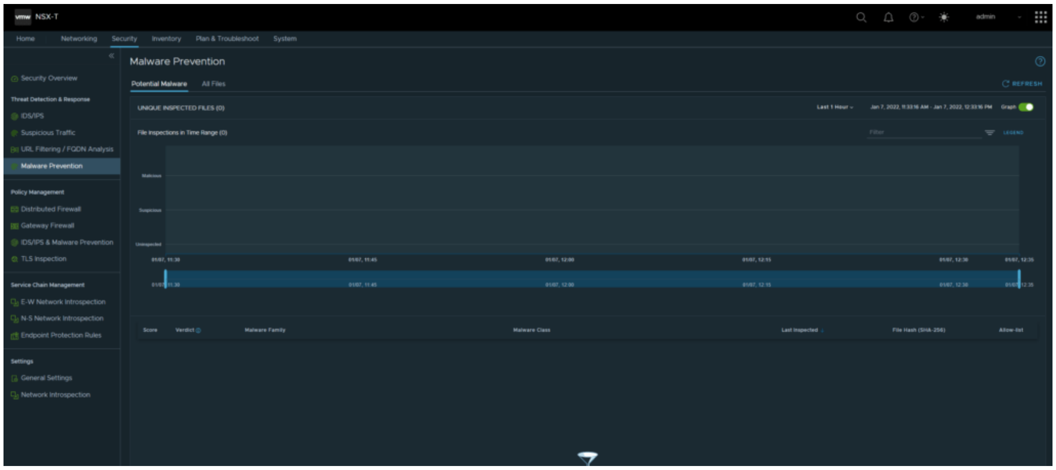
If you have already enabled NDR, one precheck will be skipped which is **NSX Advanced Threat Analyzer Cloud Eligibility**. This is because the component was already deployed during NDR enablement.

Once the prechecks have completed click **ACTIVATE**

Next navigate back to:

```
System -> NSX Application Platform
```

Click on **Go To NSX Malware Prevention** where you will be redirected to the page below to use and configure Malware Prevention

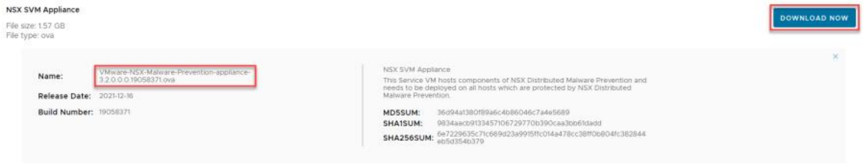


Gateway & Distributed Malware Prevention

The second stage of NSX Malware Prevention is enabling the feature for North-South Traffic and/or East-Est Traffic.

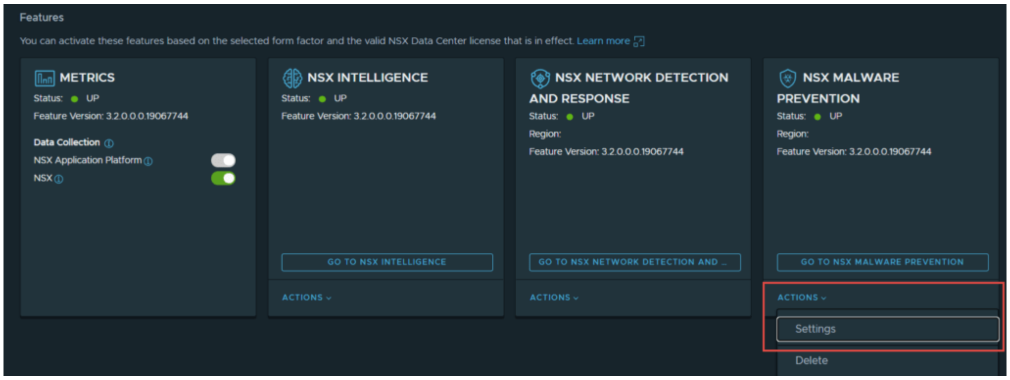
Note that each ESXi hypervisor in a cluster enabled for Distributed Malware Prevention will require an NSX Malware Prevention Service Virtual Machine (SVM).

To prepare for the installation of Malware Prevention of East-West Traffic you must download SVM the OVA from my.vmware.com. It is available under the NSX Product Page, here is an example of the OVA download from the NSX-T 3.2.0.1 release:



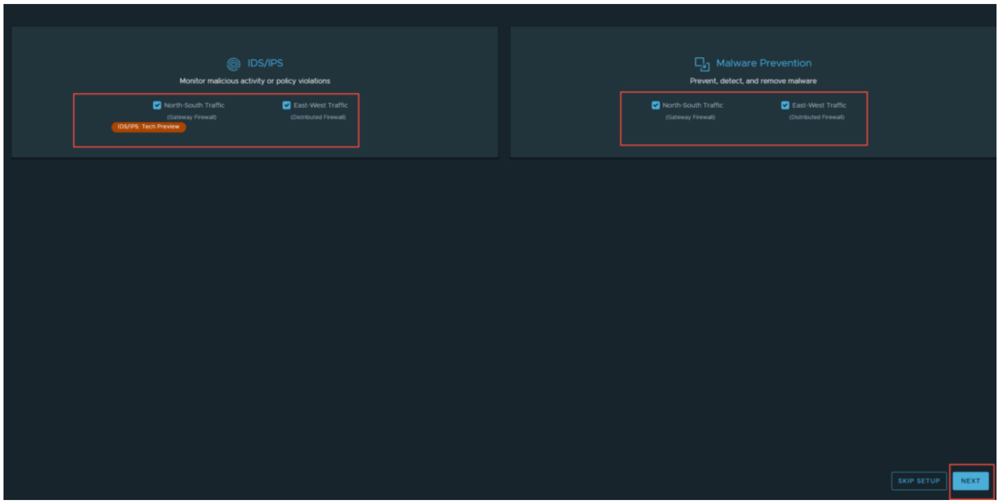
Once you have downloaded the OVA, navigate to:

```
System -> NSX Application Platform -> NSX Malware
Prevention -> Actions -> Settings
```

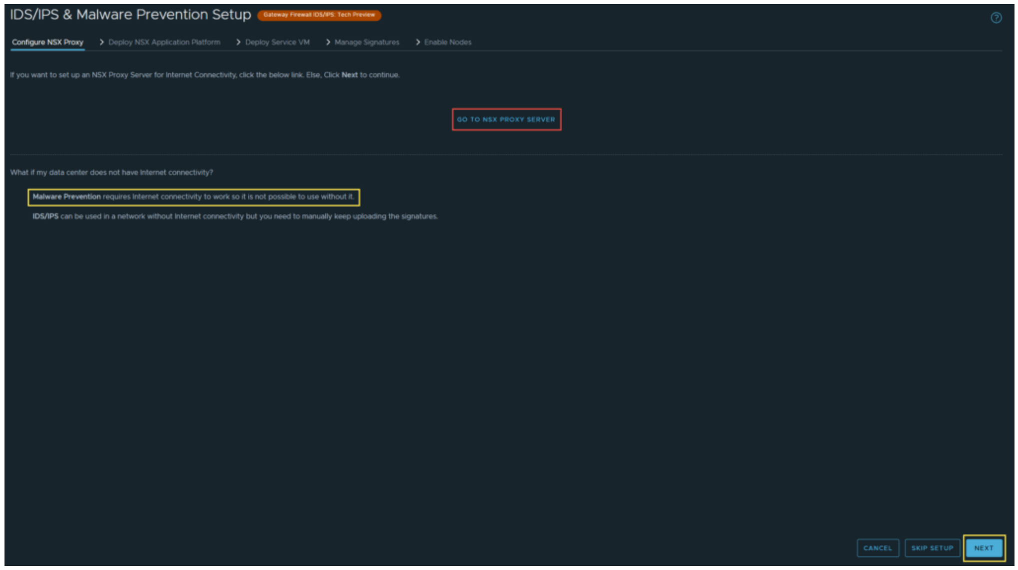


Click **START SETUP**

On the **IDS/IPS & Malware Prevention Setup** screen choose the Malware Prevention features you want enabled (**North-South Traffic** is enabled on the Gateway Firewall, while **East-West Traffic** is Distributed at the hypervisor). Enable your options and click **NEXT** when you are ready.

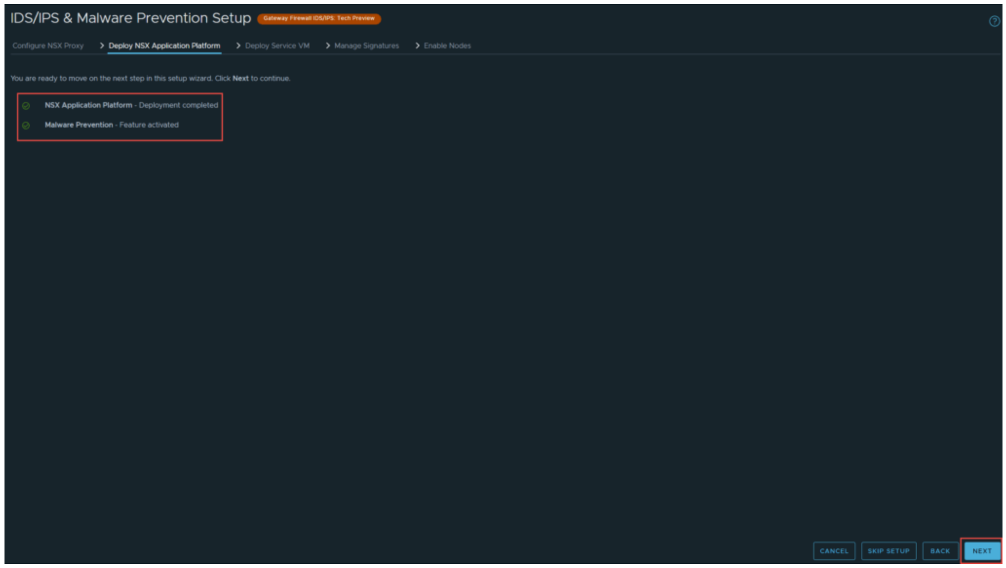


If you need to setup a proxy server for outbound internet connectivity, select **GO TO PROXY SERVER**, if not click **NEXT**



Note: As the highlighted message indicates, currently Malware Prevention only works in environments with outbound internet connectivity.

There is a validation to ensure that both the NSX Application Platform is deployed and the Malware Prevention feature is configured and enabled. If these dependencies are not completed you will receive an alert. Otherwise, if you have followed this guide and completed deployment you can click **NEXT**



Next you are required to deploy the SVM that was downloaded earlier. Click on **GO TO SERVICE DEPLOYMENT** once you have completed the following pre-requisites:

- Generate Public-Private Key Pair for SSH Access to SVMs.

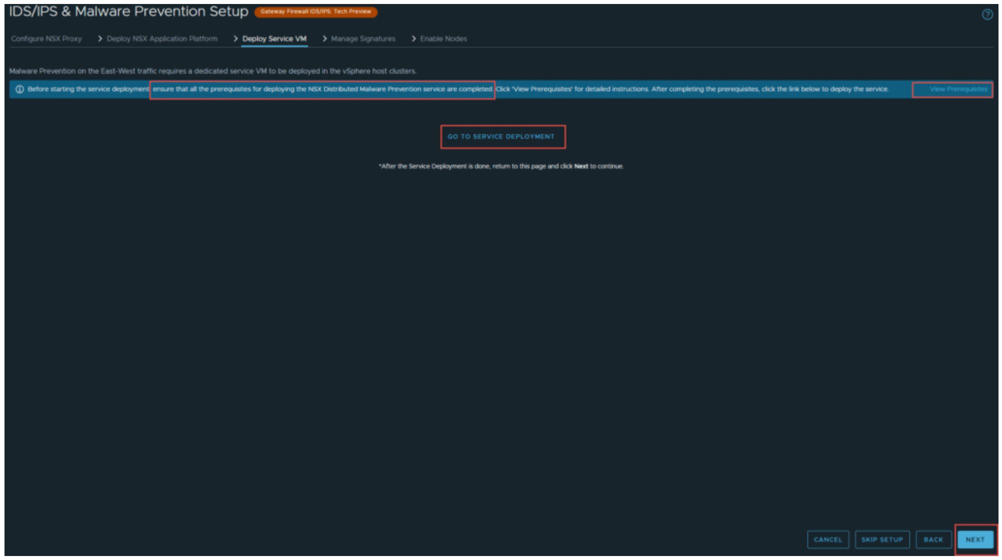
  This step is required to access logs on the SVM for troubleshooting (which is performed via SSH). You can generate the public-private key pair by using any standards based SSH key generation tool eg, ssh-keygen from OpenSSH or PuTTY Key Generator. Here are two references that will guide you through generating a Public-Private Key pair for the SVM:

  Create SSH Keys with OpenSSH
  Create SSH Keys with PuTTYgen

- Extract files from the OVA using tar/unzip

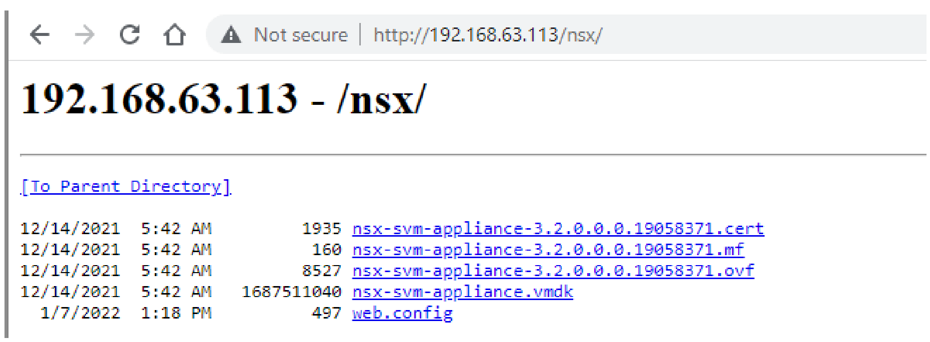
- Host the SVM files (the ovf/mf/vmdk/cert files inside the OVA) on a webserver using unauthenticated access over HTTP. This can be an existing webserver in your environment or if required you can deploy a new webserver temporarily. This webserver will be used later in the configuration process when you have to issue a POST command to register the appliance. Apache and IIS are two common webservers, but any standards-based server will work. Also ensure you have configured the required MIME types to support these file types. Refer to the following documentation for more information if required:

  NSX Malware Prevention Webserver Setup



Now you need to register the NSX Distributed Malware Prevention Service. The webserver in this example is **192.168.63.113**. A virtual directory called **nsx** has been created and the files from the NSX Malware Prevention OVA archive have been uploaded to this directory.

For reference below is the result of browsing to the webserver directory:



The POST request used to register the Malware Prevention SVM is shown below, ensure you use basic authentication and use the admin user and password. Change the password to suit your environment. Also, while the example uses CURL any REST API client can be used:

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```
curl --insecure -XPOST -H "Content-type: application/json" -d '{    "ovf_url" :  "http://{webserver-ip}/{path-to-ovf-file}/nsx-svm-appliance-3.2.0.0.0.19058371.ovf",
   "deployment_spec_name" : "Dist_Malware",
   "svm_version" : "3.2"
}' 'https://{nsx-manager-ip}/napp/api/v1/malware-prevention/svm-spec' -u "admin:{nsx-manager-admin-password}"
```

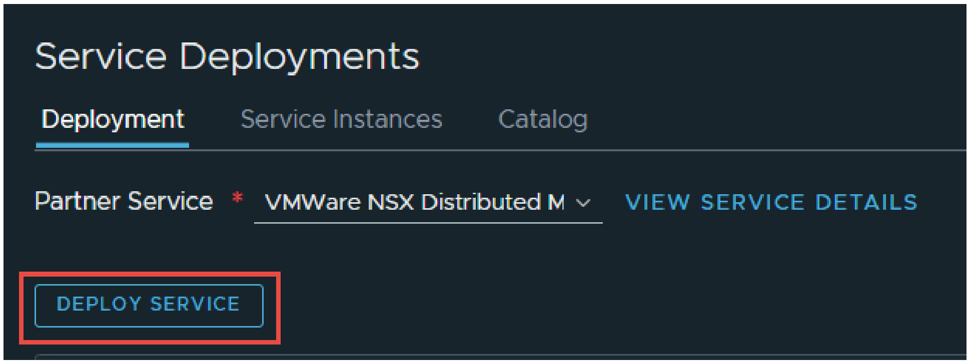
Note: Replace the values in braces with the appropriate IPs, Paths and Passwords for your environment.

Once the API request has completed successfully navigate back to:

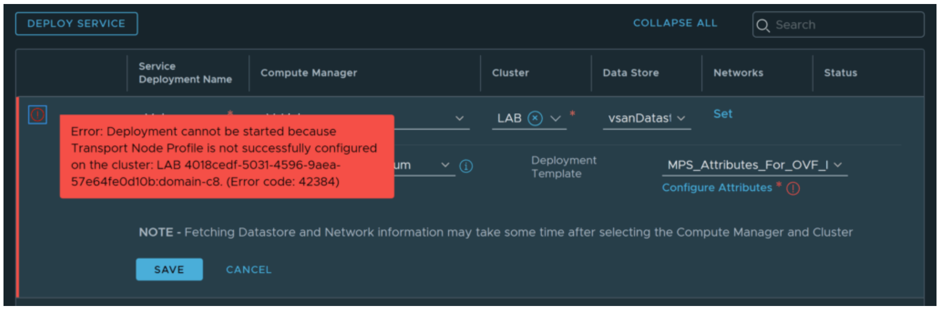
`System -> Service Deployments`

Note: You may have to refresh or log out and back in, but you should now see the partner service registered.

Click **DEPLOY SERVICE** when ready



Note: You will need to have a Transport Node Profile attached to the Transport Node cluster, before you are able to deploy the SVM's. If you don't have a Transport Node Profile attached to the cluster, you will see the below error.



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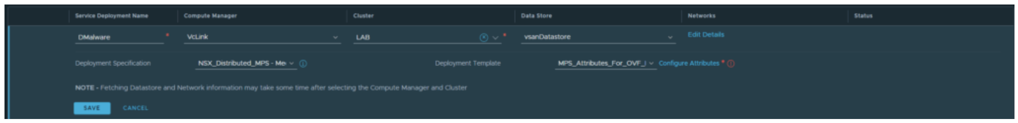
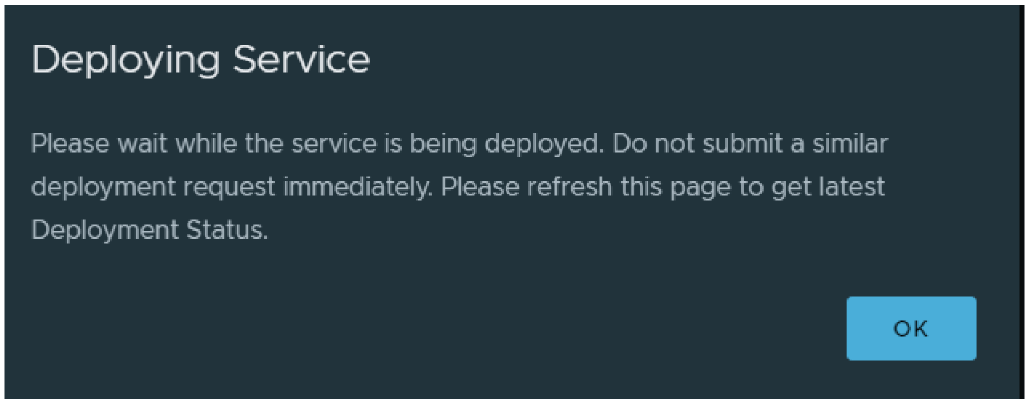
You will need to enter the Service Deployment details relevant to your environment in the fields below such as Compute Manager, Cluster, Datastore, Networks, etc. When ready click **SAVE**



And the Malware Prevention Service deployment will begin:

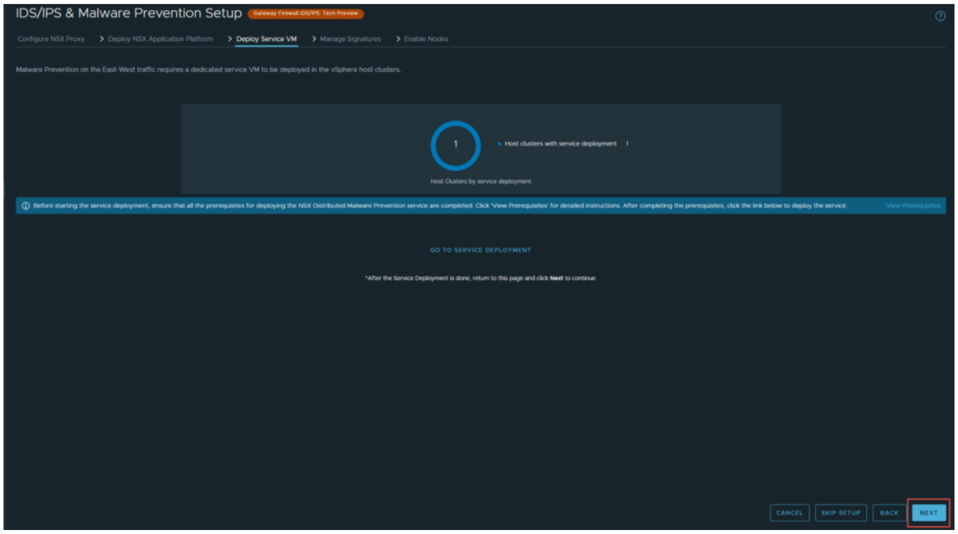


The SVMs will be deployed to each host in the designated cluster.

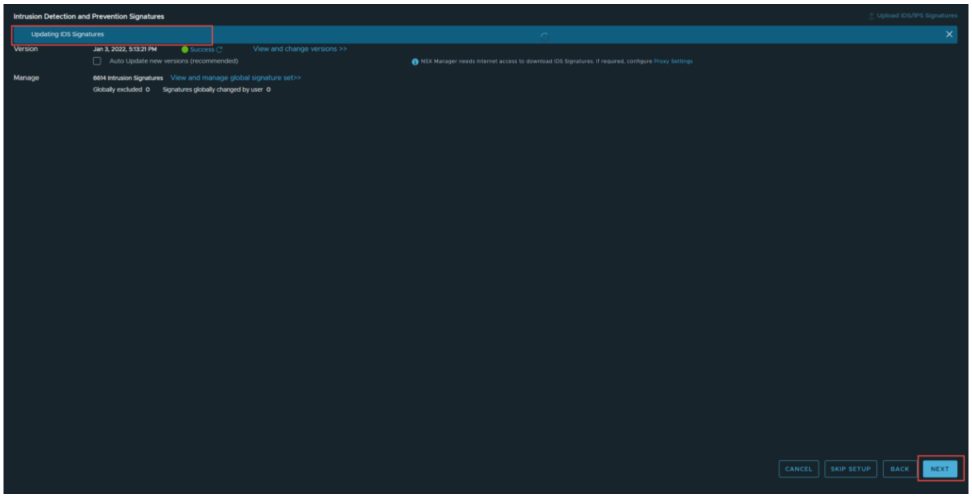
Navigate back to:

```
Security -> IDS/IPS & Malware Prevention
```

Now that the SVM's have been registered and deployed, you should see the output below where you can click **NEXT** to continue:



This page defines the Malware Prevention and IDS/IPS signatures. You can update to the latest signatures if there are any updates available. Click **NEXT** when done.



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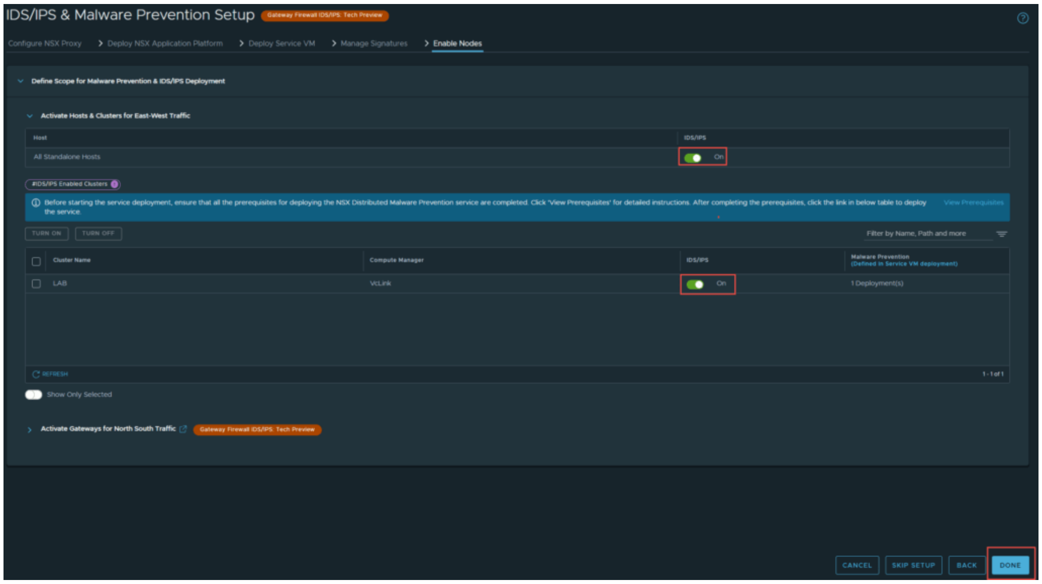
The final step is to **Activate Hosts & Clusters** for IDS & IPS and Malware Prevention. the nodes. Once you have enabled services on the appropriate hosts click **DONE** to complete the Malware Prevention configuration.



Additional Troubleshooting

```
root@nsx-demo-sc-01:~# kubectl get ns --kubeconfig=/config/vmware/napps/.kube/config
NAME                          STATUS   AGE
cert-manager                  Active   27m
default                       Active   42h
kube-node-lease               Active   42h
kube-public                   Active   42h
kube-system                   Active   42h
nsxi-platform                 Active   25m
projectcontour                Active   27m
vmware-system-auth            Active   41h
vmware-system-cloud-provider  Active   41h
vmware-system-csi             Active   41h
```

```
root@nsx-demo-sc-01:~# kubectl get ns --kubeconfig=/config/vmware/napps/.kube/config
NAME                          STATUS   AGE
cert-manager                  Active   2m34s
default                       Active   41h
kube-node-lease               Active   41h
kube-public                   Active   41h
kube-system                   Active   41h
vmware-system-auth            Active   41h
vmware-system-cloud-provider  Active   41h
vmware-system-csi             Active   41h
root@nsx-demo-sc-01:~# kubectl get pods -A --kubeconfig=/config/vmware/napps/.kube/config
NAMESPACE                     NAME                                                            READY   STATUS             RESTARTS   AGE
cert-manager                  cert-manager-5944d55cf5-vf6zw                                   0/1     ImagePullBackOff   0          2m40s
cert-manager                  cert-manager-cainjector-744b6664cc-7xjhl                        0/1     ErrImagePull       0          2m40s
cert-manager                  cert-manager-startupapicheck-hjk2l                              0/1     ErrImagePull       0          2m37s
cert-manager                  cert-manager-truststoreinjector-5cd9f7bbf5-xg6lj                0/1     ImagePullBackOff   0          2m40s
cert-manager                  cert-manager-webhook-78998ffd64-s984d                           0/1     ImagePullBackOff   0          2m40s
kube-system                   antrea-agent-7twzg                                              2/2     Running            0          41h
kube-system                   antrea-agent-c9sl5                                              2/2     Running            0          41h
kube-system                   antrea-agent-fm5v7                                              2/2     Running            0          41h
kube-system                   antrea-agent-r946b                                              2/2     Running            0          41h
kube-system                   antrea-controller-7fdf5d6b46-7xjzl                              1/1     Running            0          41h
kube-system                   antrea-resource-init-54b68499d-9pr92                            1/1     Running            0          41h
kube-system                   coredns-df8897b76-nv2dw                                         1/1     Running            0          41h
kube-system                   coredns-df8897b76-pz6p2                                         1/1     Running            0          41h
kube-system                   docker-registry-napp-advanced-01-control-plane-2m9bx            1/1     Running            0          41h
kube-system                   docker-registry-napp-advanced-01-workers-9g682-5dcddb4c6b-5gkwj 1/1     Running            0          41h
kube-system                   docker-registry-napp-advanced-01-workers-9g682-5dcddb4c6b-ftqcj 1/1     Running            0          41h
kube-system                   docker-registry-napp-advanced-01-workers-9g682-5dcddb4c6b-w4x8q 1/1     Running            0          41h
kube-system                   etcd-napp-advanced-01-control-plane-2m9bx                       1/1     Running            0          41h
kube-system                   kube-apiserver-napp-advanced-01-control-plane-2m9bx             1/1     Running            0          41h
kube-system                   kube-controller-manager-napp-advanced-01-control-plane-2m9bx    1/1     Running            0          41h
kube-system                   kube-proxy-9f67p                                                1/1     Running            0          41h
kube-system                   kube-proxy-g8b5f                                                1/1     Running            0          41h
kube-system                   kube-proxy-hcpp9                                                1/1     Running            0          41h
kube-system                   kube-proxy-s8mls                                                1/1     Running            0          41h
kube-system                   kube-scheduler-napp-advanced-01-control-plane-2m9bx             1/1     Running            0          41h
kube-system                   metrics-server-8486f89fbf-46fkd                                 1/1     Running            0          41h
vmware-system-auth            guest-cluster-auth-svc-lsv4h                                    1/1     Running            0          41h
vmware-system-cloud-provider  guest-cluster-cloud-provider-6858b8776c-t9hxp                   1/1     Running            0          41h
vmware-system-csi             vsphere-csi-controller-55bb79b978-x8w7q                         6/6     Running            0          41h
vmware-system-csi             vsphere-csi-node-ccf5j                                          3/3     Running            0          41h
vmware-system-csi             vsphere-csi-node-n547p                                          3/3     Running            0          41h
vmware-system-csi             vsphere-csi-node-wkqz6                                          3/3     Running            0          41h
vmware-system-csi             vsphere-csi-node-xdd9v                                          3/3     Running            0          41h
```

Data Migration from NSX Intelligence 1.2

If you are running NSX-T 3.1 and NSX Intelligence 1.2 you can either upgrade to NSX Intelligence 3.2 and retain your historical flow data or start with a new deployment If you chose to migrate data, please refer to the process documented at the following URL:

https://docs.vmware.com/en/VMware-NSX-Intelligence/3.2/install-upgrade/GUID-6640213F-098A-4C8E-82F6-03F3B0E2828F.html

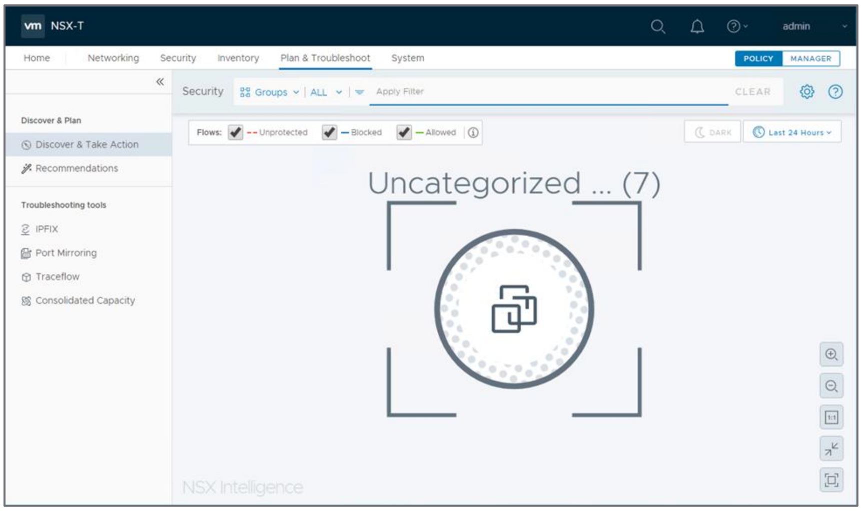
Getting started with NSX Intelligence

You can access the NSX Intelligence home page by clicking **Plan & Troubleshoot** > **Discover & Take Action** within the NSX Manager UI.

After you install and configure NSX Intelligence for the first time, when you click **Discover & Take Action**, NSX Intelligence will display visualizations. These visualizations are based on flow and additional context data that has been received from NSX Transport Nodes and configuration data received from NSX Manager. NSX Intelligence will correlate both sources of information prior to visualizing them.

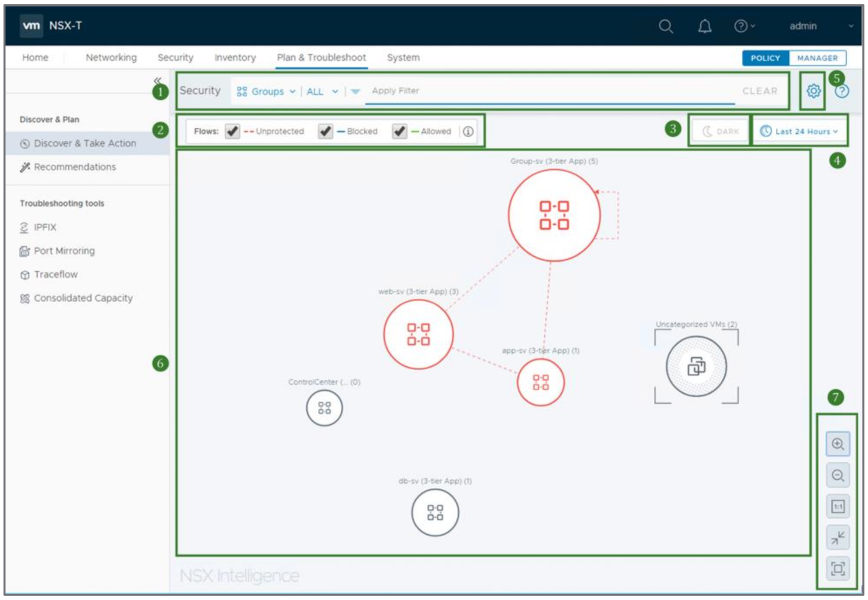
By default, you can see the visualization of the security status of all the groups in your on-premise NSX-T Data Center environment that has allowed, blocked, and unprotected traffic flows between VM, Physical Server or IP Addresses in the last 24 hours.

If there are VMs, but they do not belong to any groups, you will see the icon for the Uncategorized VMs group, as shown in the following screenshot.



If you already have groups defined and network traffic data has been captured, you will see a visualization of their communication.

The default view that is shown in the NSX Intelligence home page is the Groups view. The table that follows describe the numbered sections in the screenshot.



NSX-T
admin
Home
Networking
Security
Inventory
Plan & Troubleshoot
System
POLICY
MANAGER
Security
Groups
ALL
Apply Filter
CLEAR
Discover & Plan
Discover & Take Action
Recommendations
Troubleshooting tools
IPFIX
Port Mirroring
Traceflow
Consolidated Capacity
Flows:
Unprotected
Blocked
Allowed
DARK
Last 24 Hours
NSX Intelligence

Table 8. NSX Intelligence Section Description

Section	Description
1	The Security view section area is where you select the type of security visualization to display. There are two types of Security view available: **Groups** and **VMs**. To select the type, click the down arrow next to **Groups** or **VMs**. Select the specific groups or VMs to include in the view, click the down arrow next to **ALL** and select the specific groups or VMs. To clear your selection filters, click **CLEAR** on the right of the Security view section. From the drop-down list of the **Apply Filter**, you can select the criteria to use for the visualization, such as VM members, tags, flow types, source IP, destination IP, rule ID or name. You can define multiple filters to apply by clicking **Apply Filter** again.
2	The Flows section is where you can select which traffic flow type to include in the visualization during the selected time period. The color used in the visualization for the flow types are also shown in this section: a **red**-hued dashed line for Unprotected flows, solid **blue**-hued line for Blocked flows, and solid **green**-hued line for the Allowed flows. By default, the Unprotected traffic flow type is selected for the current NSX Intelligence visualization.
3	There are two options for the Display mode, Light theme and Dark theme. Light theme is the default mode used. To use the dark theme mode, go into full screen mode, click **full-screen** icon in the viewing control section and click the **DARK** icon.
4	In this section, you select the time period to use to determine which network flow data is used to generate the desired visualization and recommendation. To change the selected time period, click the currently-selected time period and select **Last 1 hr**, **Last 12 hrs**, **Last 24 hrs**, **Last 1 week**, or **Last 1 month**. Last 24 hours is the default time range used.
5	When you click the gear icon, the **Private IP Range Settings for NSX Intelligence** dialog box is displayed. NSX Intelligence categorizes an IP address belonging to one of the CIDR notations listed in the dialog box as a private IP address. Any IP address that does not belong to any of these CIDR notations is classified as a public IP address. If your

	VM's IP address does not fall into one of these CIDR notations, consider adding your CIDR notation using this dialog box.
6	This section is the visualization of the security status of the Groups or VMs in your on-premise NSX-T Data Center, and the network traffic flows that occurred during the selected time period. In this section, you can point to a node or flow arrow to obtain details about that entity. For more information, see the next part of this guide or see Getting Familiar with NSX Intelligence Graphic Elements and Understanding Views and Flows.
7	This section includes the viewing controls to **Zoom in**, **Zoom out**, apply **1:1 aspect ratio**, resize to **Fit To View**, and go into or out of **Full Screen** viewing mode.

For more details, refer to Using and Managing VMware NSX Intelligence for Understanding NSX Intelligence View and Flows and Working recommendations

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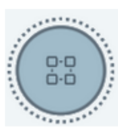
NSX Intelligence Visualizations

The NSX Intelligence user interface provides several graphic elements to help with the visualization of the data center entities, traffic flows, and certain activities in your NSX-T Data Center environment.

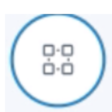
Table 9.NSX Intelligence Graphic Element and Description

Graphic Element	Description
	**Regular Group**. This icon represents any collection of NSX object in your NSX-T Data Center environment. Those NSX objects are VMs only and so NSX Intelligence supports Regular Groups with only VM member types. An NSX object can belong to more than one group and so a VM can appear in more than one group node.
	**Uncategorized Group**. This icon is used for the group of VMs that do not belong to a group.
	**Unknown Group**. This icon represents a set of miscellaneous objects that were not found in your NSX-T Data Center inventory. However, these objects are communicating to one or more NSX objects in your NSX-T Data Center environment.
	**Public IPs Group** or **Public IP Node**. This icon represents the public IP address, either an IPv4 or IPv6 , that is communicating to or from your NSX-T Data Center environment.
	**Regular VM Node**. This icon represents a virtual machine (VM) that is part of your NSX-T Data Center. A VM can belong to more than one group.
	**IP Node**. An IP address, such as a unicast, broadcast, or multicast IP, that participated in the network traffic activities during the selected time period.

  	An arrow represents a network traffic flow that occurred between two VMs or between a VM and any IP during a selected time period. A dashed red-hued arrow indicates that the system detected that the traffic flow encountered a rule (Source: Any | Destination: Any | Action: Allow or Reject or Drop) and that more granular security policies are needed. A solid blue-hued arrow indicates that the system detected that the traffic flow encountered a 'Reject' or 'Drop' rule that is more granular than the one mentioned in the 'Unprotected' flow definition. A solid green-hued arrow indicates that the system detected that the traffic flow encountered an 'Allow' rule that is more granular than the one mentioned in the 'Unprotected' flow definition.
	A node that has been selected as the current node in focus is surrounded with a dashed circle. It is the pinned node during the selection mode and the current view being displayed.

The color of the node's border indicates the types of traffic flows that have occurred with the VMs belonging to that group.

Table 10.NSX Intelligence Node Type and Description

Node Type	Description
	A node with a red-hued border indicates that at least one unprotected flow occurred with a VM in the group or a VM, regardless of how many blocked or allowed flows were detected during the selected time period.
	A blue-hued border on a node means that no unprotected traffic flows were detected, but at least one blocked flow was detected, regardless of how many allowed flows were detected during the selected time period.

	A node with a green-hued border indicates that there were no unprotected or blocked flows detected during the selected time period, and at least one allowed flow was detected.
	A node with a gray-hued border means that there were no traffic flows detected for the VMs belonging to that group during the selected time period.

For more details, refer to Getting Started with NSX Intelligence and Understanding NSX Intelligence View and Flows

Working with NSX Intelligence Recommendations

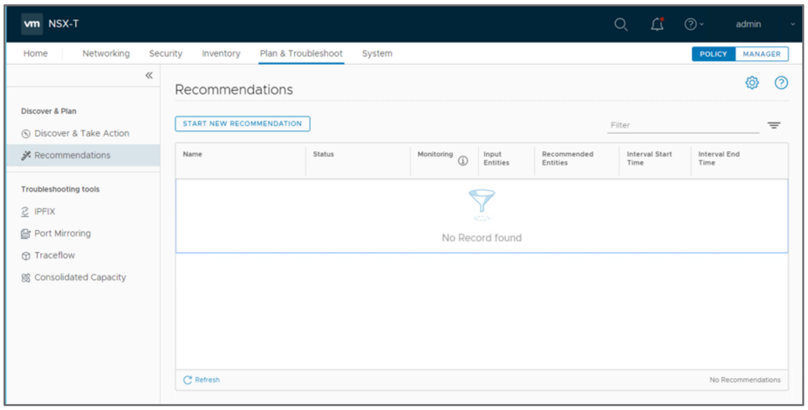
NSX Intelligence can provide micro-segmentation recommendations that are based on the patterns of the real network traffic flows that have occurred between the VMs in your NSX-T Data Center environments during a selected time period.

The recommendations that NSX Intelligence generates cover Distributed Firewall policy rules, security groups, and services for applications.

The security policy recommendations are of the East-West Distributed Firewall Security Policies of Application category. The policy security group recommendations consist of a list of VMs that are seen in the network traffic flows that were analyzed for the time period and the VM boundary you had specified. The service recommendations are service objects that were used in certain ports by applications in the VMs you had specified, but the services are not yet defined in the NSX-T Data Center inventory.

Creating Recommendations

1. As one of the options to start a recommendation, select **Plan & Troubleshoot** > **Recommendations** tab and click **START NEW RECOMMENDATION**.



2. In the Start New Recommendation wizard, optionally change the default value for the Recommendation Name and Description.

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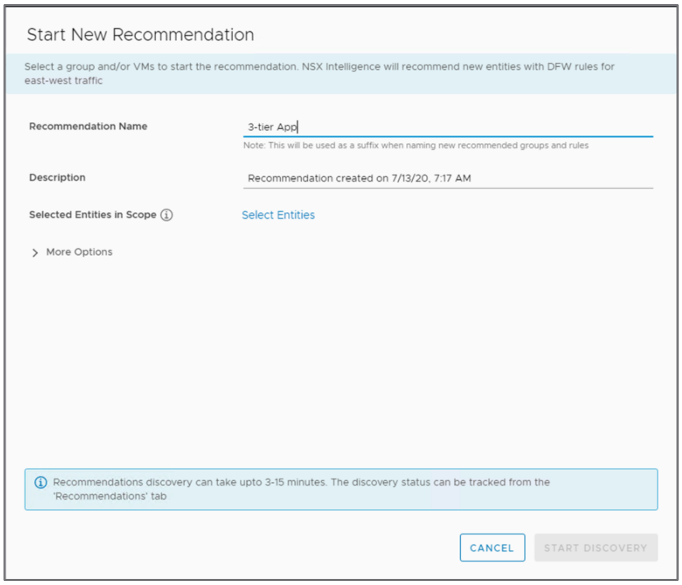
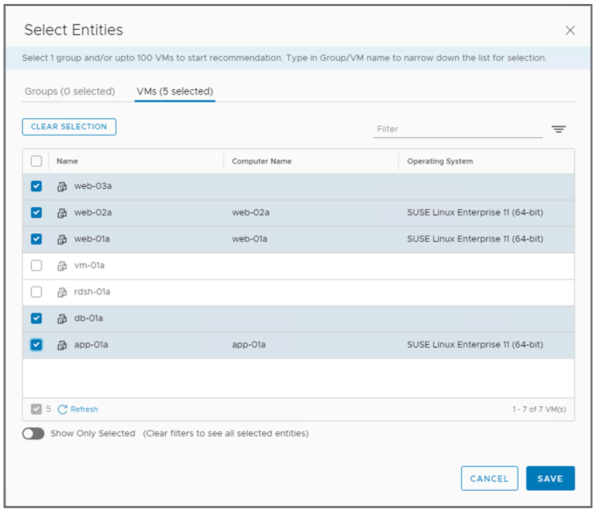


3. Define or modify the VMs that are to be used as the boundary for the security policy recommendation.

    a. Click **Select Entities** in Select Entities in Scope**.** In the Select Entities dialog box, select the Groups or VMs that you want to use as the boundary for the analysis.

    In NSX Intelligence 1.1, you can select up to one group and/or up to 100 VMs to use for the recommendation boundary.

    b. Click **SAVE**.

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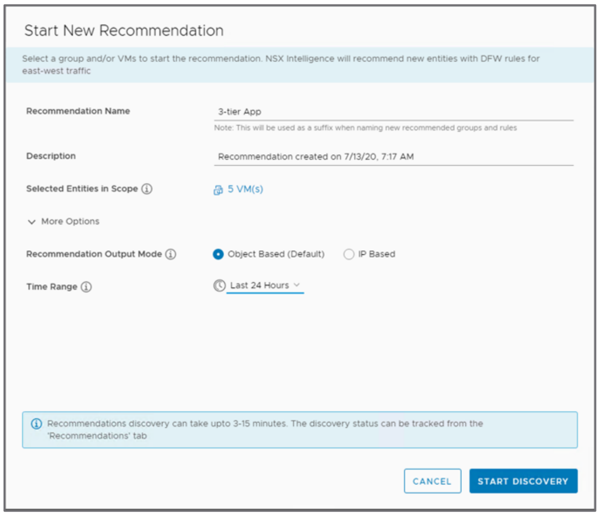
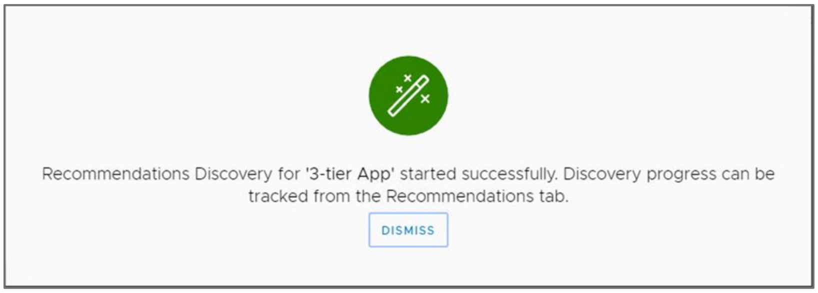
4. Expand **More Options** and change the default value for the **Recommendation Output Mode** and **Time Range**, if necessary. The default Time Range value is Last 1 Month, which means the network traffic flows that occurred in the last one month between the selected Groups or VMs are used during the recommendation analysis.

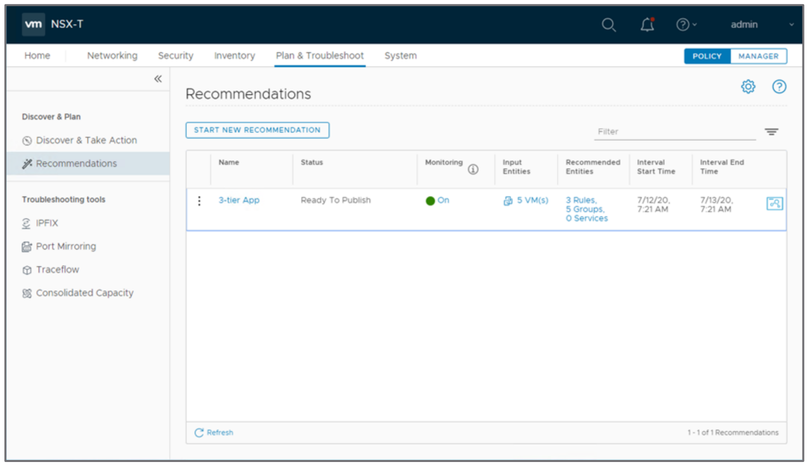


5. Click **START DISCOVERY**. The pop-up message is displayed.



6. Click **DISMISS**

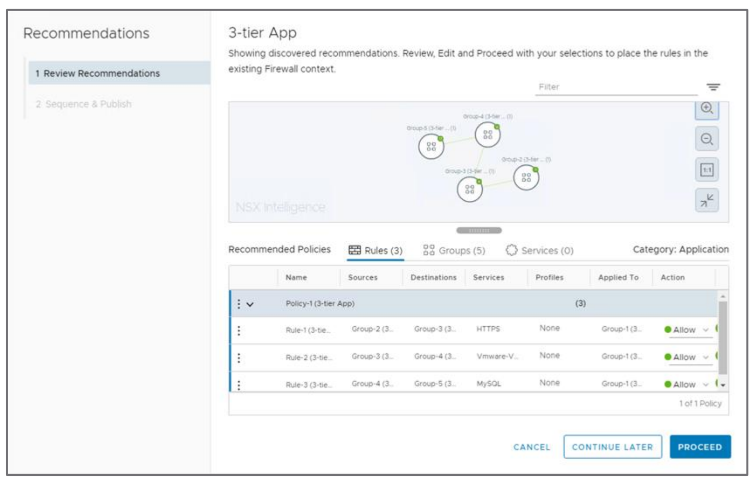
7. Recommendations are processed serially. On average, it can take anywhere from 1 to 15 minutes to finish each recommendation, depending on the number of VMs per recommendation and whether there are other recommendations that are pending to be processed. The status can be tracked from the Recommendation tab. The status progresses from **Waiting**, to **Discovery in Progress**, and finally to **Ready to Publish**.



8. After the generated NSX Intelligence recommendation reaches to **Ready To Publish** status, you can review the recommendation, modify it if necessary, and decide whether to publish it.

9. Review and manage the details of the recommendation.

    a. Click the recommendation’s name. The **Recommendation** wizard is displayed.



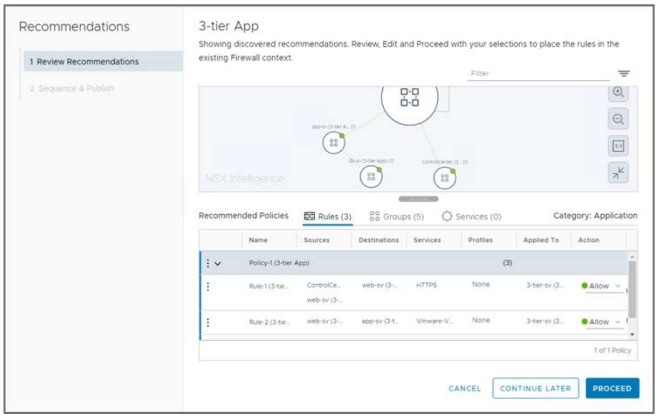
    b. In the **Rules** tab, review the firewall rule details. To modify any of the details, click the value in the appropriate column and select the edit (pencil) icon.

    c. In the **Groups** tab, click the link in the Members column to review the details about the VMs and IPs that were set for the group recommendation. Click the **menu icon** (three-dots) next to the group’s name and select **Edit** to modify the group recommendation.

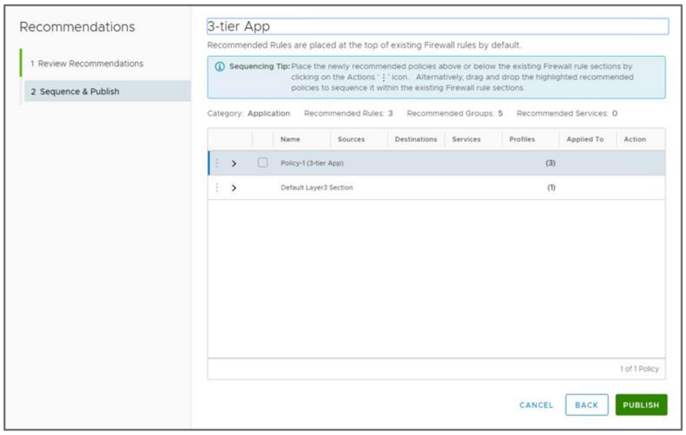
d. In the **Recommended Groups** tab, review the details and click the **menu icon** (three-dots) next to the service's name and select **Edit** to modify the name or description.

10. In the Place rules in FW context pane, you can change the order in which the rule recommendation is to be applied with the existing firewall rules. Drag the rule, and drop to move to the ideal place.

11. Click **PROCEED**.

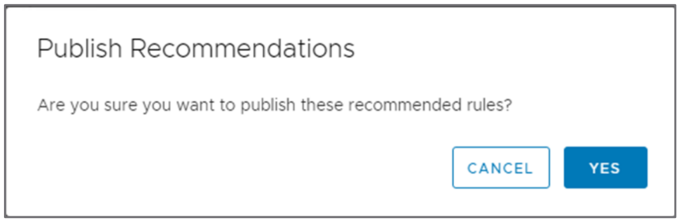


12. You can choose the newly-recommended policies above or below the existing firewalls rules sections.

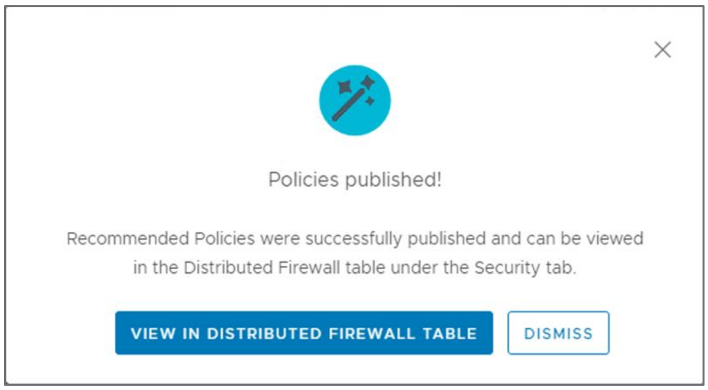


13. Click **PUBLISH**.

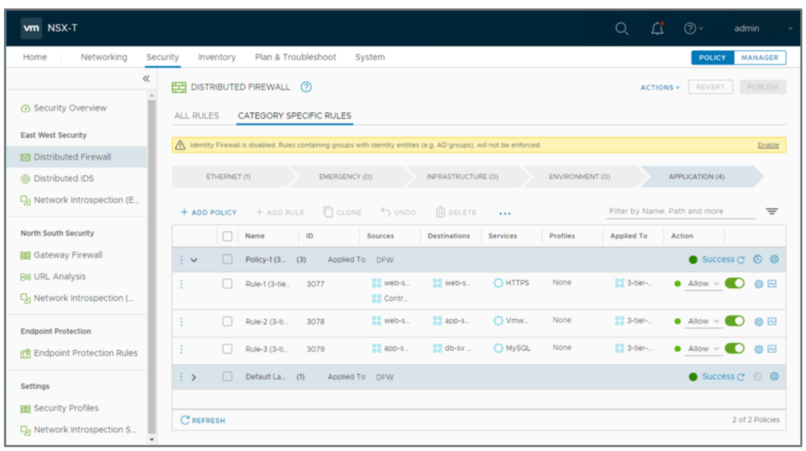
14. In the **Publish Recommendations** dialog box, click **YES**.



15. To verify that the security policies have been published successfully, click **VIEW IN DISTRIBUTED FIREWALL TABLE**.



16. The published rules can be viewed in the Distributed Firewall tables.



For more details, refer to [Working with NSX Intelligence Recommendations]

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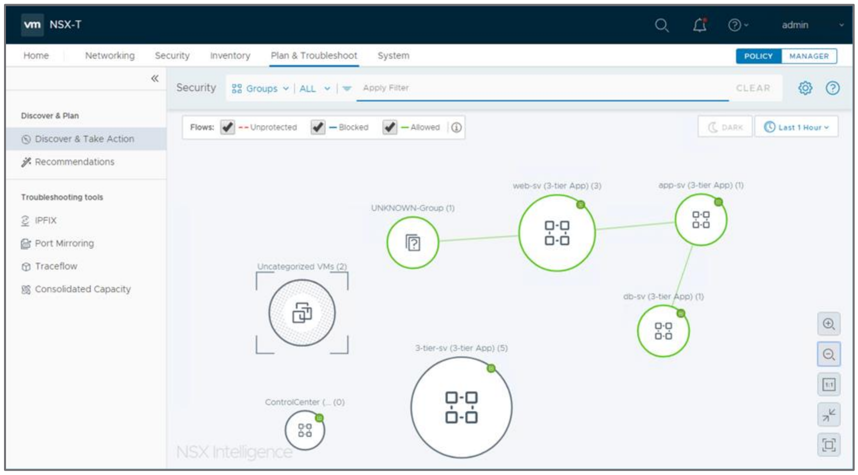
Validating Recommendations

After you have published the rule recommendations, the visualization continues to display the affected flows between the VMs as red-hued arrows (Unprotected Flows) until new flows are generated between the affected VMs.



The visualization only reports traffic flows based on the time when they occurred on the host and does not reflect the rule set published after those traffic flows occurred. After the rule set is published and new traffic flows are generated, the new flows are displayed as green-hued arrows (Allowed Flows).

To view the flows in detail, click on a group to the which the VM with the allowed flows belongs. VMs and allowed flows are displayed.

For more information about using and managing NSX Intelligence, refer to Working with NSX Intelligence Recommendations

Appendix

Troubleshooting NSX Intelligence

If you encounter any issues installing or working with NSX Intelligence, please refer to the following documents:
Troubleshooting NSX Intelligence Appliance Installation
Troubleshooting NSX Intelligence Usage Issues

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